Signals and Systems

Lecture 8: Feedback and Control

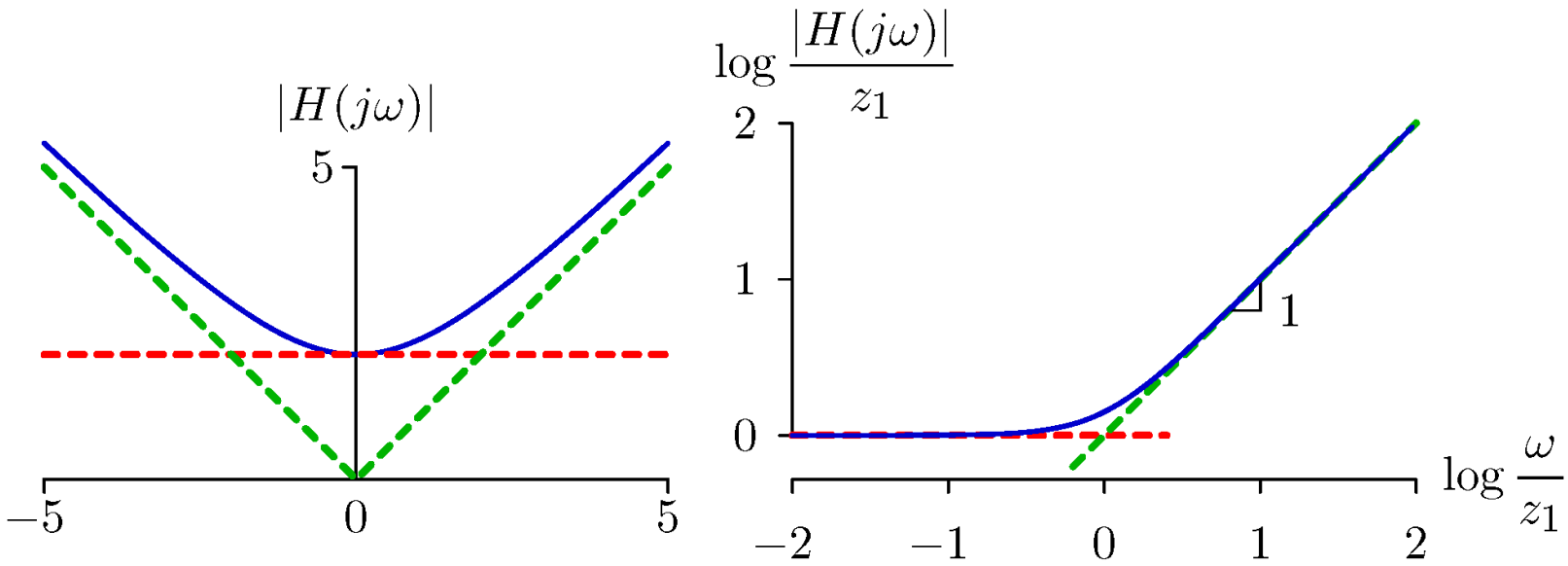
Instructor: Dr. Richeng Jin
Zhejiang University

04/03/2025
Partly adapted from the materials provided on
the MIT OpenCourseWare

Asymptotic Behavior: Isolated Zero

Two asymptotes provide a good approxmation on log-log axes.

$$H(s) = s - z_1$$



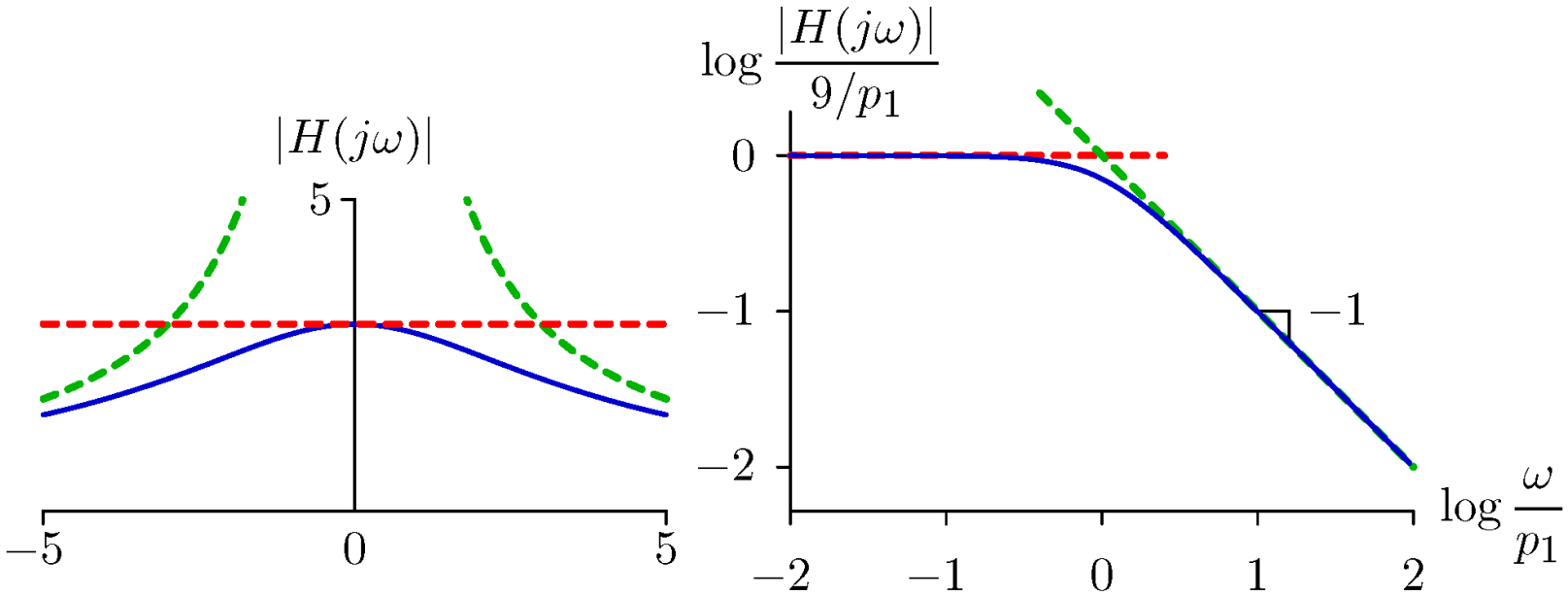
$$\lim_{\omega \to 0} |H(j\omega)| = z_1$$

$$\lim_{\omega \to \infty} |H(j\omega)| = \omega$$

Asymptotic Behavior: Isolated Pole

Two asymptotes provide a good approxmation on log-log axes.

$$H(s) = \frac{9}{s - p_1}$$



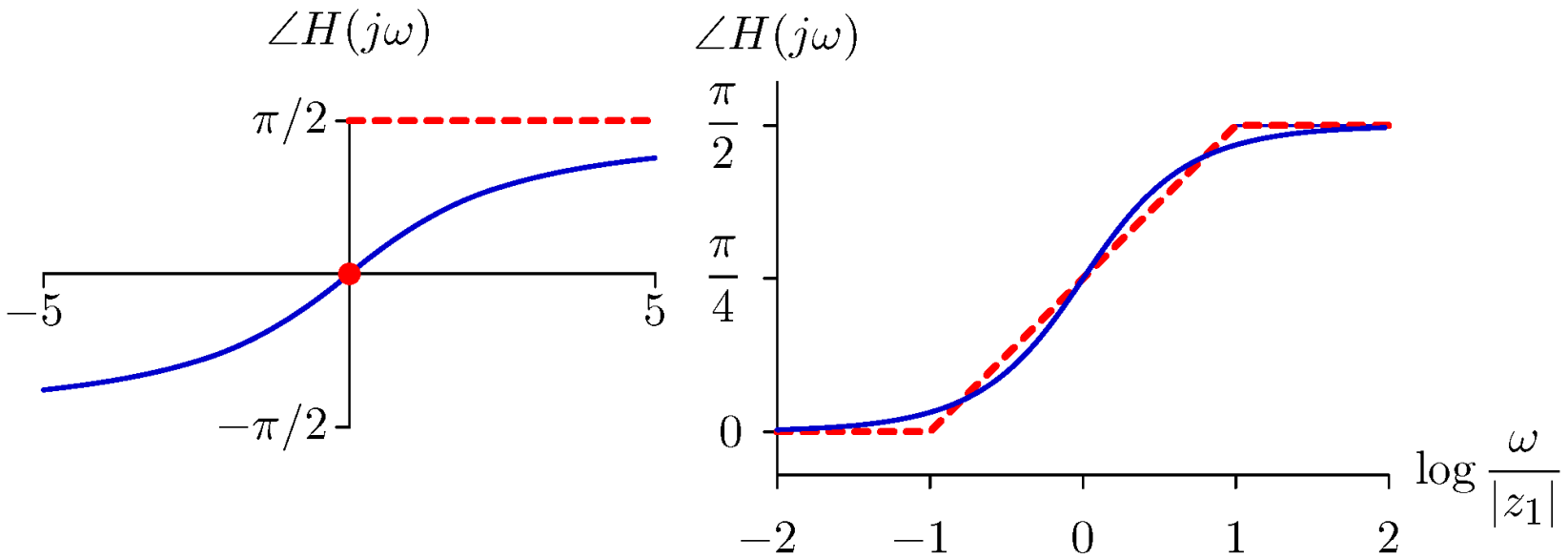
$$\lim_{\omega \to 0} |H(j\omega)| = \frac{9}{p_1}$$

$$\lim |H(j\omega)| = \frac{9}{}$$

Asymptotic Behavior: Isolated Zero

Three straight lines provide a good approxmation versus log ω.

$$H(s) = s - z_1$$



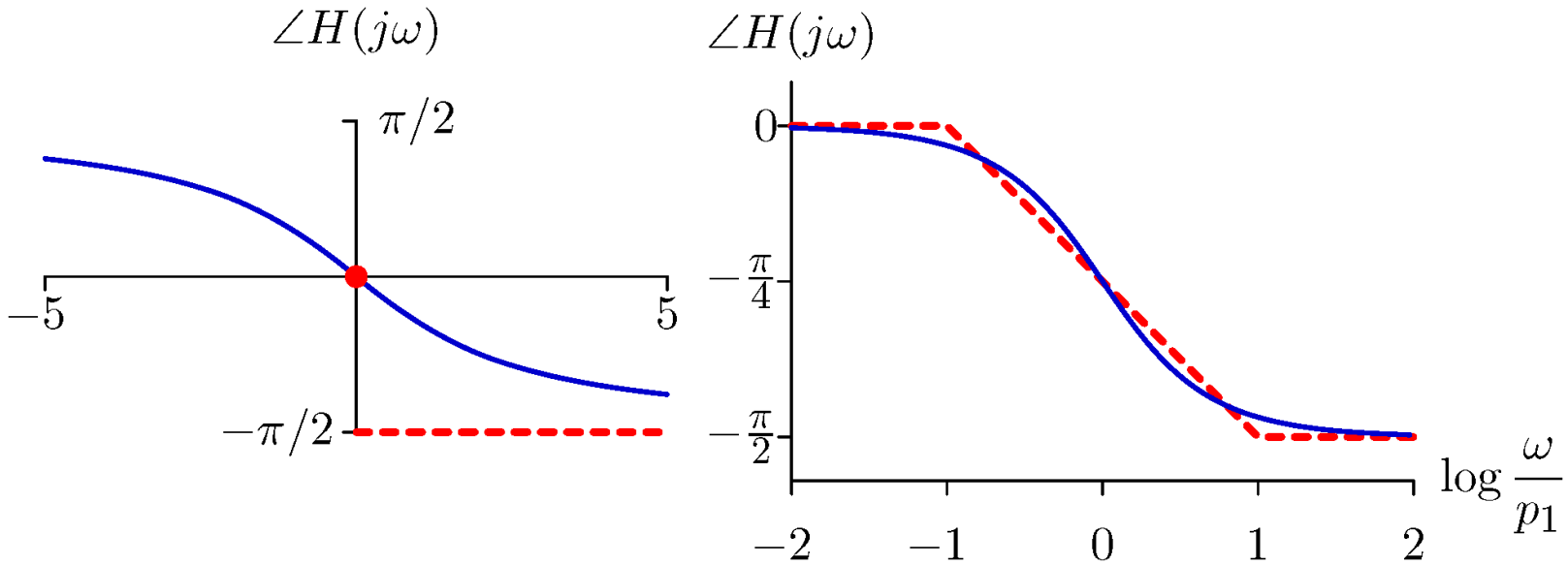
$$\lim_{\omega \to 0} \angle H(j\omega) = 0$$

$$\lim_{\omega \to \infty} \angle H(j\omega) = \pi/2$$

Asymptotic Behavior: Isolated Pole

Three straight lines provide a good approxmation versus log ω.

$$H(s) = \frac{9}{s - p_1}$$



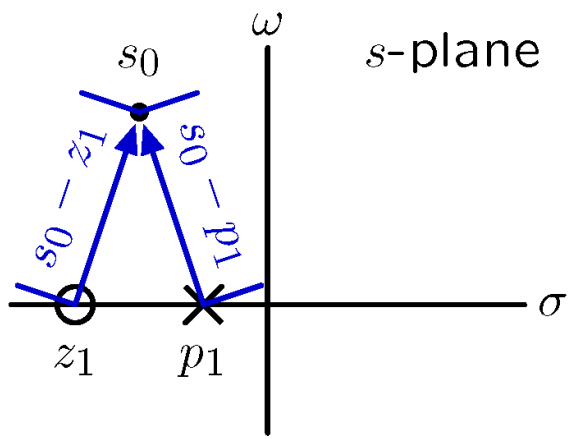
$$\lim_{\omega \to 0} \angle H(j\omega) = 0$$

$$\lim_{\omega \to \infty} \angle H(j\omega) = -\pi/2$$

Asymptotic Behavior of More Complicated Systems

The magnitude of a product is the product of the magnitudes.

$$|H(s_0)| = \left| K \frac{\prod_{q=1}^{Q} (s_0 - z_q)}{\prod_{p=1}^{P} (s_0 - p_p)} \right| = |K| \frac{\prod_{q=1}^{Q} |s_0 - z_q|}{\prod_{p=1}^{P} |s_0 - p_p|}$$



Bode Plot

The log of the magnitude is a sum of logs.

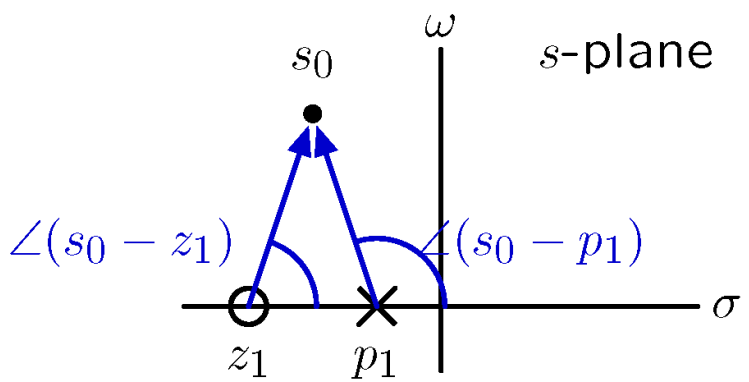
$$|H(s_0)| = \left| K \frac{\prod_{q=1}^{Q} (s_0 - z_q)}{\prod_{p=1}^{P} (s_0 - p_p)} \right| = |K| \frac{\prod_{q=1}^{Q} |s_0 - z_q|}{\prod_{p=1}^{P} |s_0 - p_p|}$$

$$\log |H(j\omega)| = \log |K| + \sum_{q=1}^{Q} \log |j\omega - z_q| - \sum_{p=1}^{P} \log |j\omega - p_p|$$

Bode Plot

The angle of a product is the sum of the angles.

$$\angle H(s_0) = \angle \left(K \frac{\prod_{q=1}^{Q} (s_0 - z_q)}{\prod_{p=1}^{P} (s_0 - p_p)} \right) = \angle K + \sum_{q=1}^{Q} \angle (s_0 - z_q) - \sum_{p=1}^{P} \angle (s_0 - p_p)$$



The angle of K can be 0 or π for systems described by linear differential equations with constant, real-valued coefficients.

From Frequency Response to Bode Plot

The magnitude of $H(j\omega)$ is a product of magnitudes.

$$|H(j\omega)| = |K| \frac{\prod_{q=1}^{Q} |j\omega - z_q|}{\prod_{p=1}^{P} |j\omega - p_p|}$$

The log of the magnitude is a sum of logs.

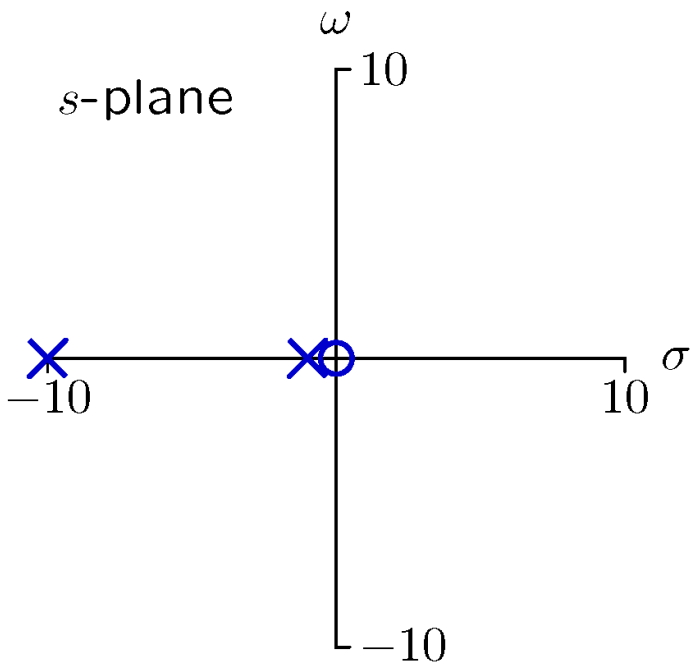
$$\log|H(j\omega)| = \log|K| + \sum_{q=1}^{Q} \log\left|j\omega - z_q\right| - \sum_{p=1}^{P} \log\left|j\omega - p_p\right|$$

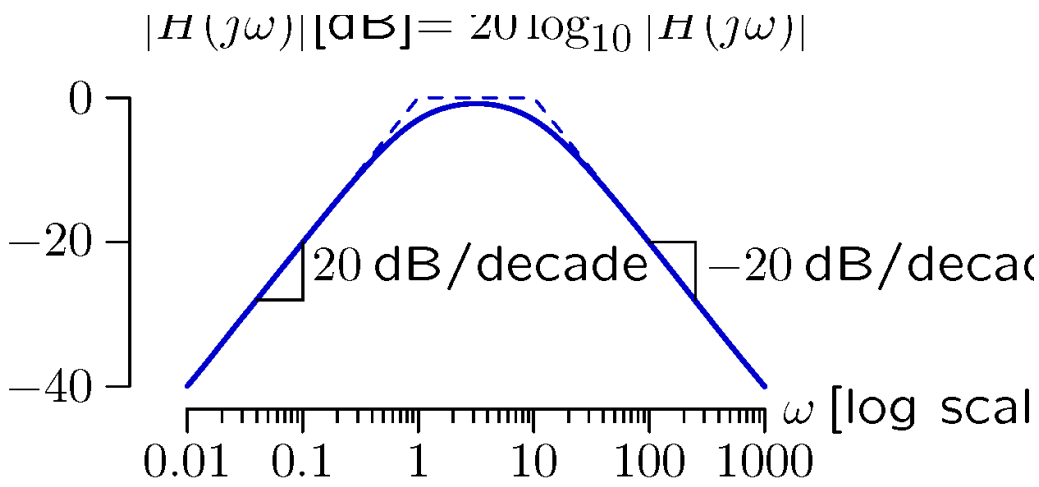
The angle of $H(j\omega)$ is a sum of angles.

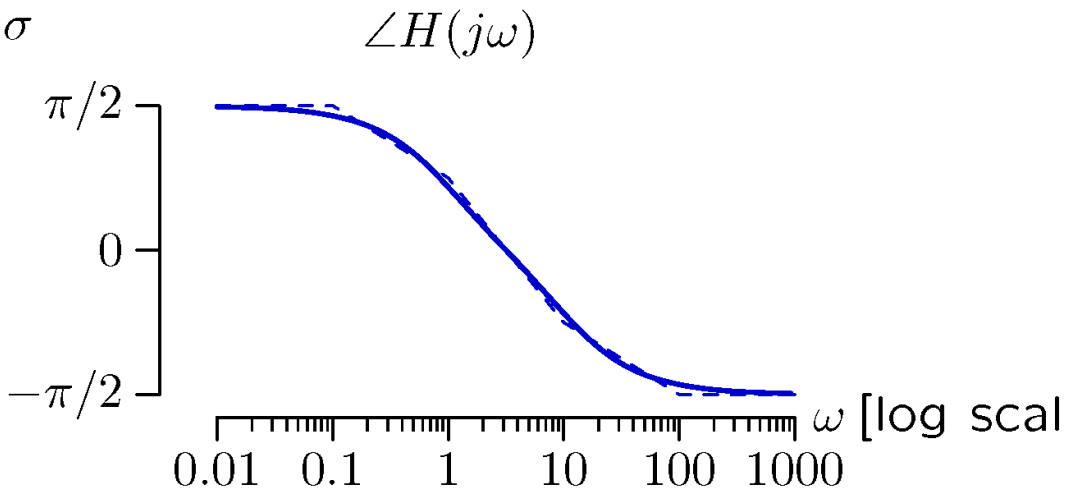
$$\angle H(j\omega) = \angle K + \sum_{q=1}^{Q} \angle\left(j\omega - z_q\right) - \sum_{p=1}^{P} \angle\left(j\omega - p_p\right)$$

Bode Plot: dB

$$H(s) = \frac{10s}{(s+1)(s+10)}$$

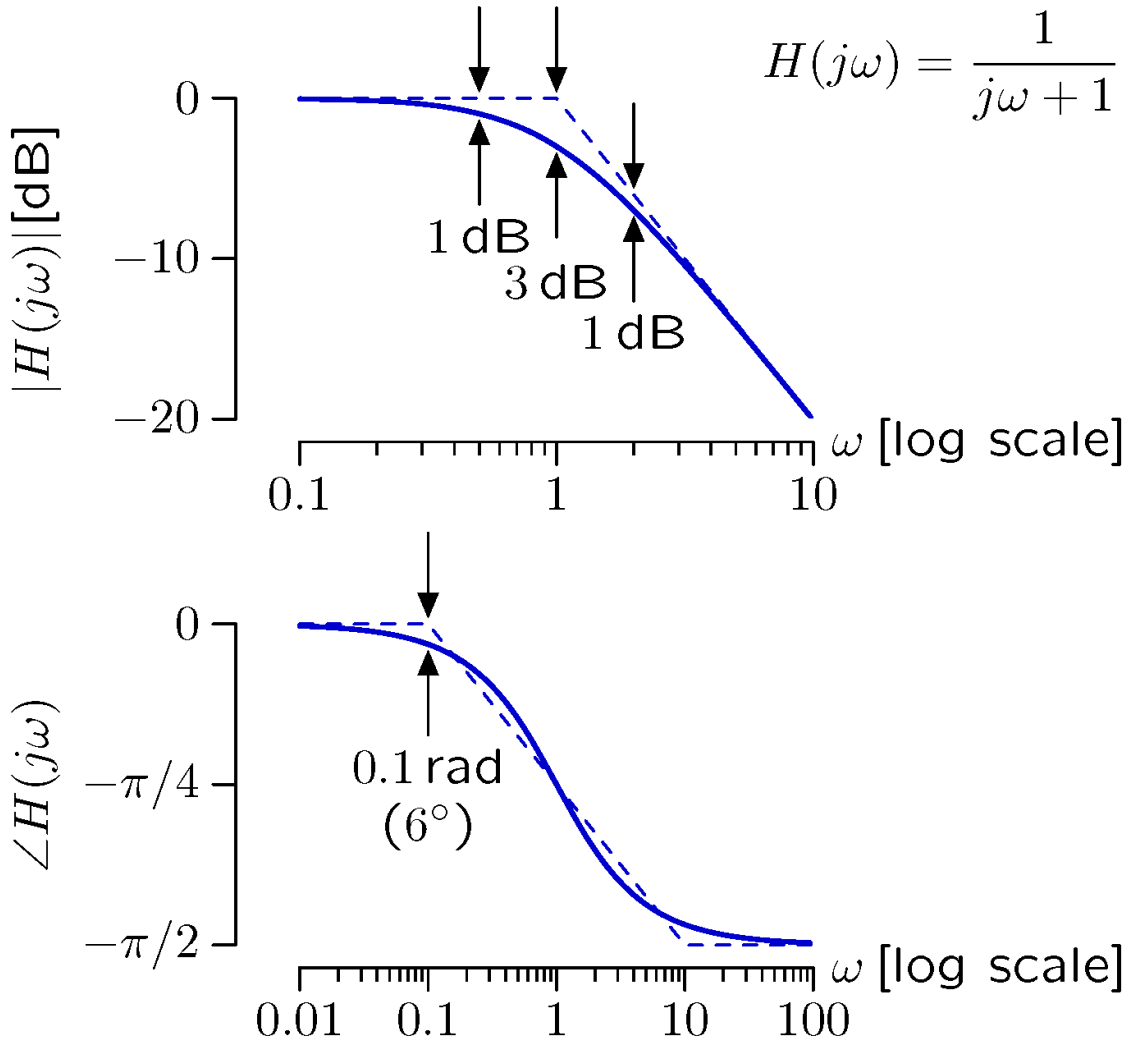






Bode Plot: Accuracy

The straight-line approximations are surprisingly accurate.

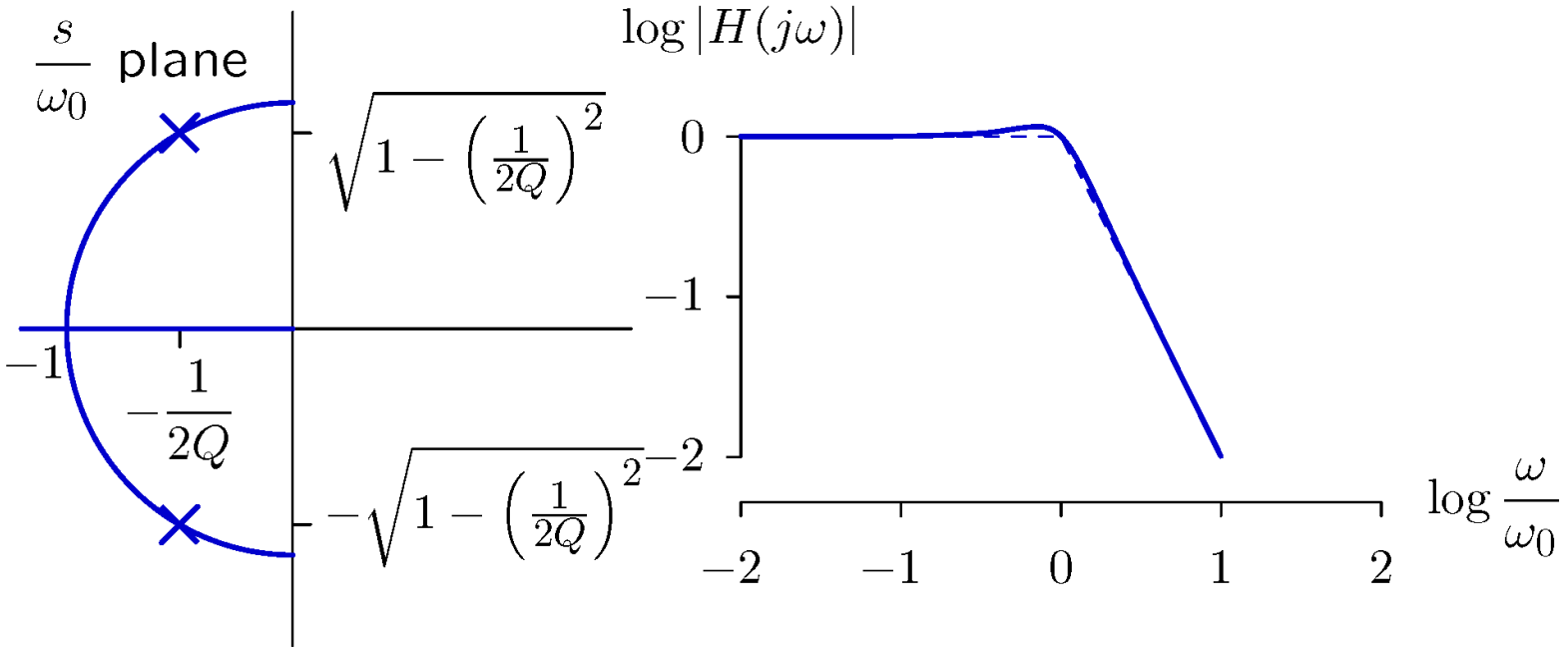


X	$20 \log_{10} X$
1	0 dB
$\sqrt{2}$	≈ 3 dB
2	≈ 6 dB
10	20 dB
100	40 dB

Frequency Response of a High-Q System

The frequency-response magnitude of a high-Q system is peaked.

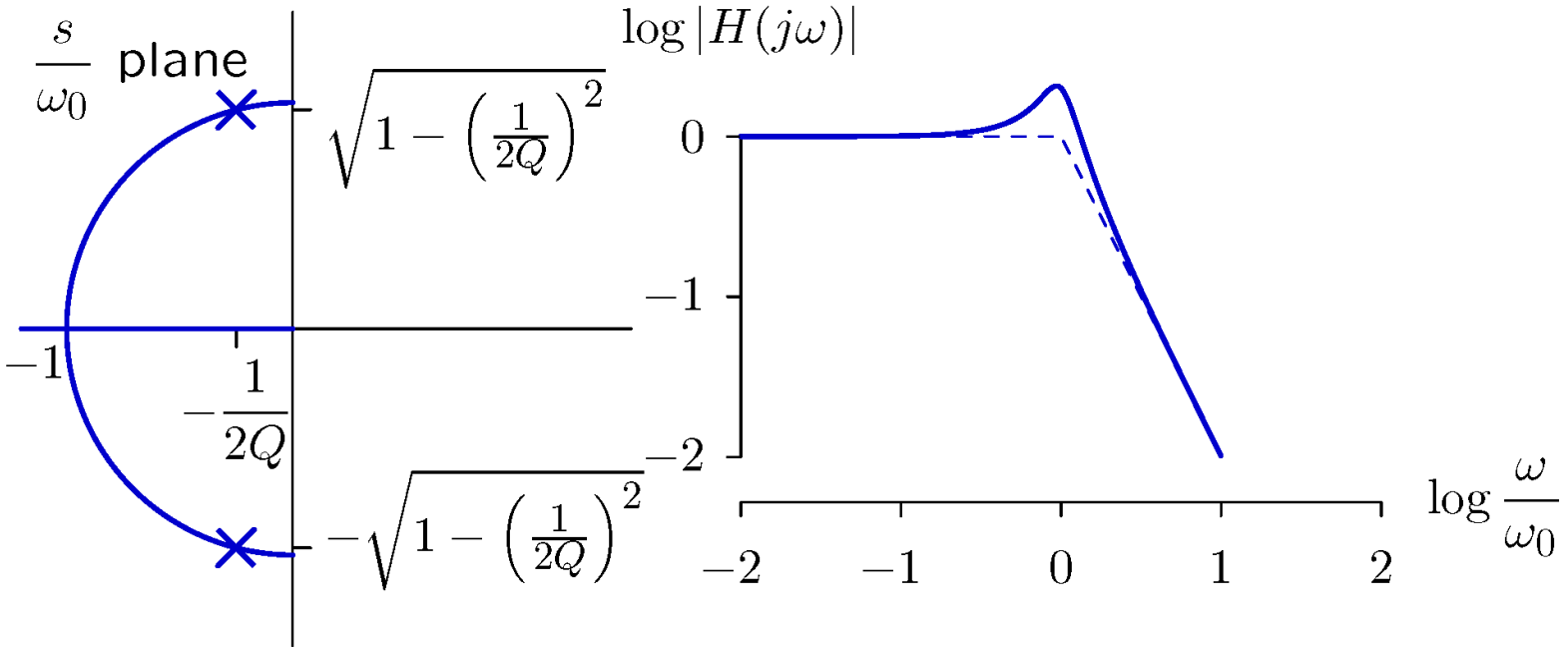
$$H(s) = \frac{1}{1 + \frac{1}{Q}\frac{s}{\omega_0} + \left(\frac{s}{\omega_0}\right)^2}$$



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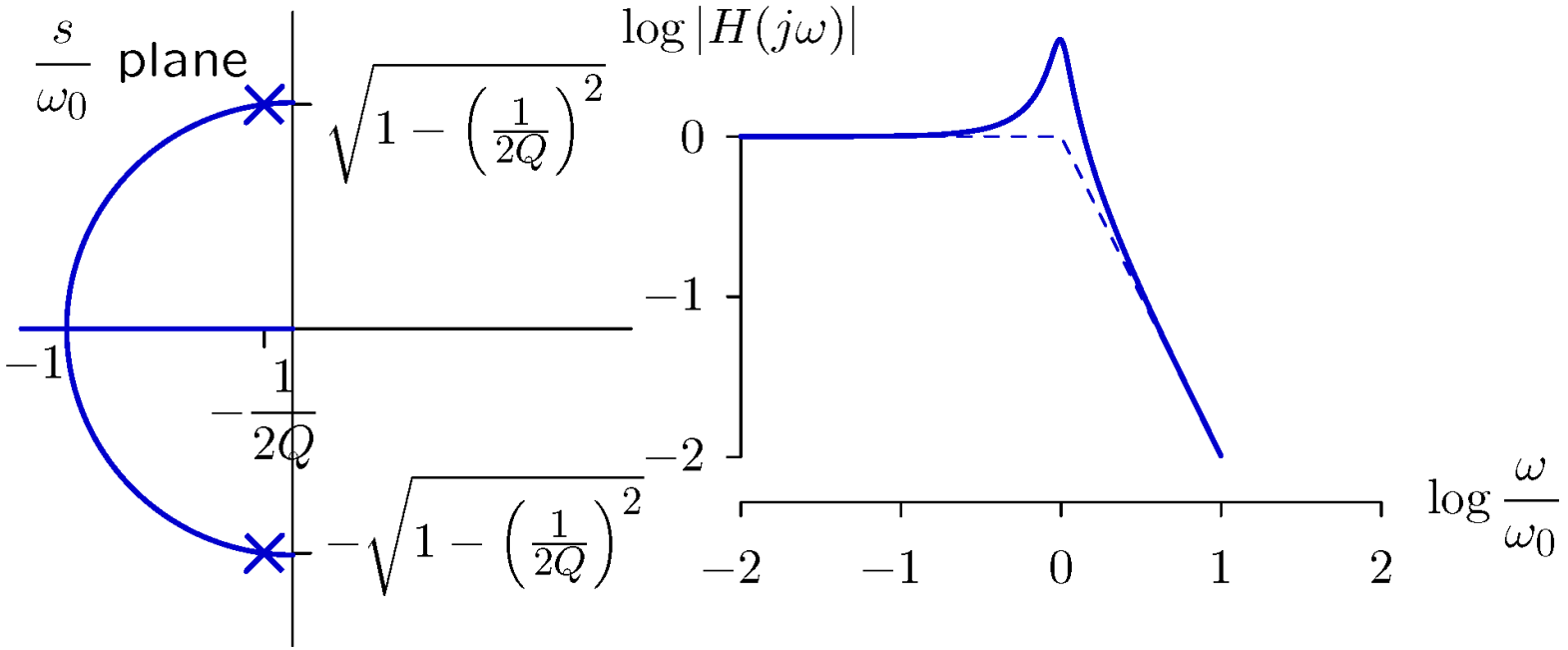
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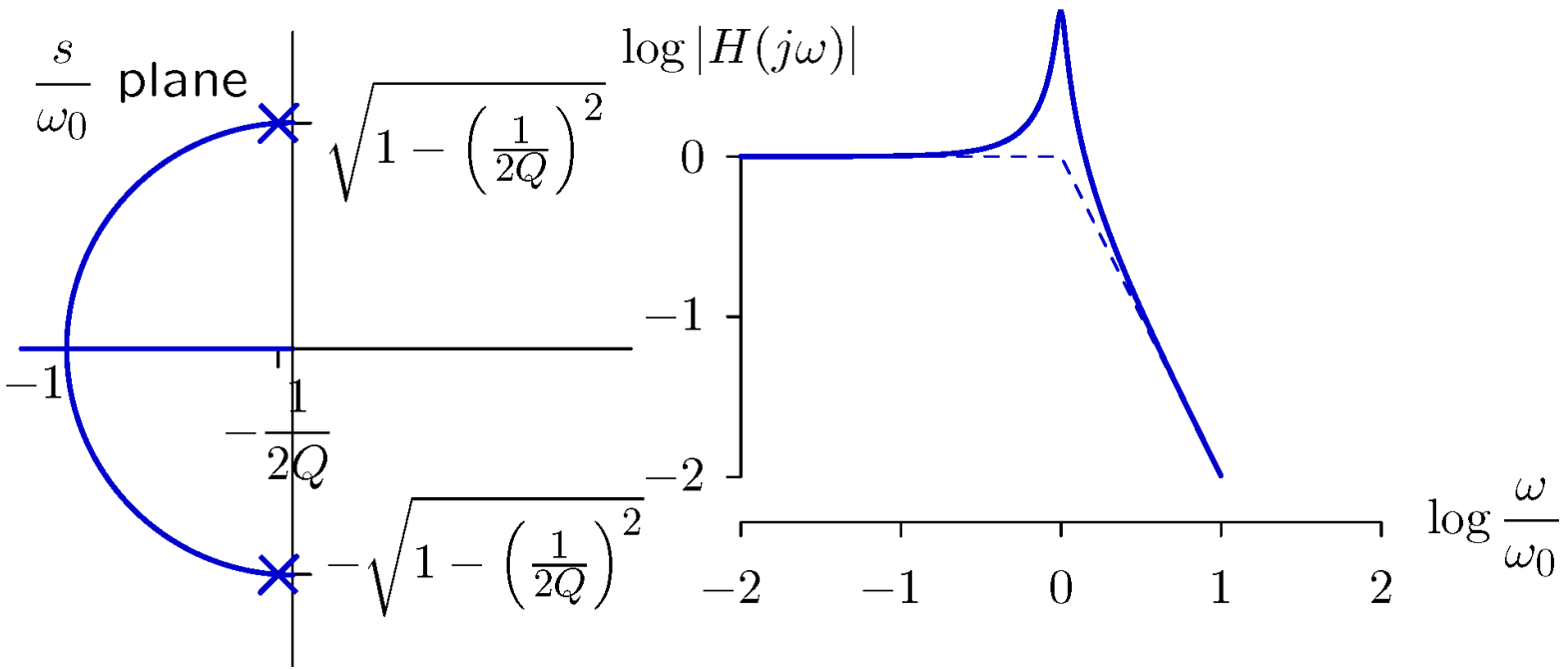
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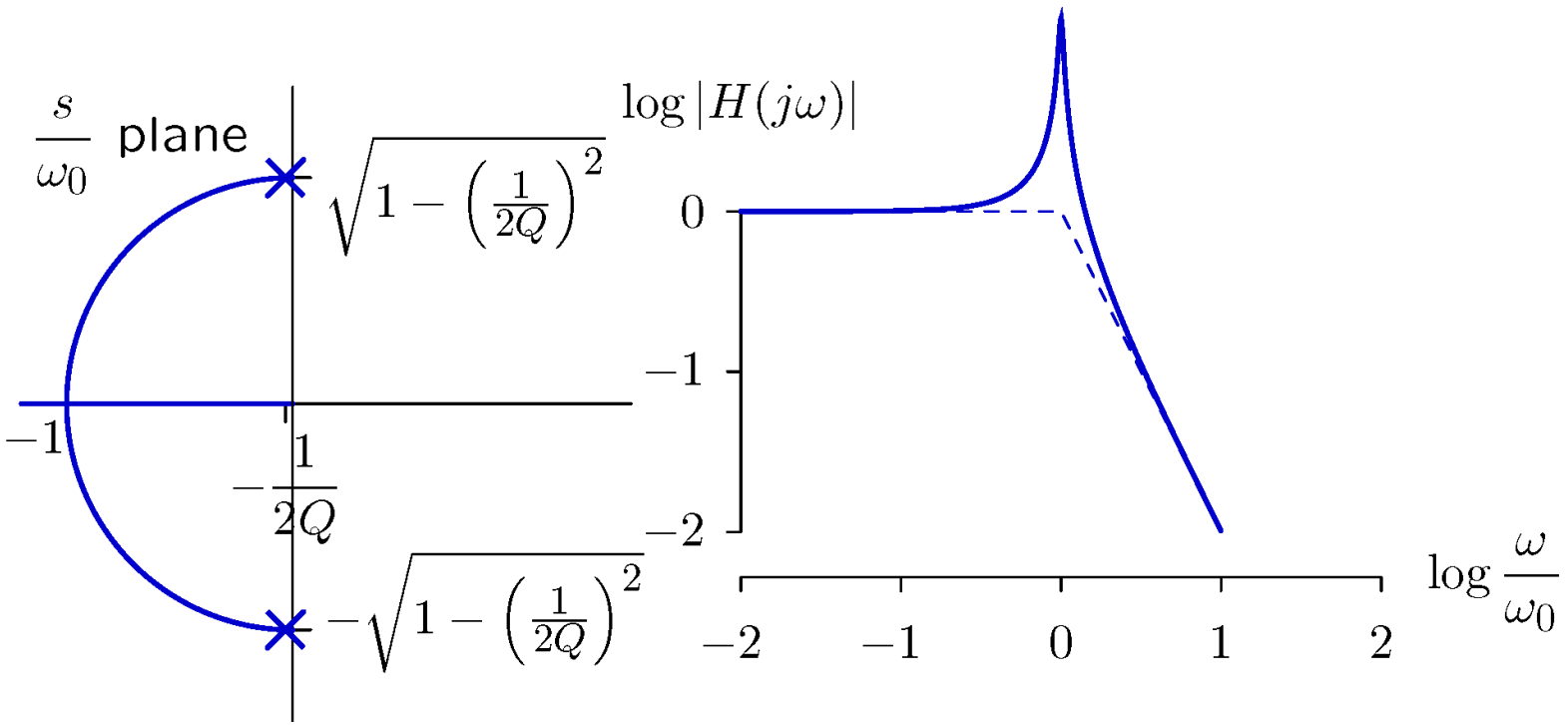
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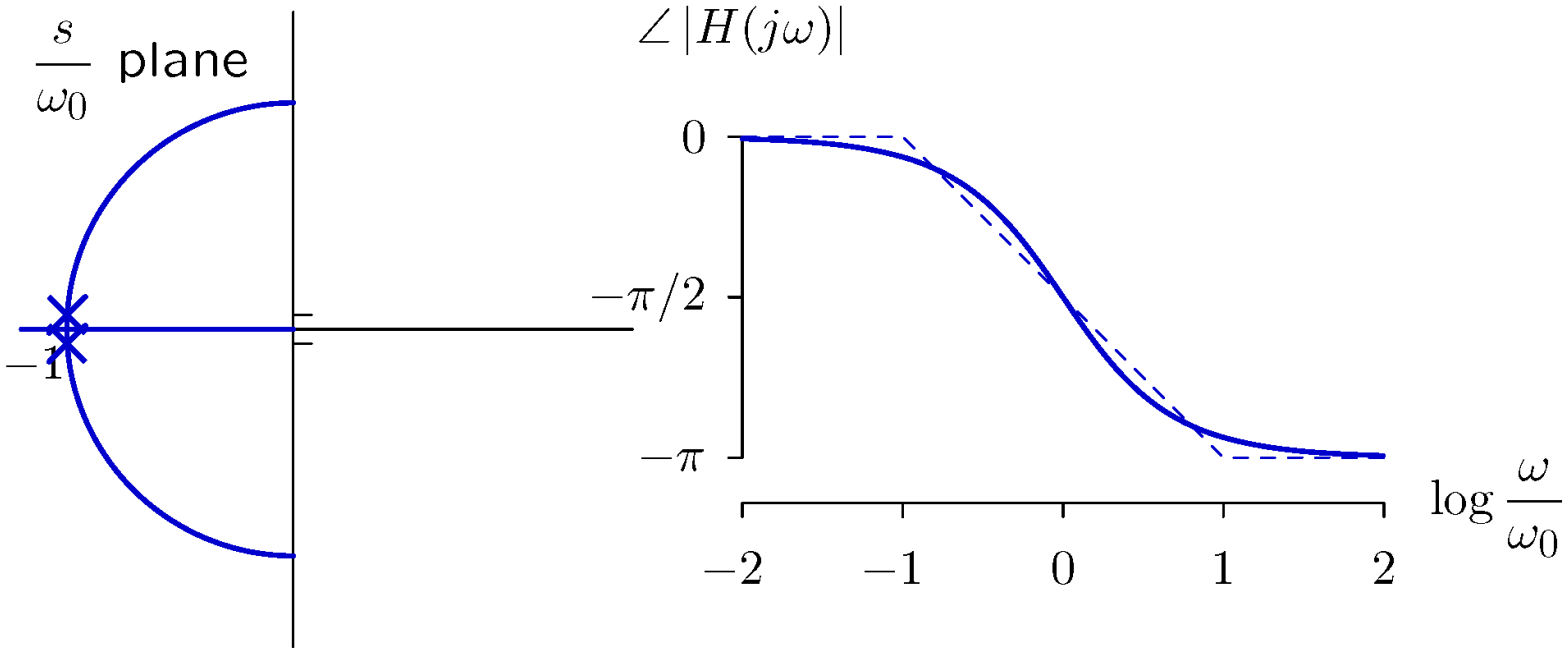
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Frequency Response of a High-Q System

As Q increases, the phase changes more abruptly with ω.

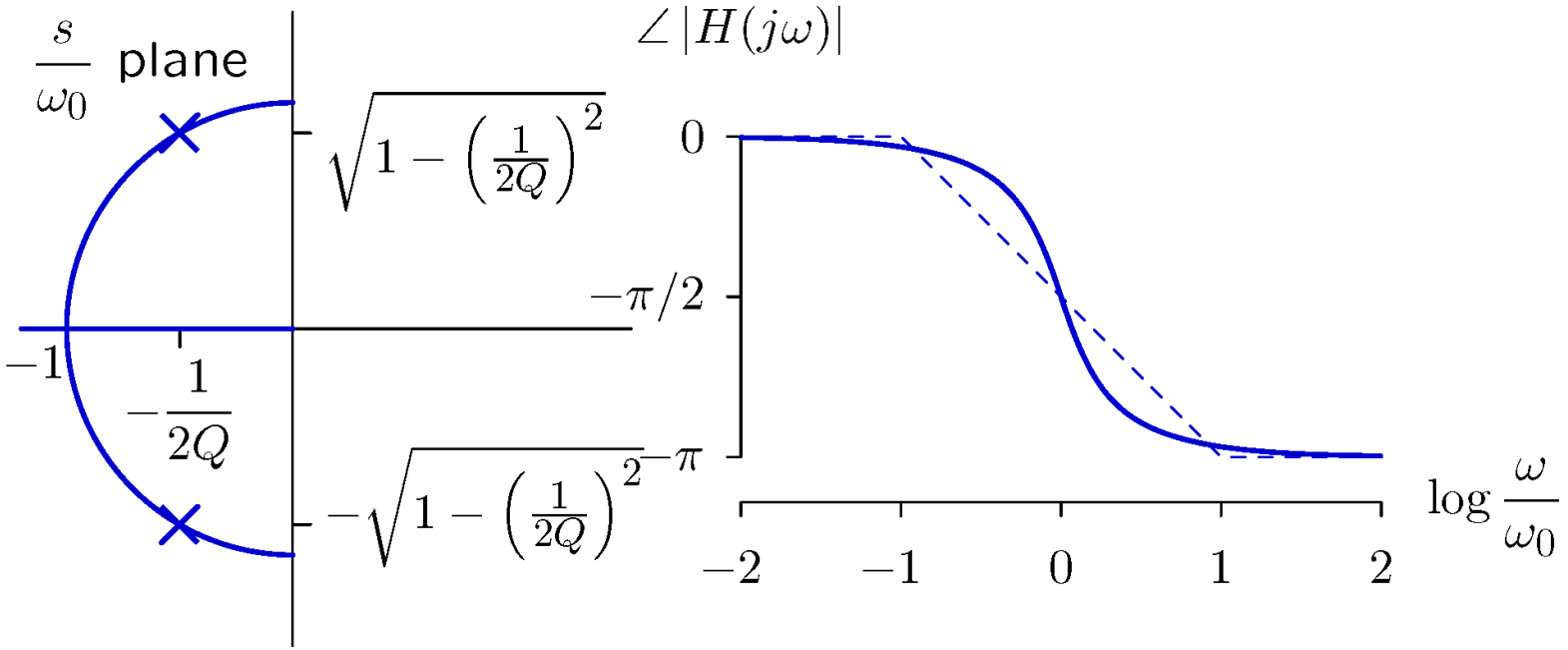
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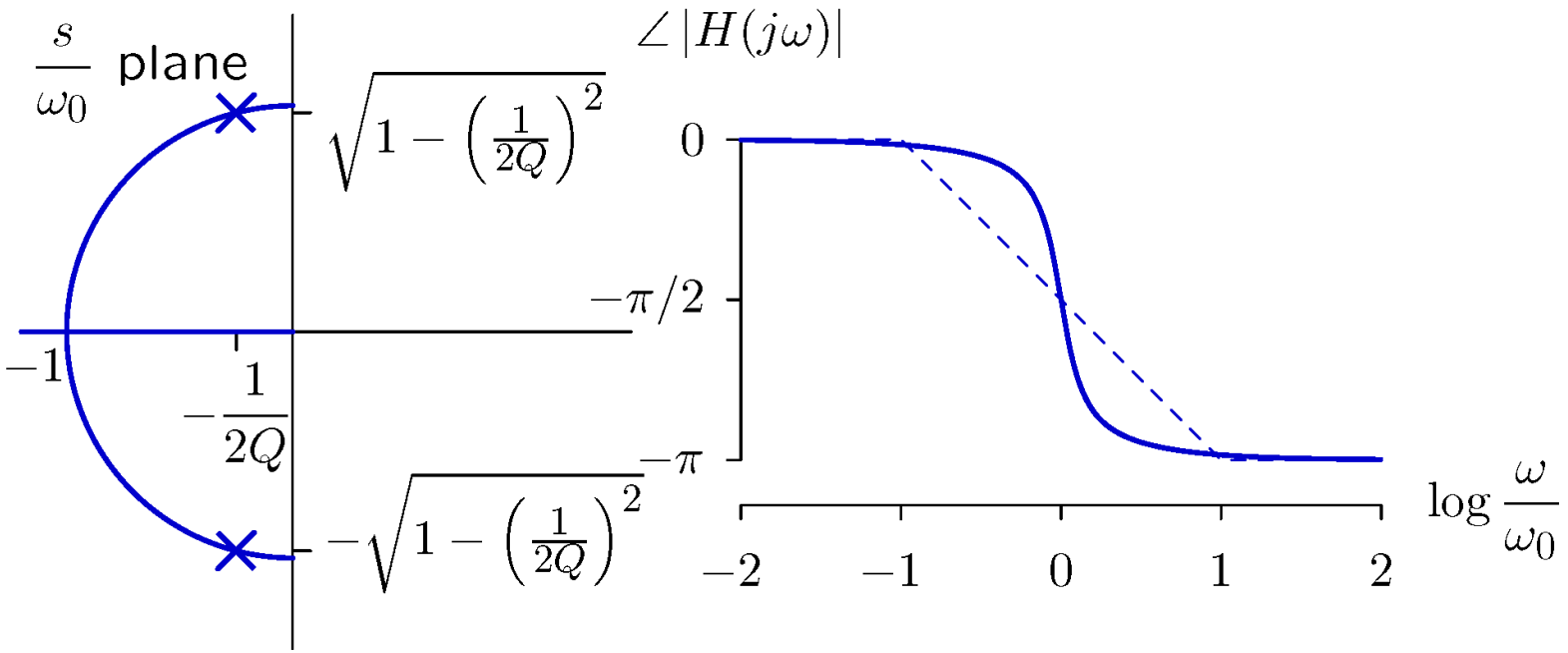
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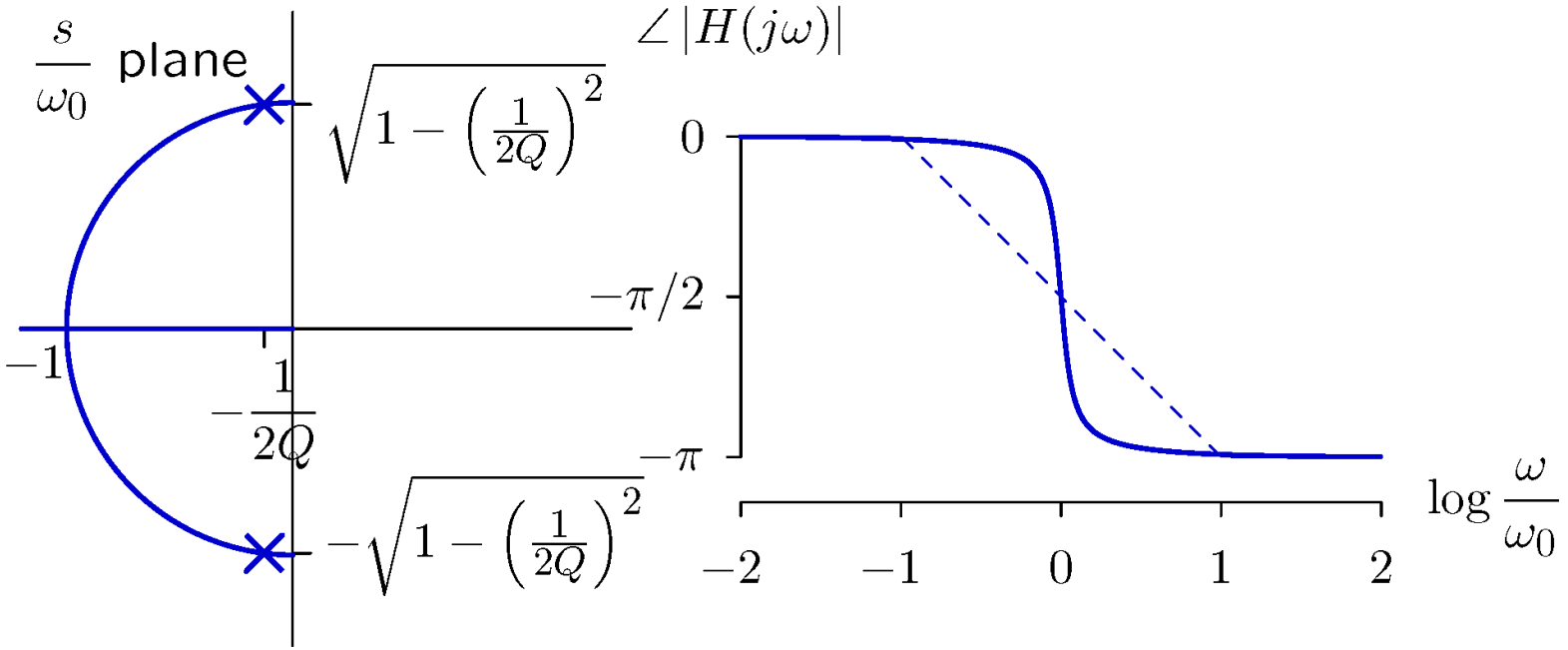
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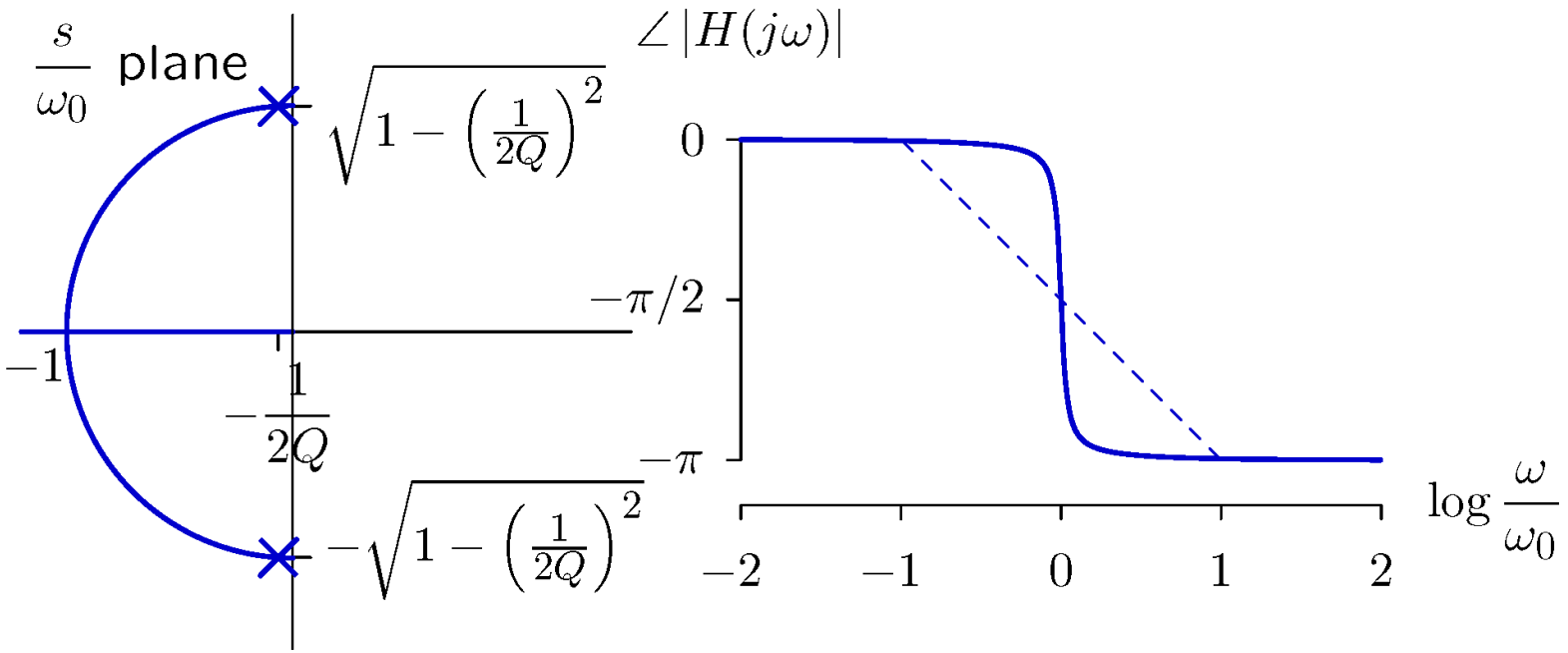
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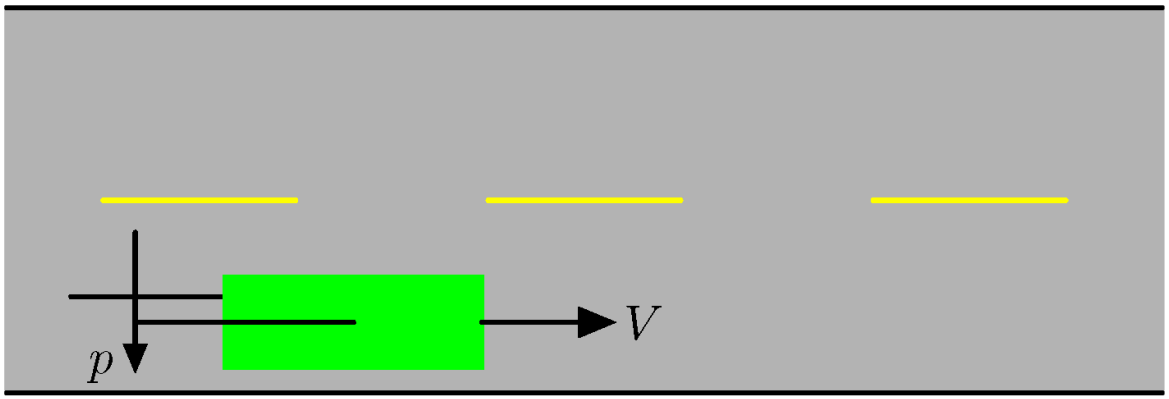


Today's goal

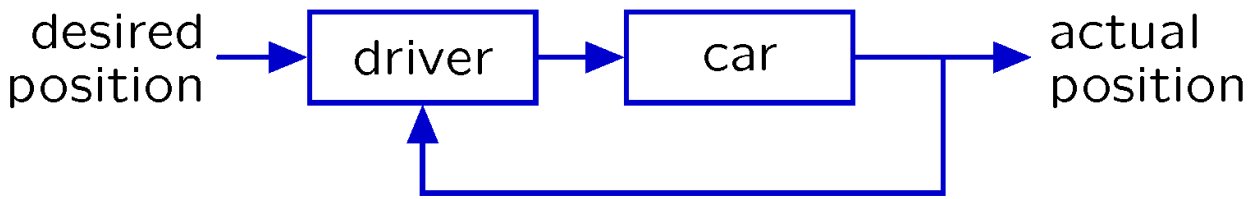
Use systems theory to gain insight into how to control a system.

Feedback and Control

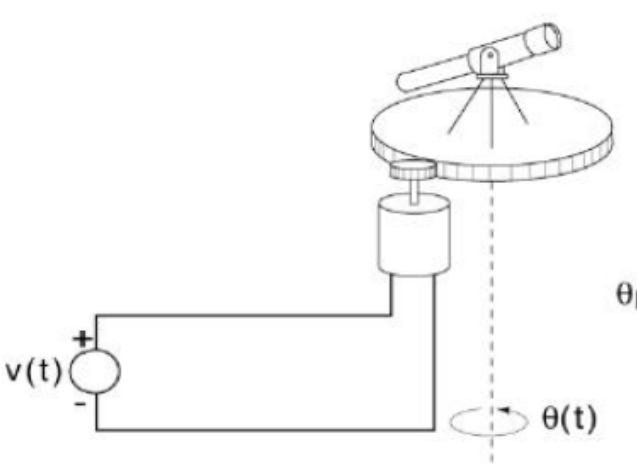
Feedback is pervasive in natural and artificial systems.

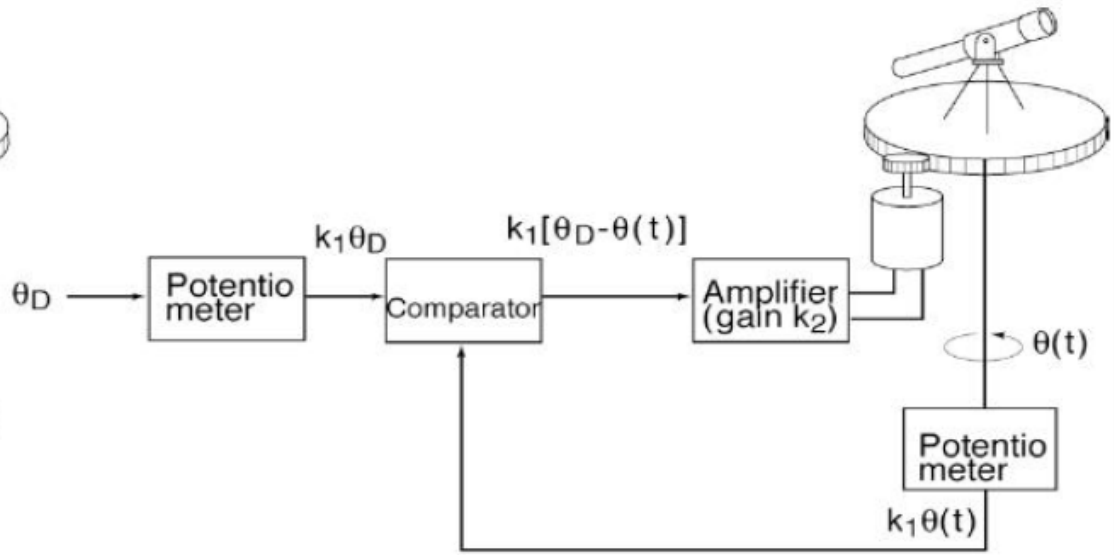


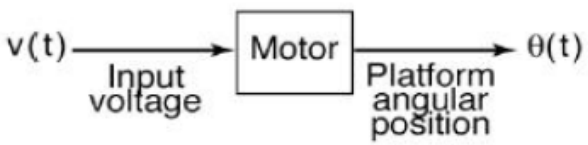
Turn steering wheel to stay centered in the lane.

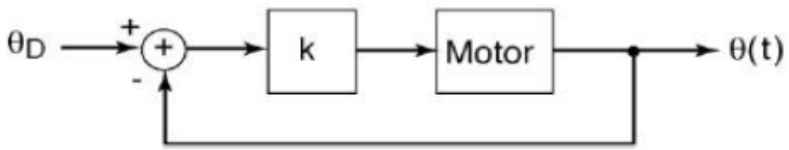


One Motivating Example — pointing a telescope







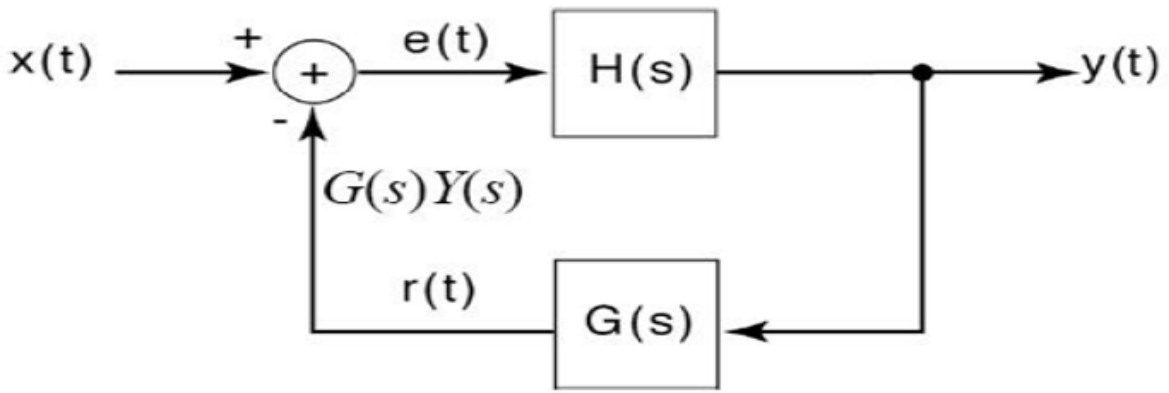


Open-loop System
Aim — shoot

Closed-loop Feedback System
Not done until it is pointed

System Function of a Closed-loop System

Example: A basic Feedback System — By its nature, we are dealing with real physical systems. ⇒ They are all *causal*.



$$E(s) = X(s) - R(s) = X(s) - G(s)Y(s)$$

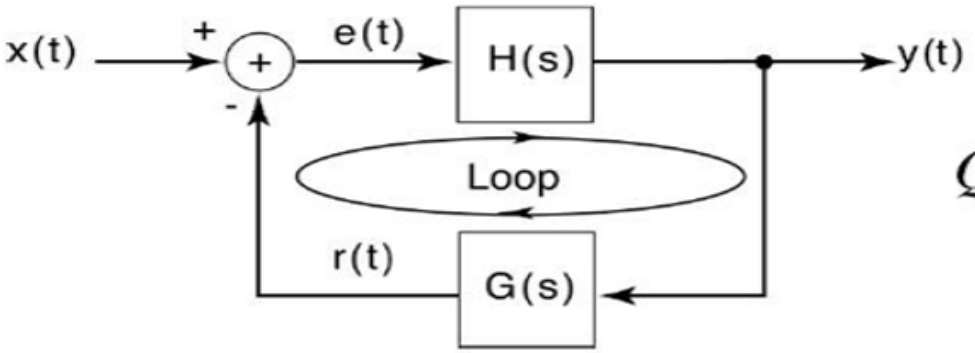
$$Y(s) = H(s)E(S) = H(s)[X(s) - G(s)Y(s)]$$

$$\Downarrow$$

$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$ — System function of the close-loop

General formula for a closed-loop system:

Black's Formula



$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$

Can show for any closed-loop systems, the system function is given by Black's formula (H. S. Black in the 1920's, along with Nyquist and Bode):

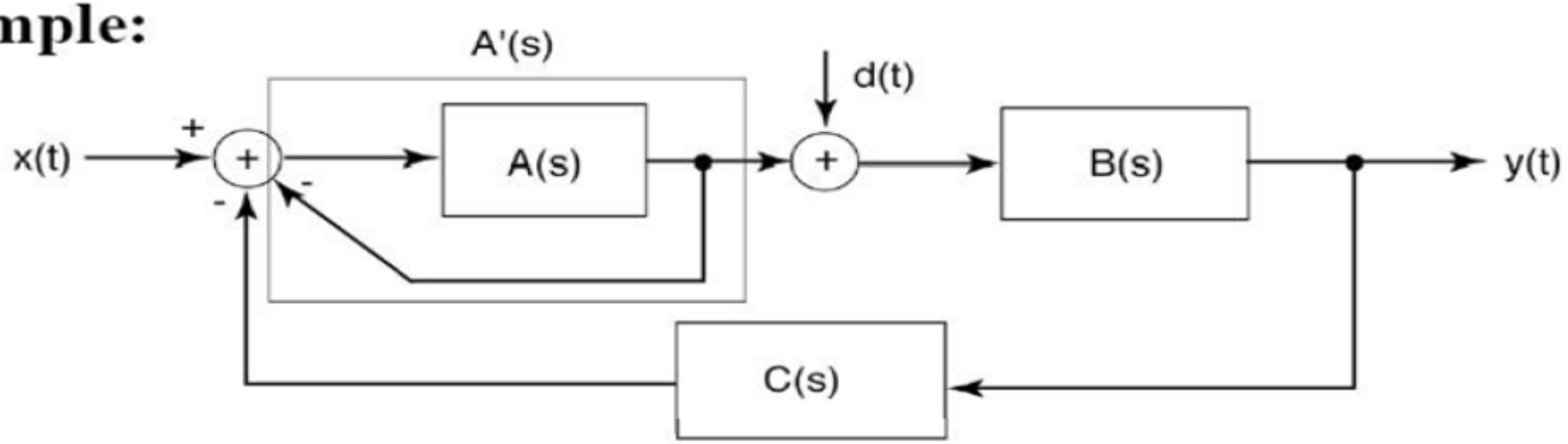
$$\text{Closed - loop system function} = \frac{\text{forward gain}}{1 \text{ - loop gain}}$$

Forward gain — total gain along the forward path from the *input* to the *output*
the gain of an adder is ≡ 1

Loop gain — total gain along the closed loop — shared by all the system functions

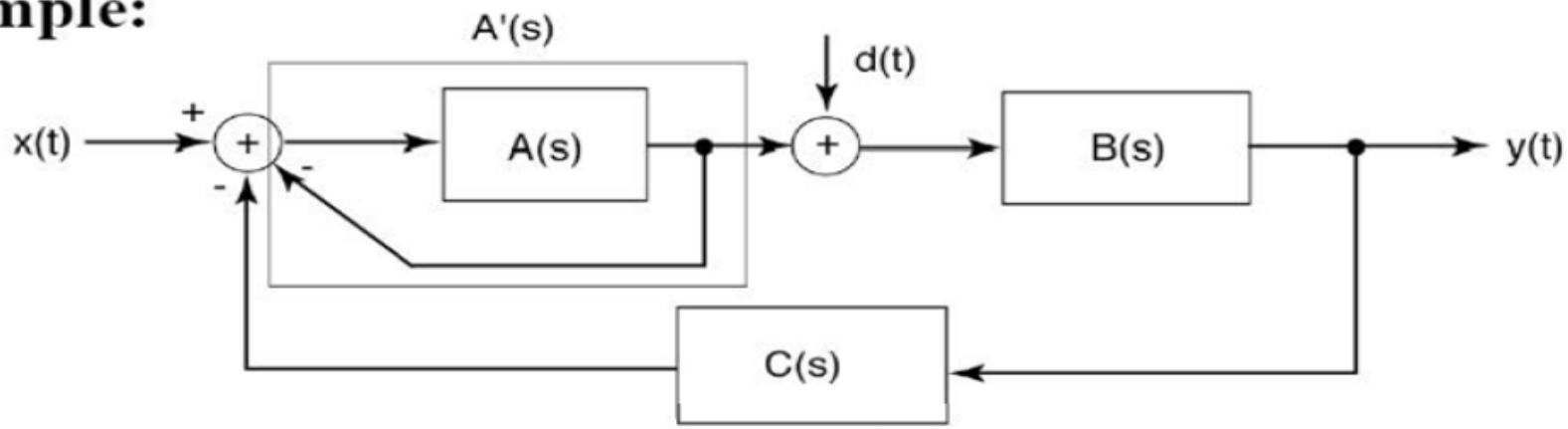
Applications of Black's Formula

Example:



Applications of Black's Formula

Example:



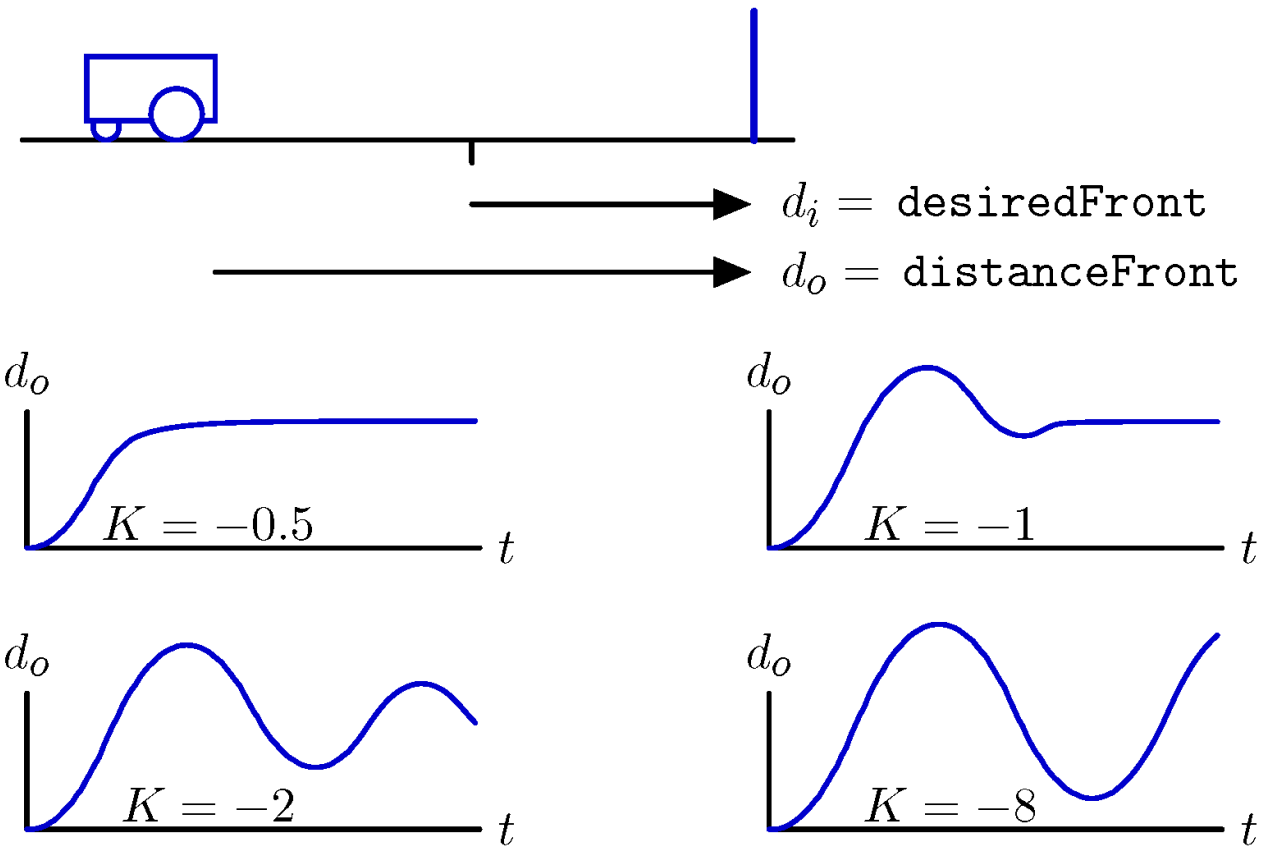
1) $$\frac{Y(s)}{X(s)} = ? = \frac{\text{Forward gain}}{1 \text{ - loop gain}} = \frac{A'B}{1 + A'BC}$$

$$A' = \frac{A}{1+A} \quad \Rightarrow \quad \frac{Y(s)}{X(s)} = \frac{AB}{1 + A + ABC}$$

2) $$\frac{Y(s)}{D(s)} = ? = \frac{\text{Forward gain}}{1 \text{ - loop gain}} = \frac{B}{1 + A'BC} = \frac{B(1+A)}{1 + A + ABC}$$

Example: wallFinder System

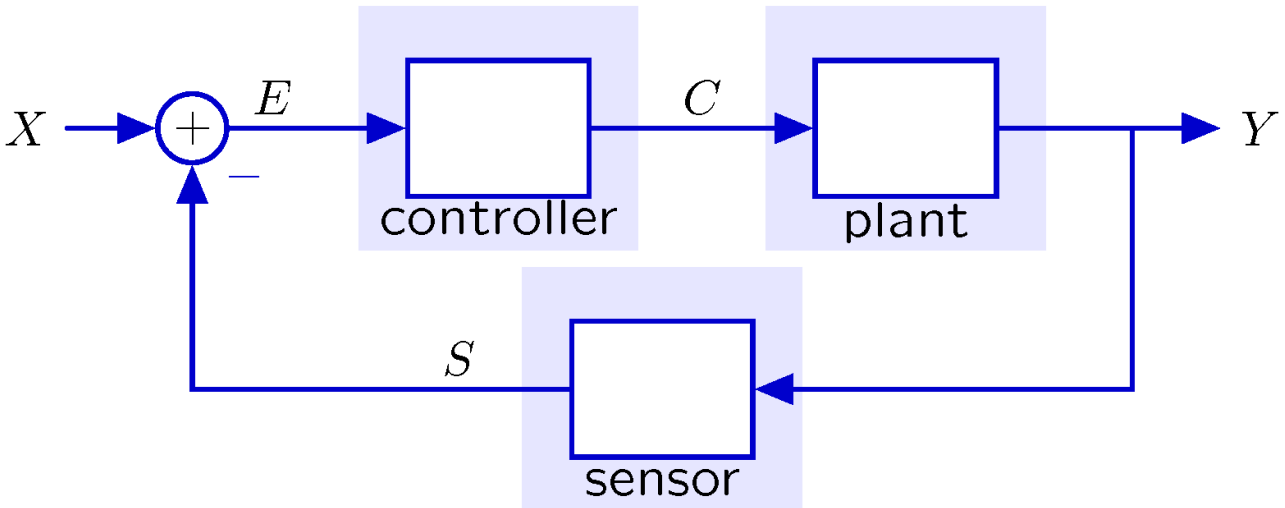
Approach a wall, stopping a desired distance d_i in front of it.



What causes these different types of responses?

Structure of a Control Problem

(Simple) Control systems have three parts.



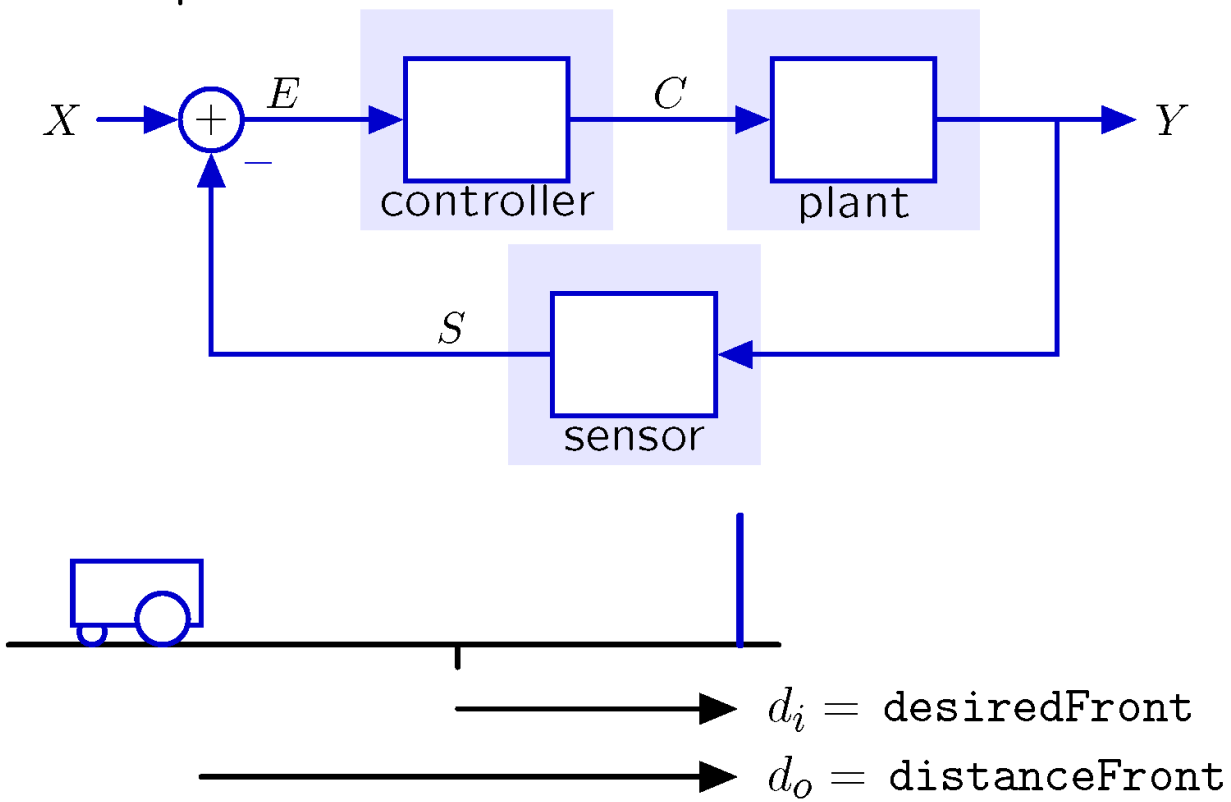
The **plant** is the system to be controlled.

The **sensor** measures the output of the plant.

The **controller** specifies a command C to the plant based on the *difference* between the input X and sensor output S.

Analysis of wallFinder System

Cast wallFinder problem into control structure.



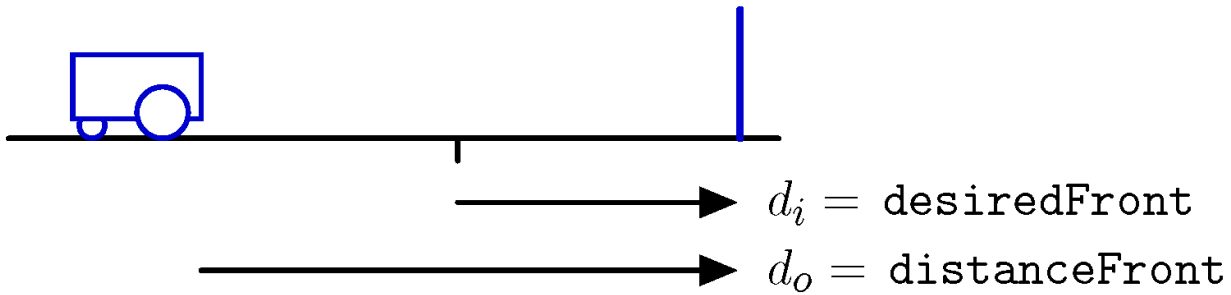
proportional controller: $v[n] = Ke[n] = K\big(d_i[n] - d_s[n]\big)$

locomotion: $d_o[n] = d_o[n-1] - Tv[n-1]$

sensor with no delay: $d_s[n] = d_o[n]$

Analysis of wallFinder System: Block Diagram

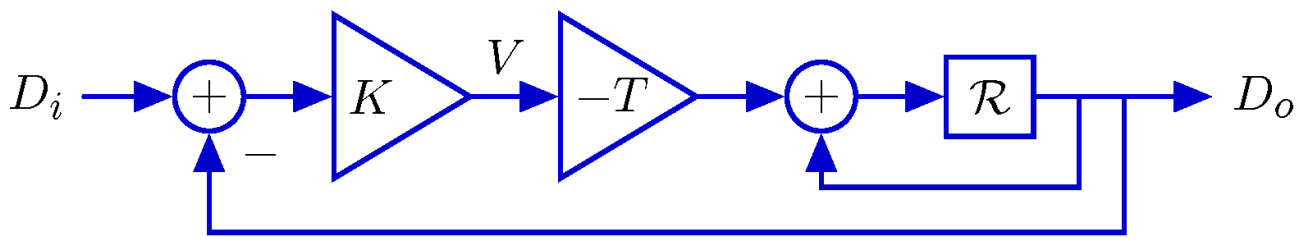
Visualize as block diagram.



proportional controller: $v[n] = Ke[n] = K\big(d_i[n] - d_s[n]\big)$

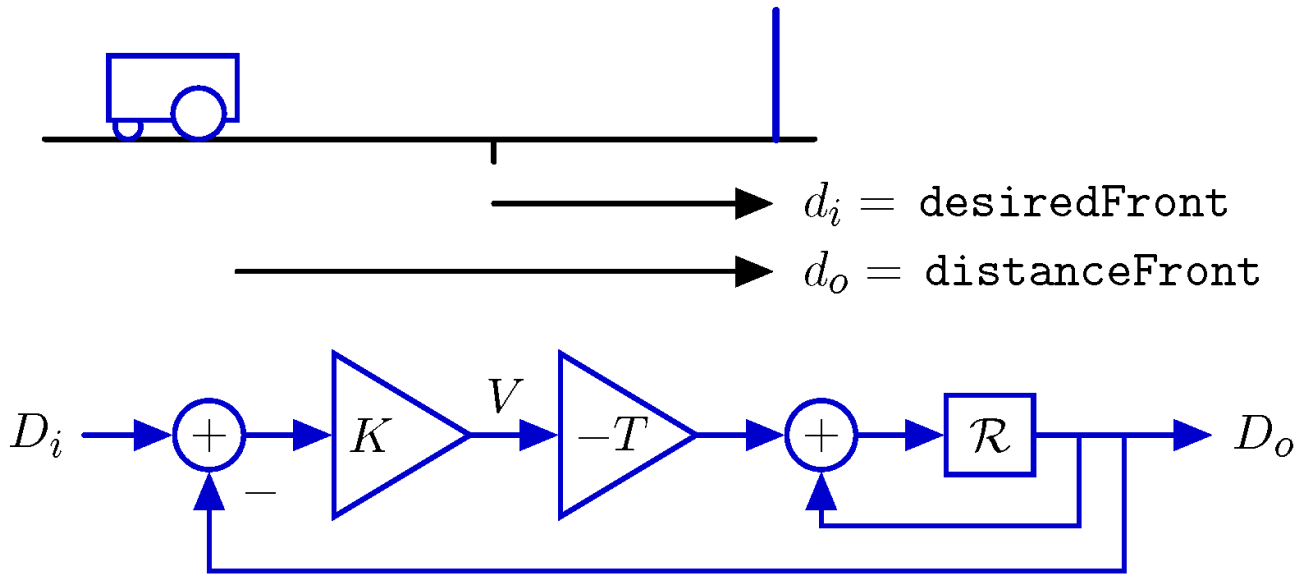
locomotion: $d_o[n] = d_o[n-1] - Tv[n-1]$

sensor with no delay: $d_s[n] = d_o[n]$



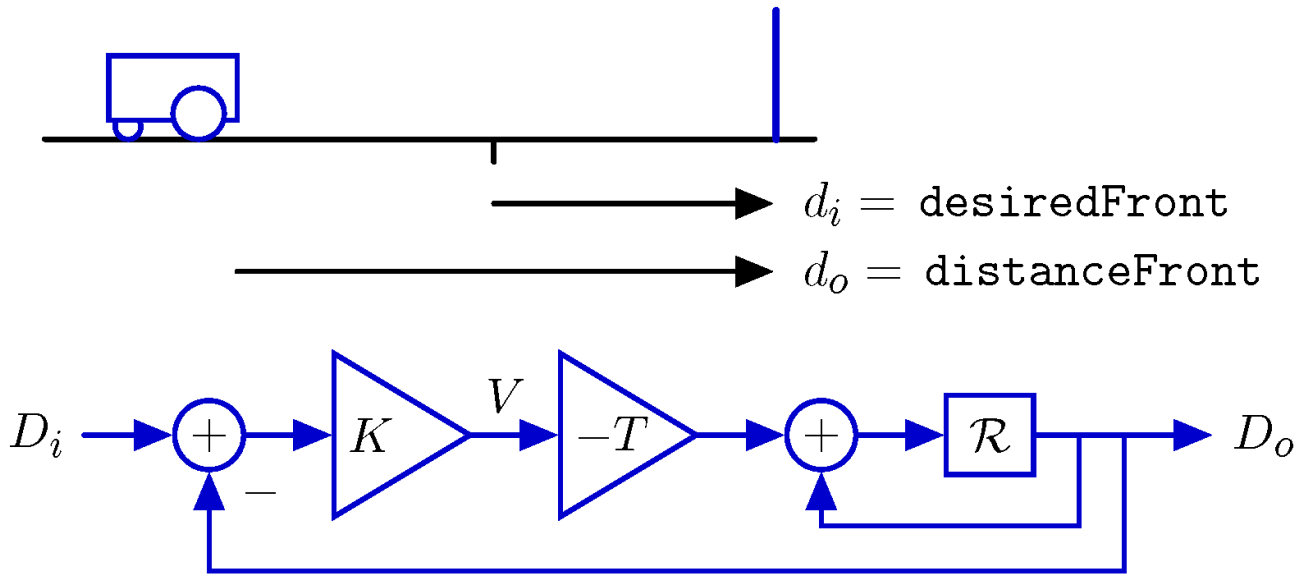
Analysis of wallFinder System: System Function

Solve.



Analysis of wallFinder System: System Function

Solve.



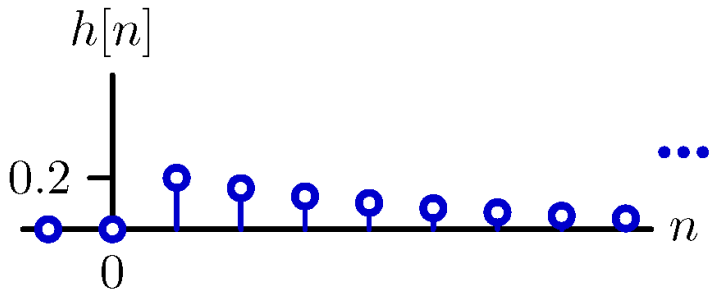
$$\frac{D_o}{D_i} = \frac{\frac{-KT\mathcal{R}}{1-\mathcal{R}}}{1+\frac{-KT\mathcal{R}}{1-\mathcal{R}}} = \frac{-KT\mathcal{R}}{1-\mathcal{R}-KT\mathcal{R}} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}}$$

Analysis of wallFinder System: Poles

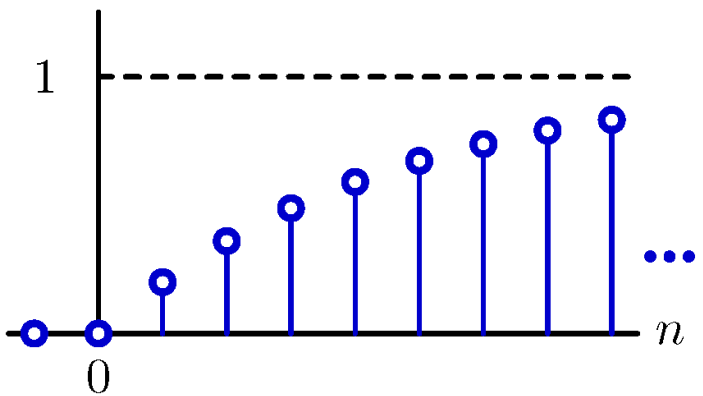
The system function contains a single **pole** at $z = 1 + KT$.

$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1 - (1 + KT)\mathcal{R}}$$

Unit-sample response for $KT = -0.2$:



Unit-step response $s[n]$ for $KT = -0.2$:



What determines the speed of the response? Could it be faster?

Check Yourself

Find KT for fastest convergence of unit-sample response.

$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}}$$

1. $KT = -2$
2. $KT = -1$
3. $KT = 0$
4. $KT = 1$
5. $KT = 2$
0. none of the above

Check Yourself

Find KT for fastest convergence of unit-sample response.

$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}}$$

If $KT = -1$ then the pole is at $z = 0$.

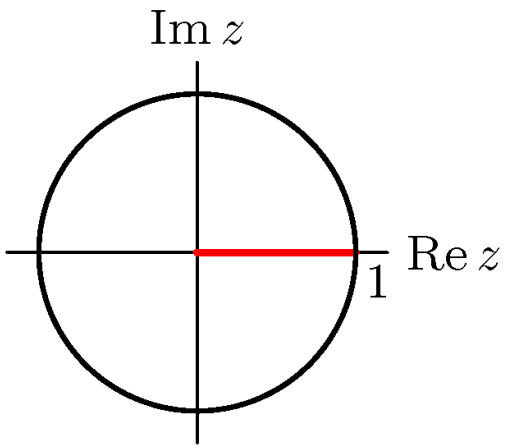
$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}} = \mathcal{R}$$

Unit-sample response has a single non-zero output sample, at $n = 1$.

Analysis of wallFinder System: Poles

The poles of the system function provide insight for choosing K.

$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}} = \frac{(1-p_o)\mathcal{R}}{1-p_o\mathcal{R}} \; ; \quad p_0 = 1+KT$$

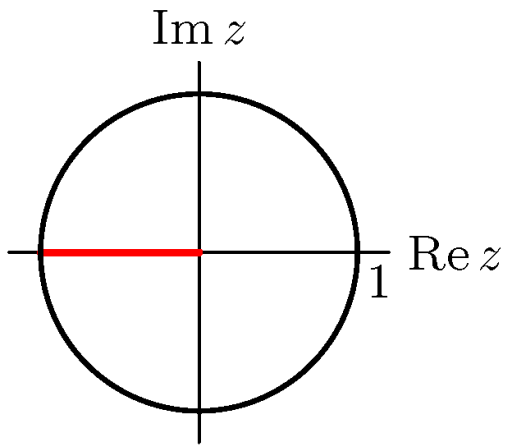


$0 < p_0 < 1$

$-1 < KT < 0$

monotonic converging

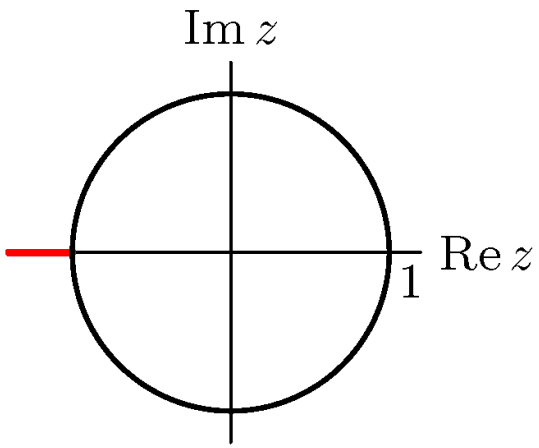
单调



$-1 < p_0 < 0$

$-2 < KT < -1$

alternating converging



$p_0 < -1$

$KT < -2$

alternating diverging

Check Yourself

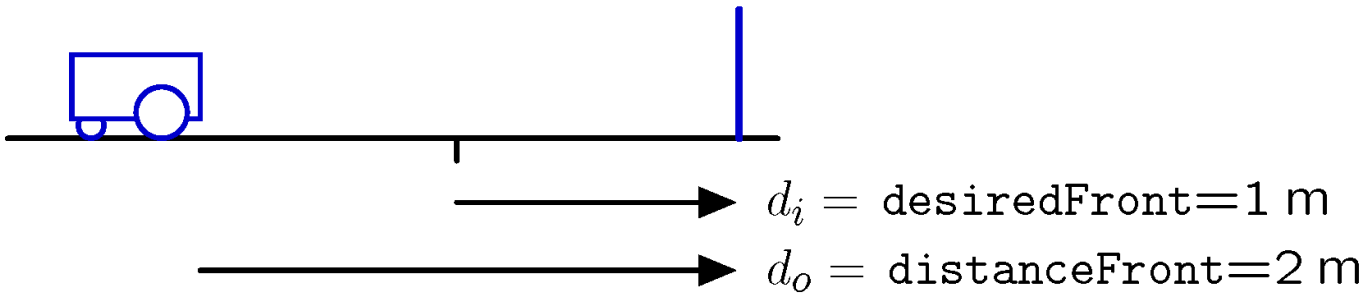
Find KT for fastest convergence of unit-sample response.

$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1-(1+KT)\mathcal{R}}$$

1. $KT = -2$
2. $KT = -1$
3. $KT = 0$
4. $KT = 1$
5. $KT = 2$
0. none of the above

Analysis of wallFinder System

The optimum gain K moves robot to desired position in **one** step.



$$KT = -1$$

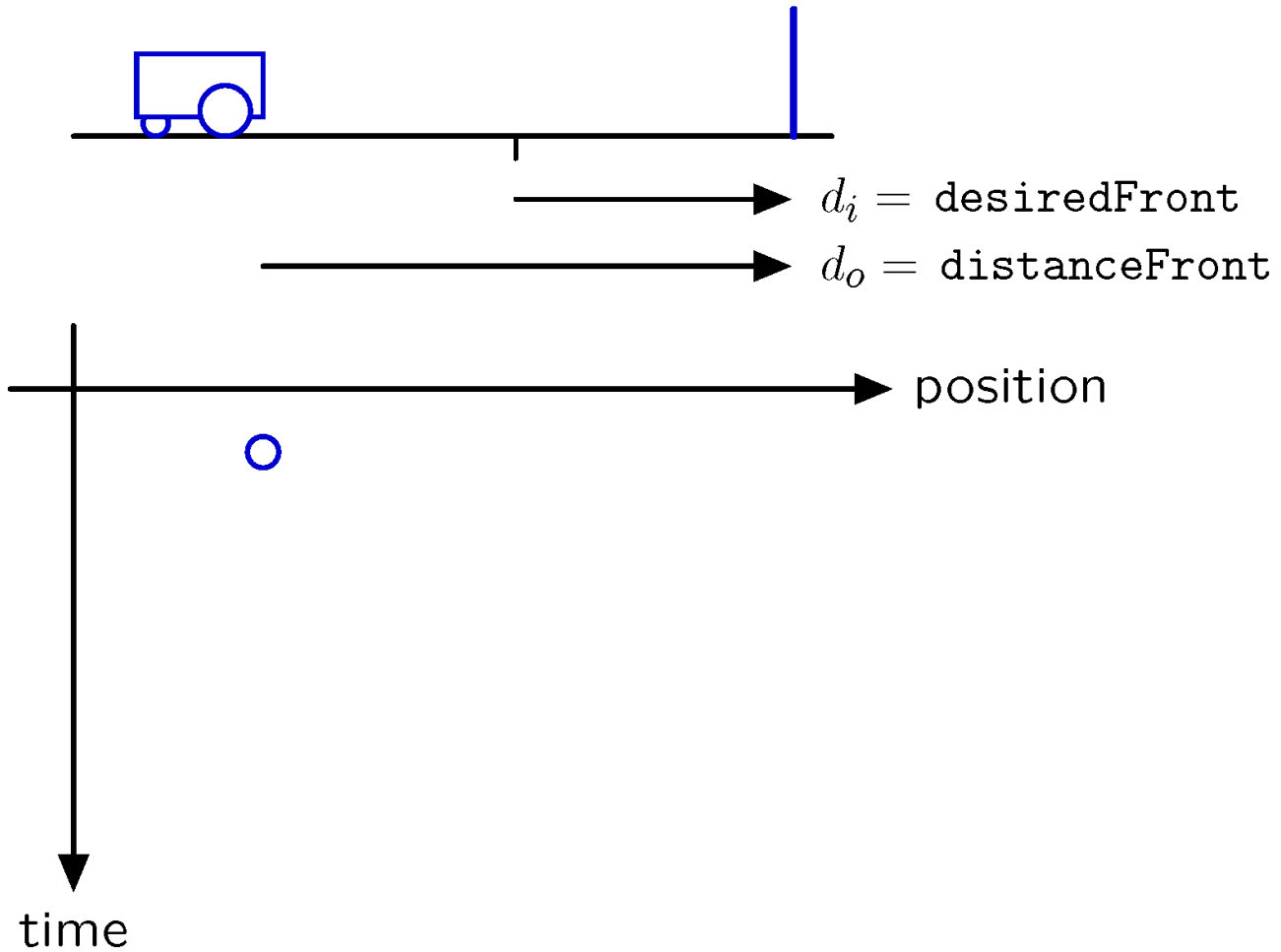
$$K = -\frac{1}{T} = -\frac{1}{1/10} = -10$$

$$v[n] = K\big(d_i[n] - d_o[n]\big) = -10\big(1-2\big) = 10 \text{ m/s}$$

exactly the right speed to get there in one step!

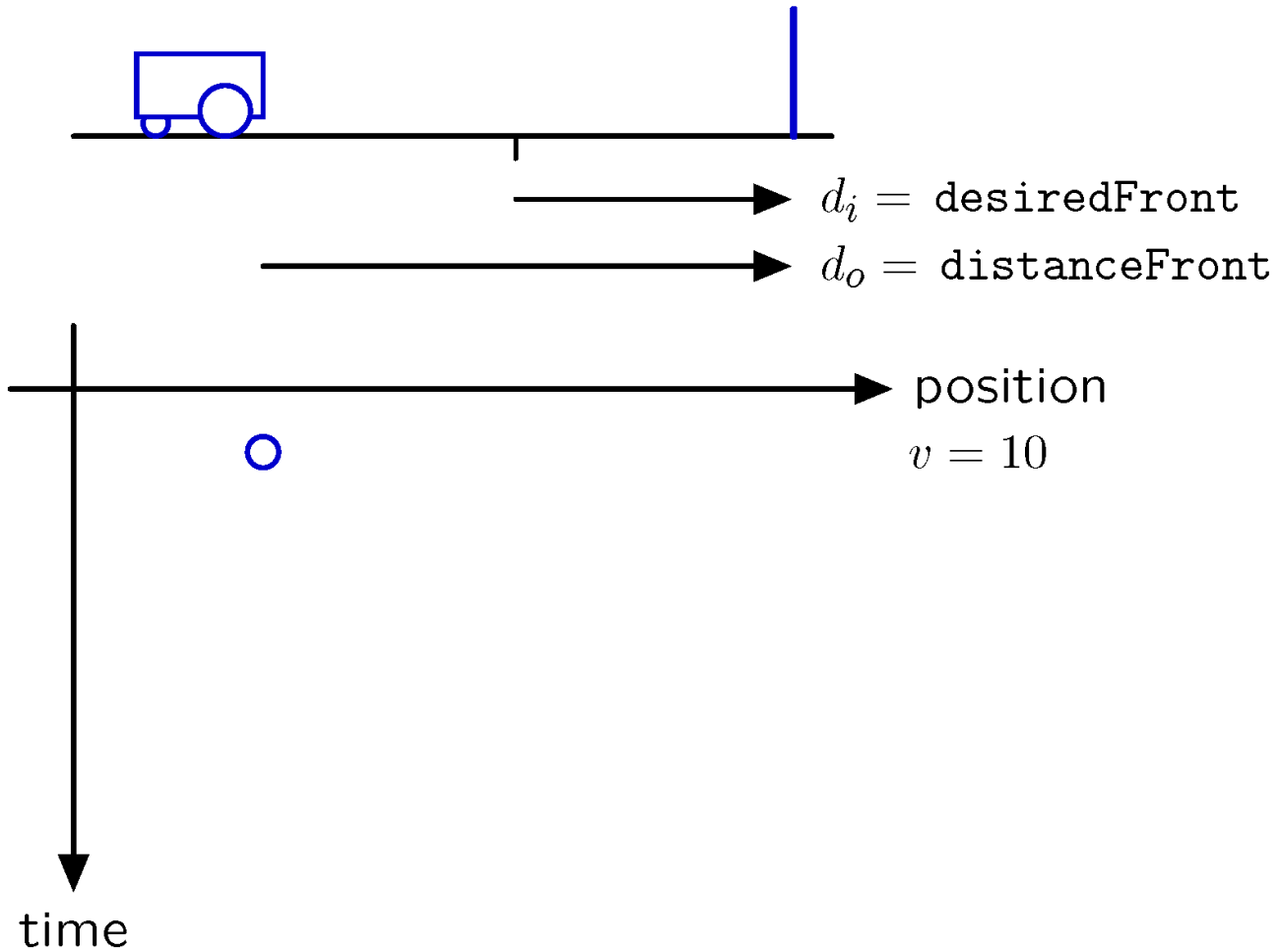
Analyzing wallFinder: Space-Time Diagram

The optimum gain K moves robot to desired position in **one** step.



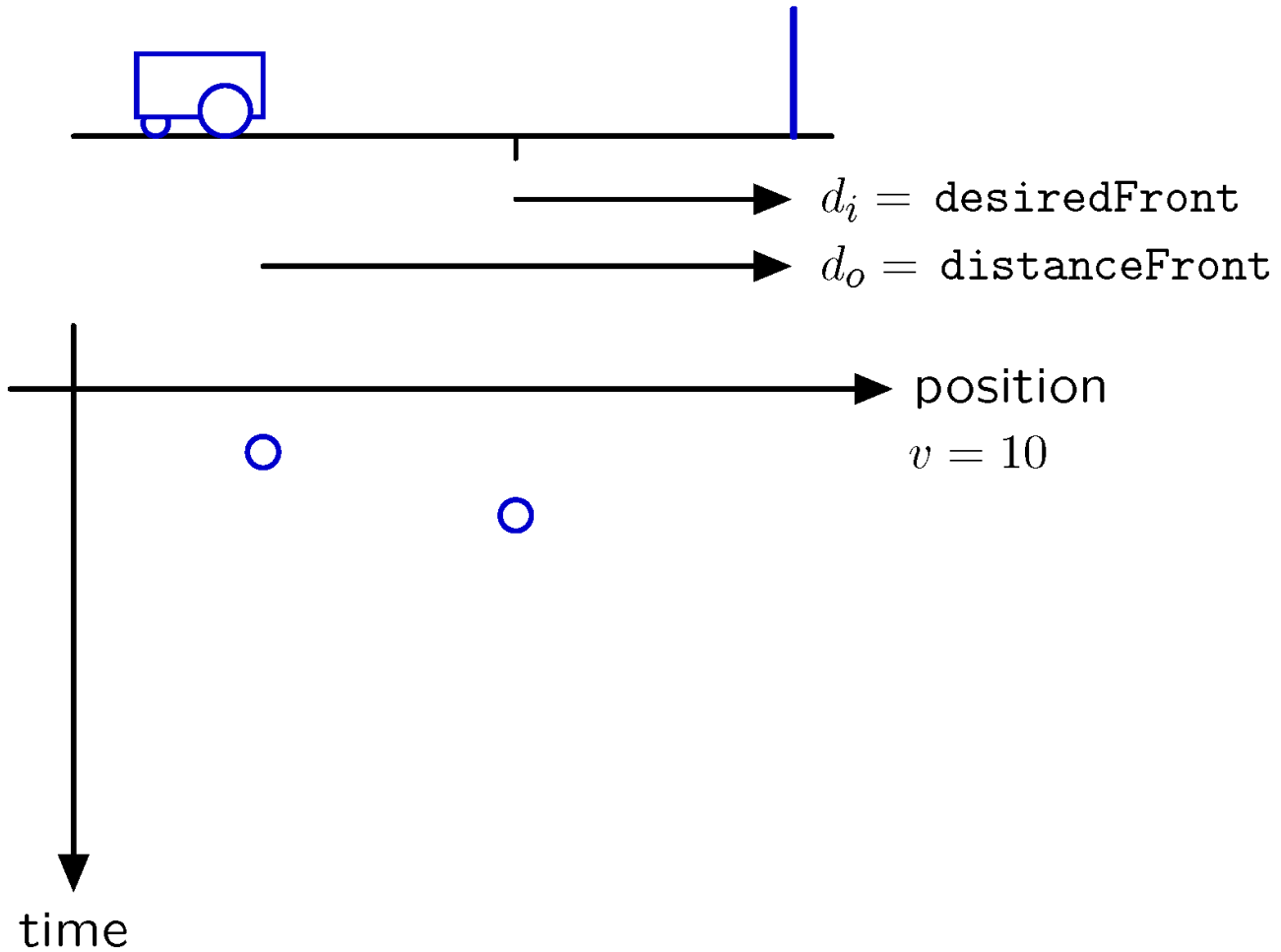
Analyzing wallFinder: Space-Time Diagram

The optimum gain K moves robot to desired position in **one** step.



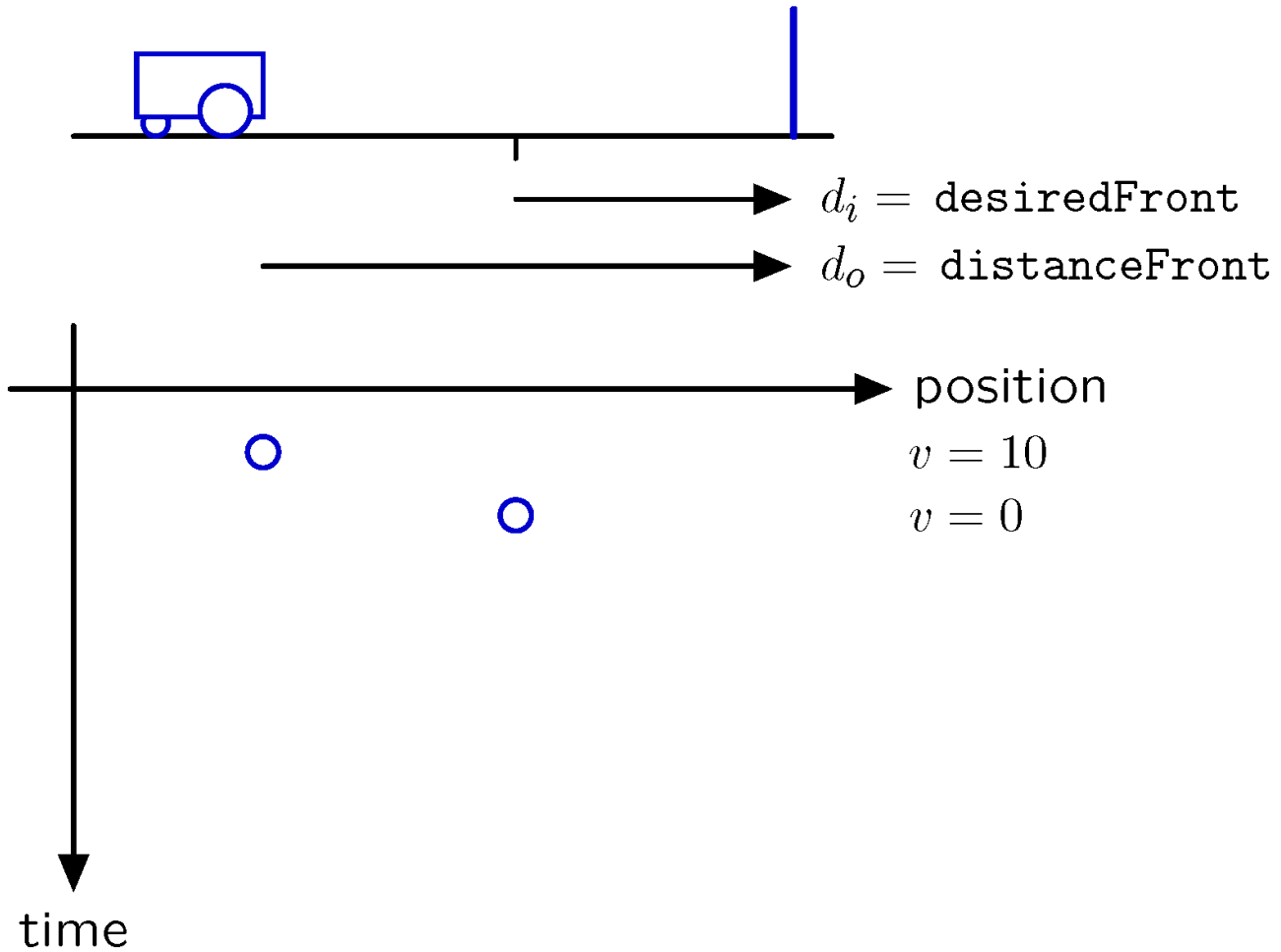
Analyzing wallFinder: Space-Time Diagram

The optimum gain K moves robot to desired position in **one** step.



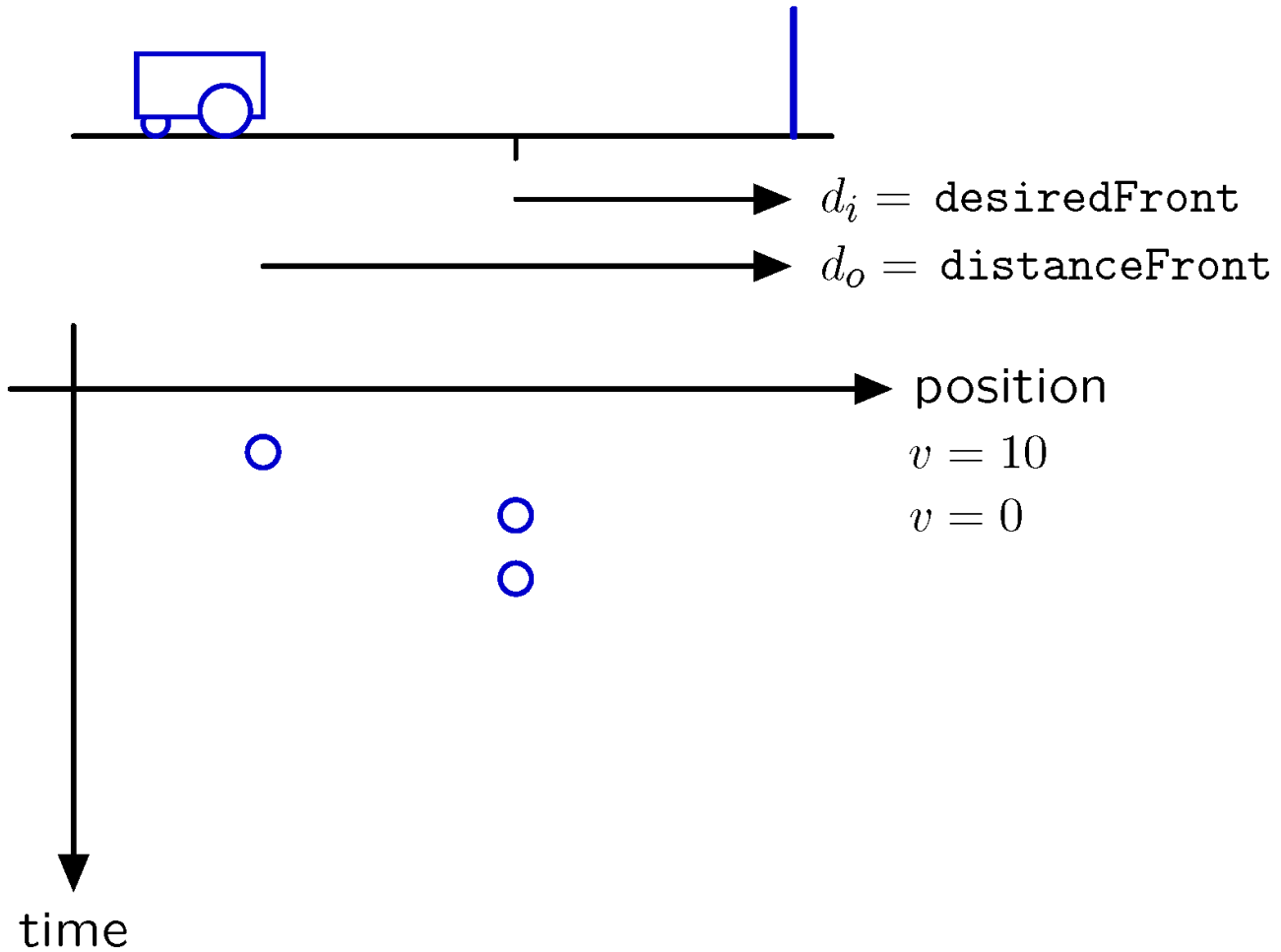
Analyzing wallFinder: Space-Time Diagram

The optimum gain K moves robot to desired position in **one** step.



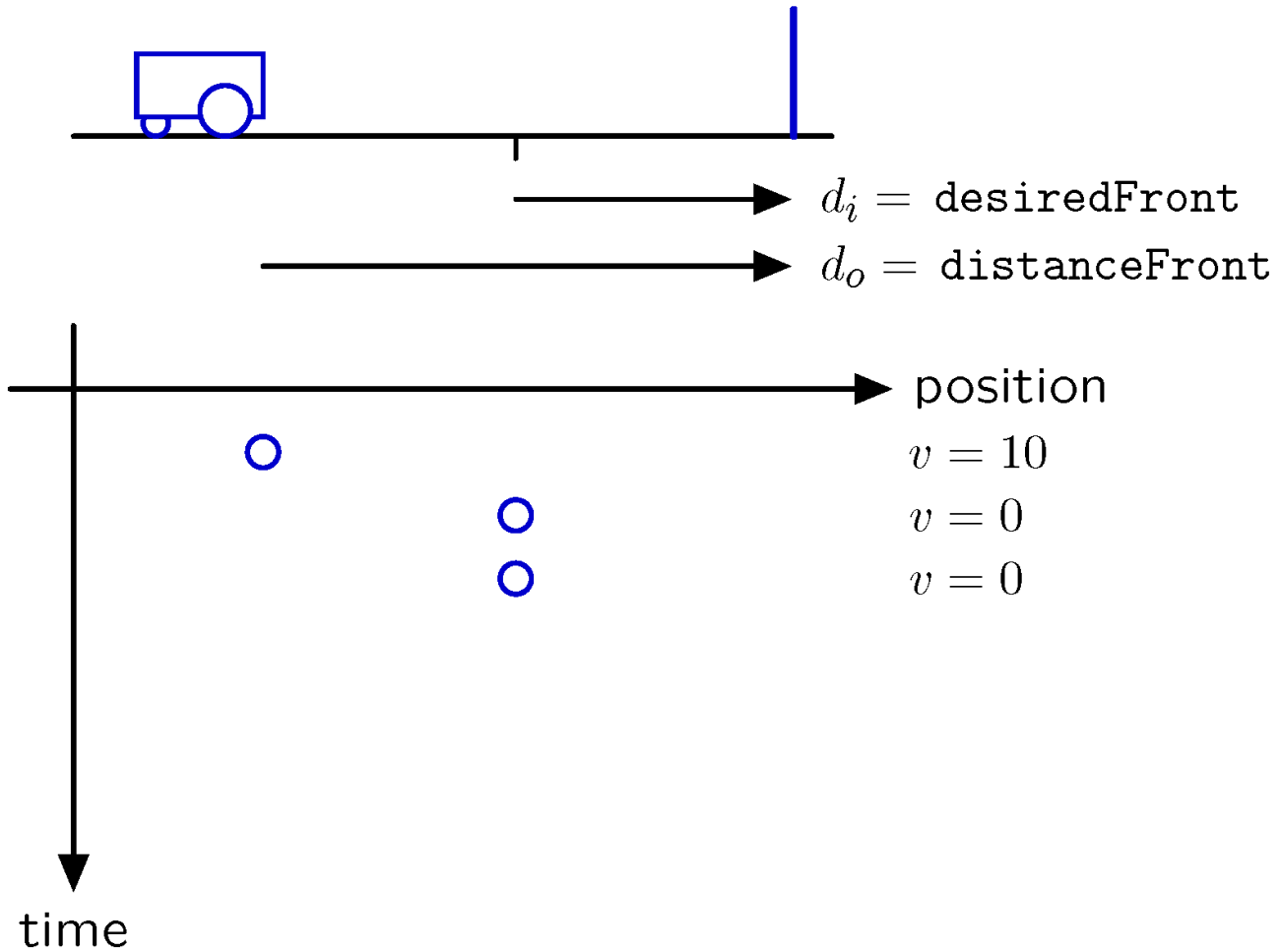
Analyzing wallFinder: Space-Time Diagram

The optimum gain K moves robot to desired position in **one** step.



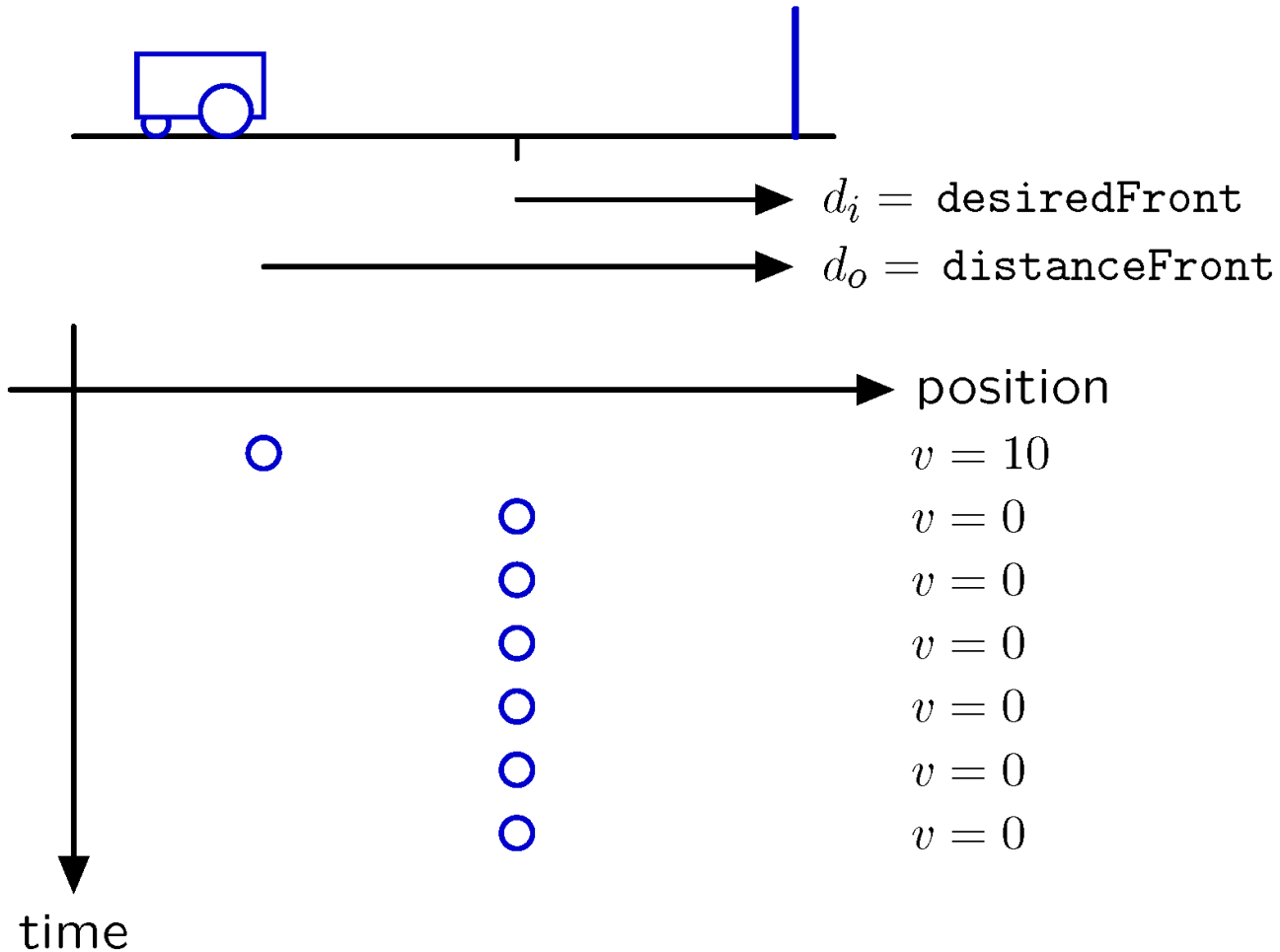
Analyzing wallFinder: Space-Time Diagram

The optimum gain K moves robot to desired position in **one** step.



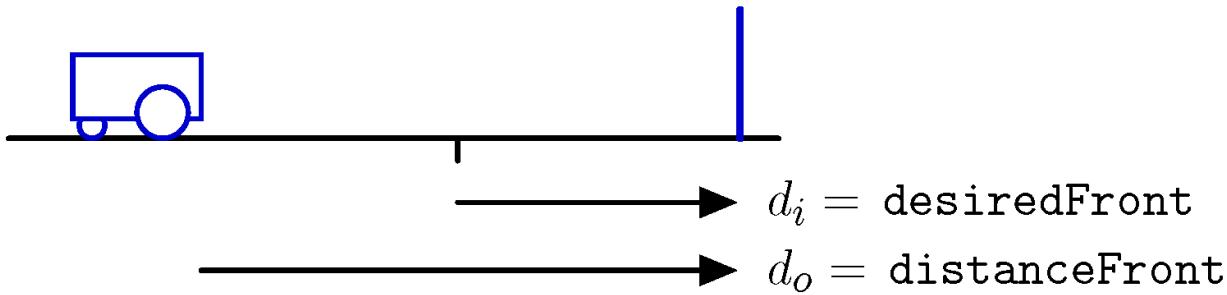
Analyzing wallFinder: Space-Time Diagram

The optimum gain K moves robot to desired position in **one** step.



Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



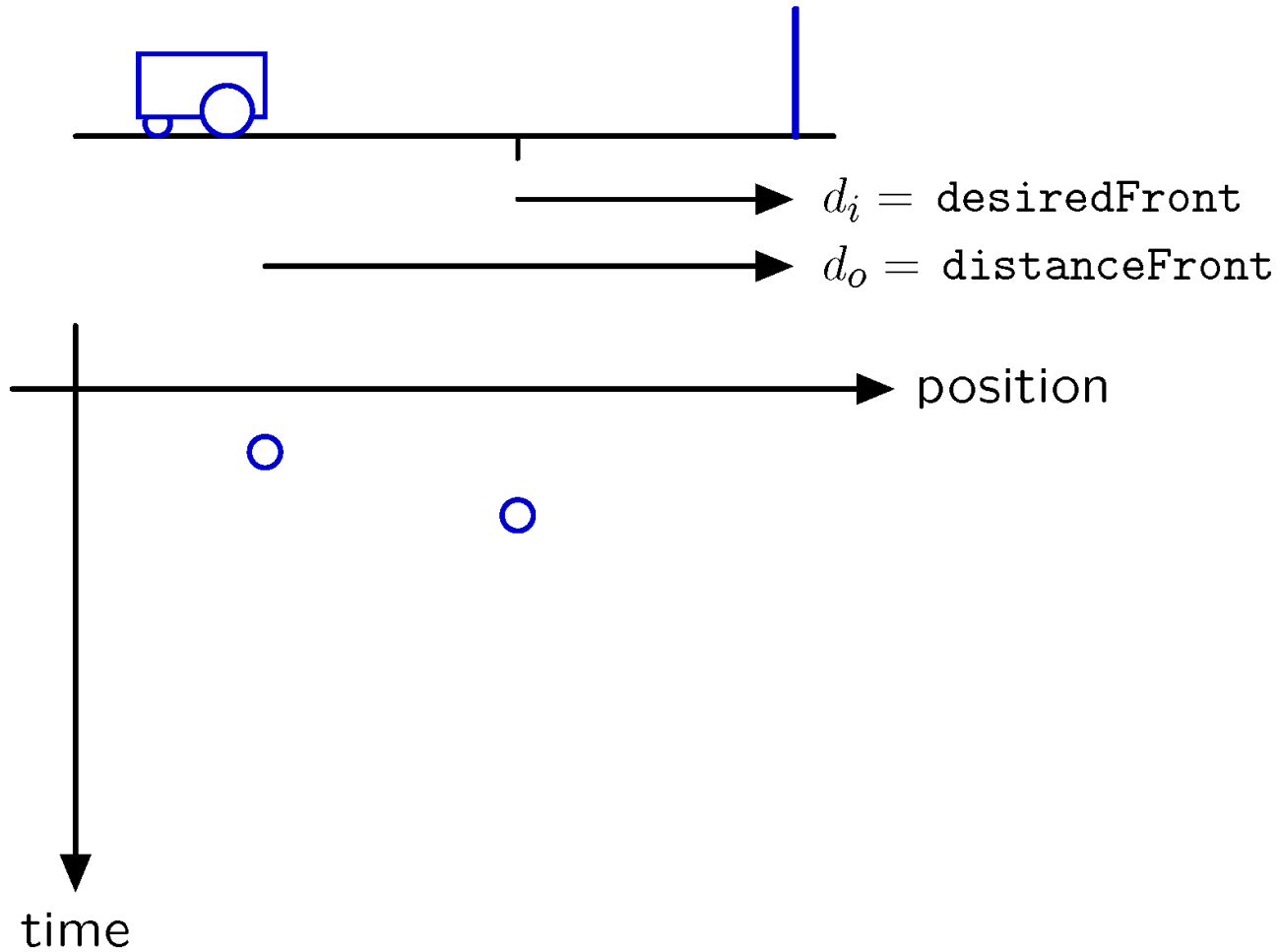
proportional controller: $v[n] = Ke[n] = K\big(d_i[n] - d_s[n]\big)$

locomotion: $d_o[n] = d_o[n-1] - Tv[n-1]$

sensor **with delay**: $d_s[n] = d_o[\mathbf{n-1}]$

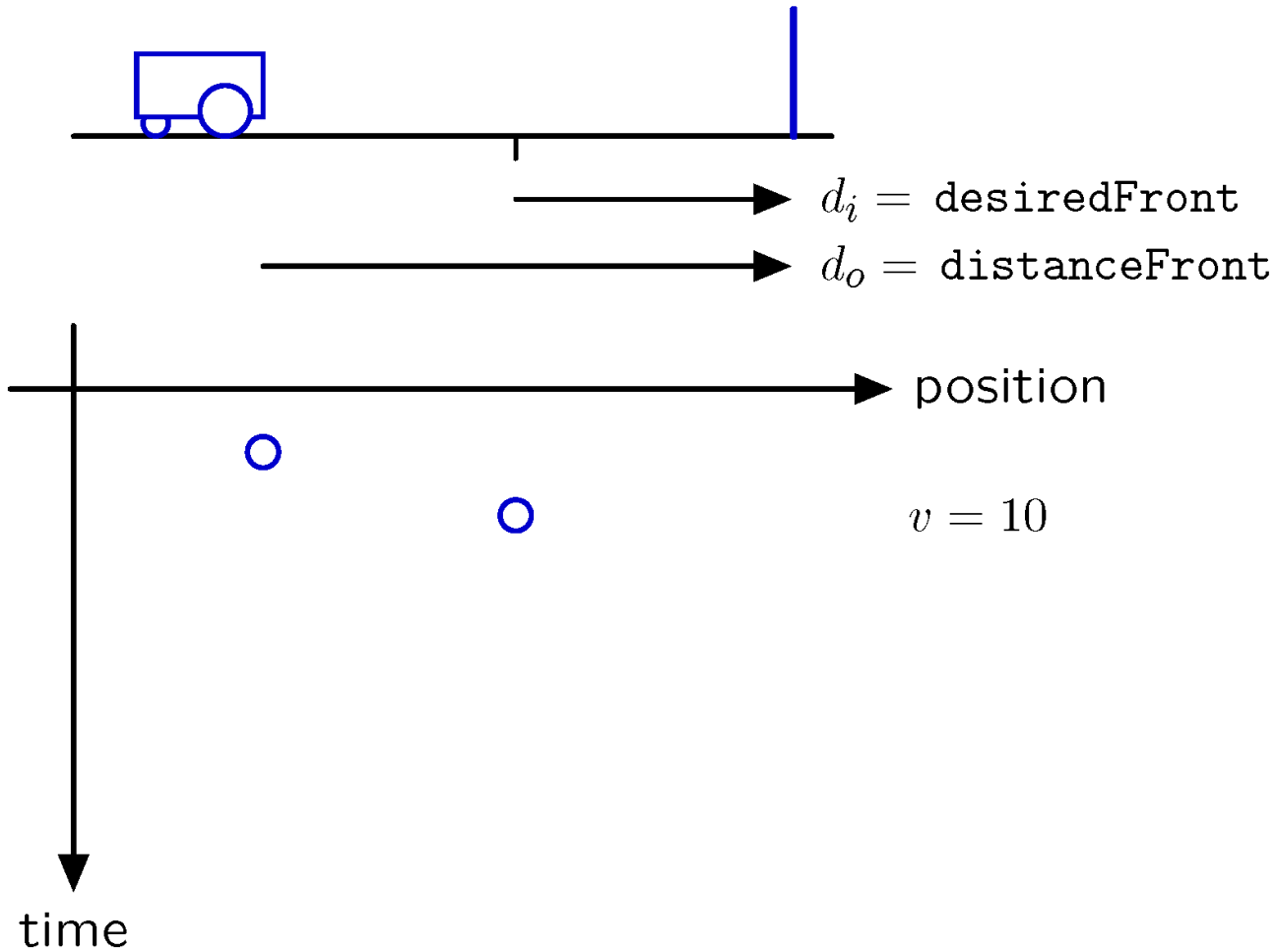
Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



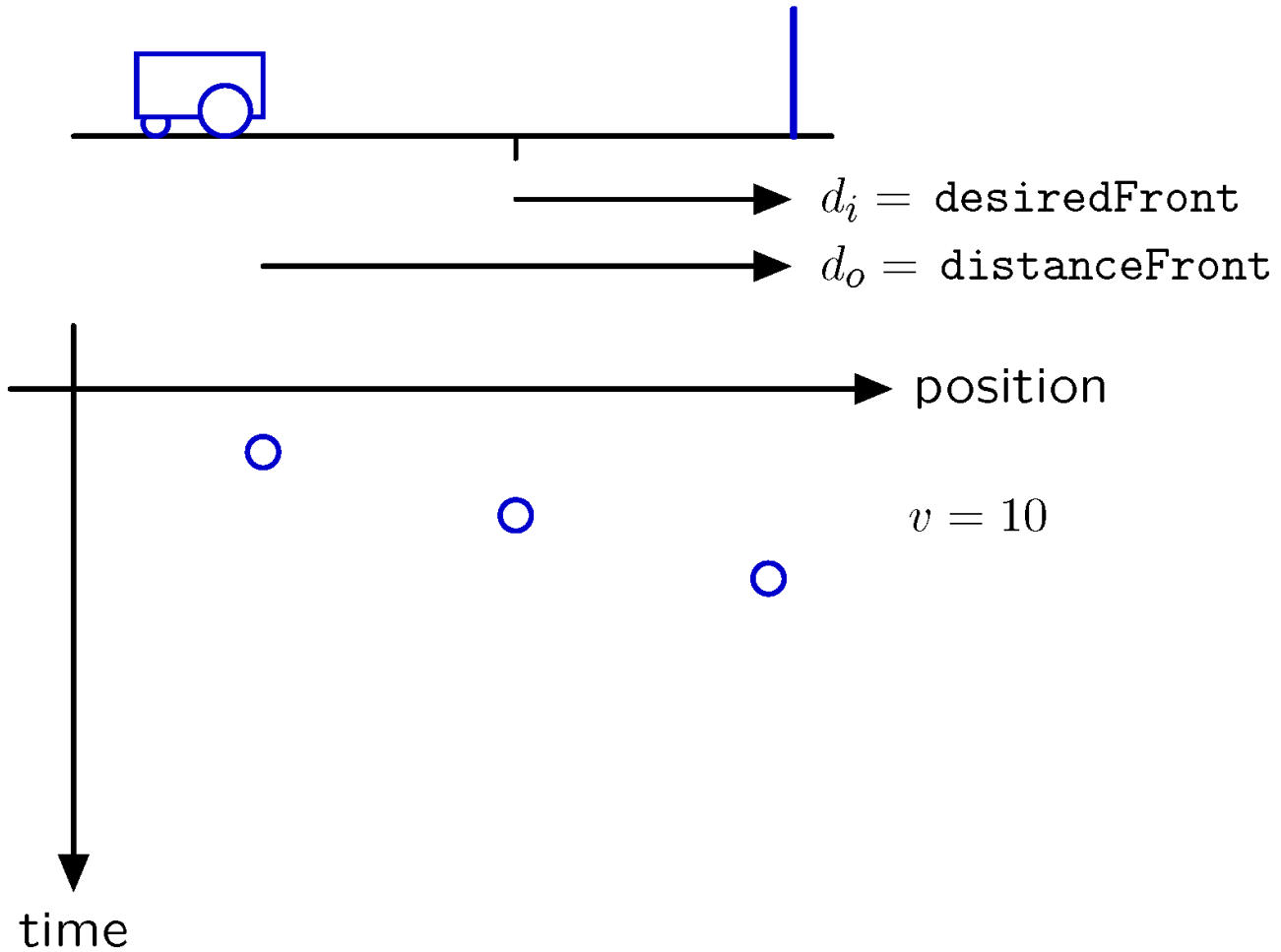
Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



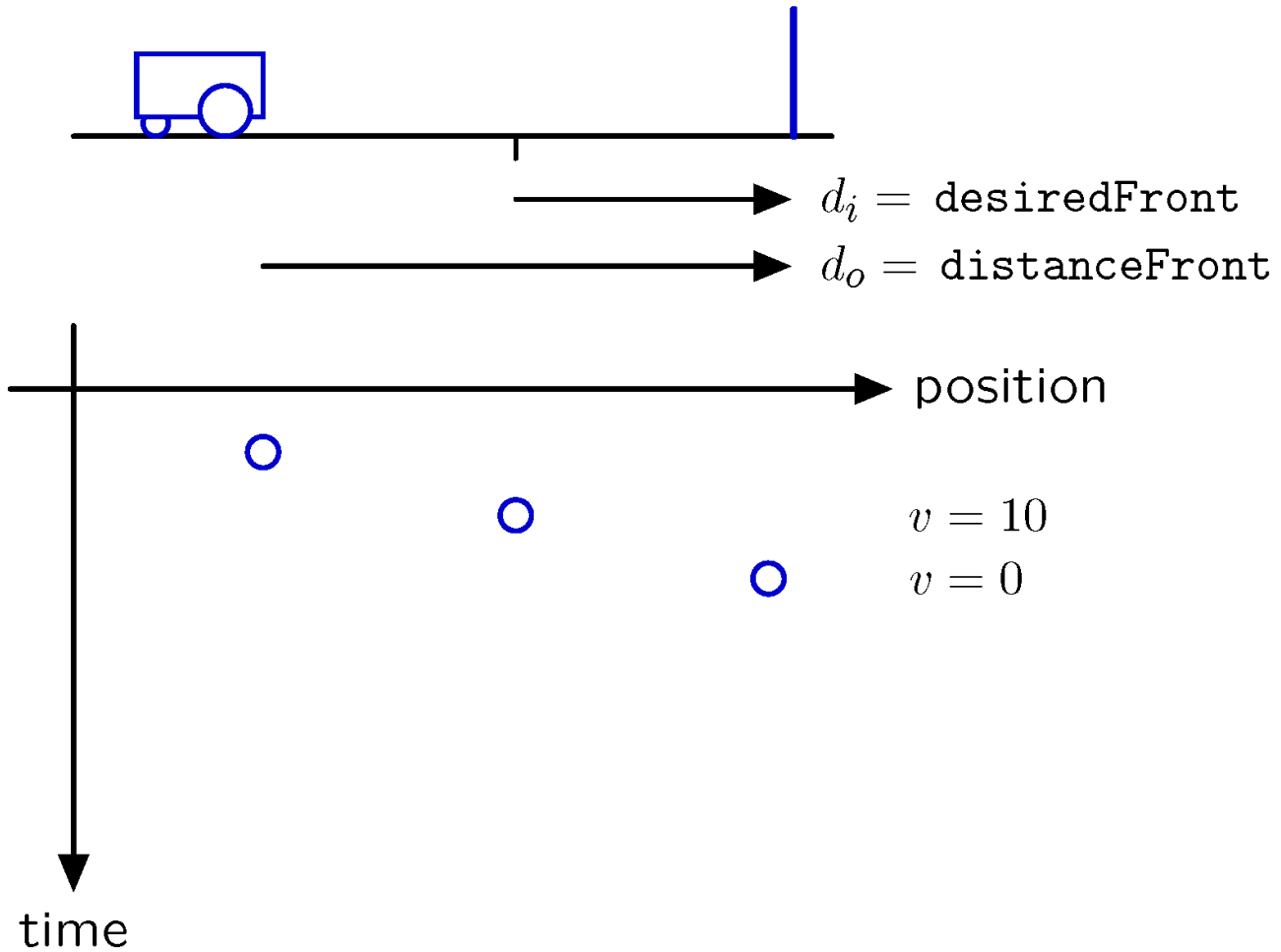
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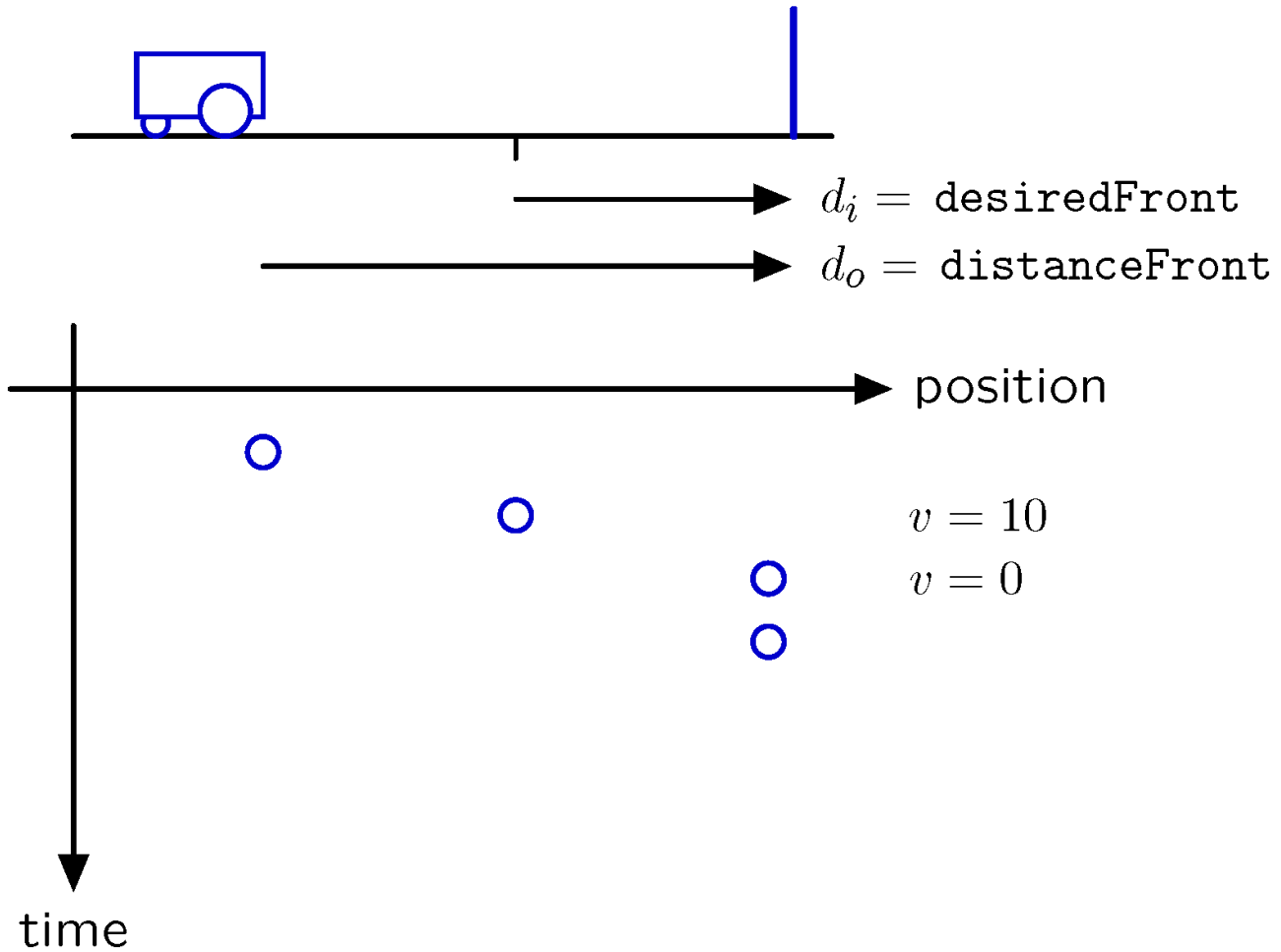
Analysis of wallFinder System: Adding Sensor Delay

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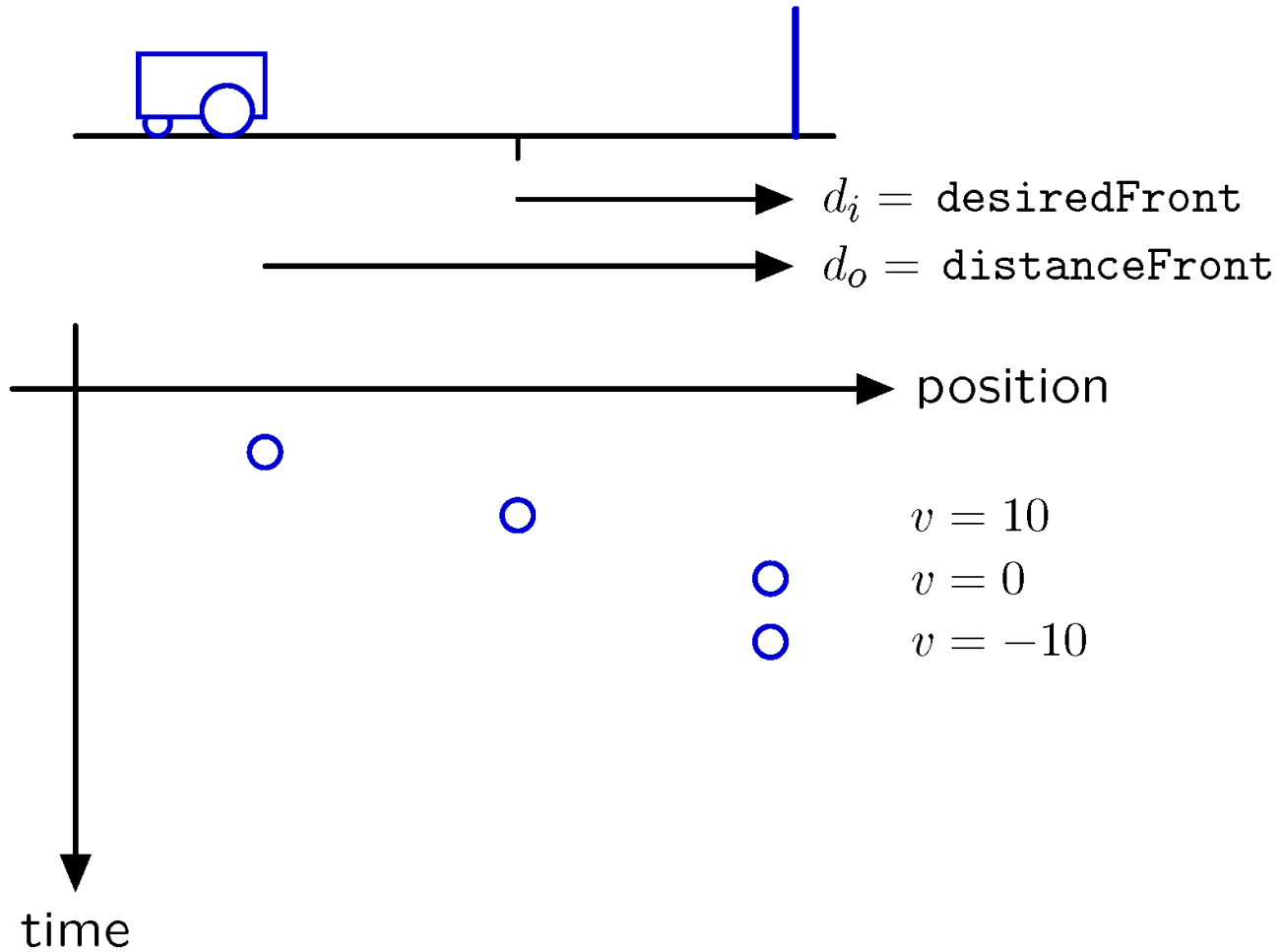
Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



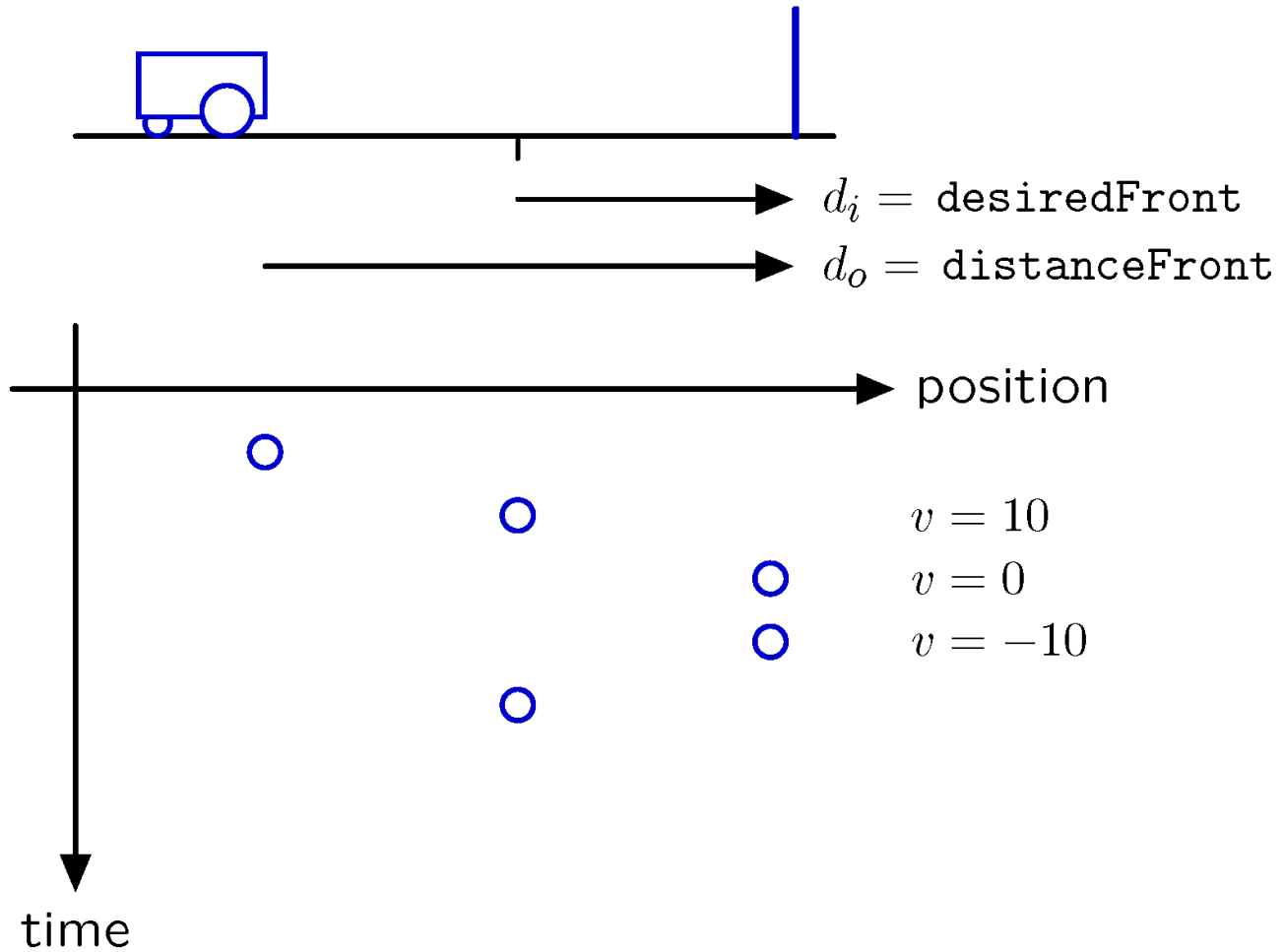
Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



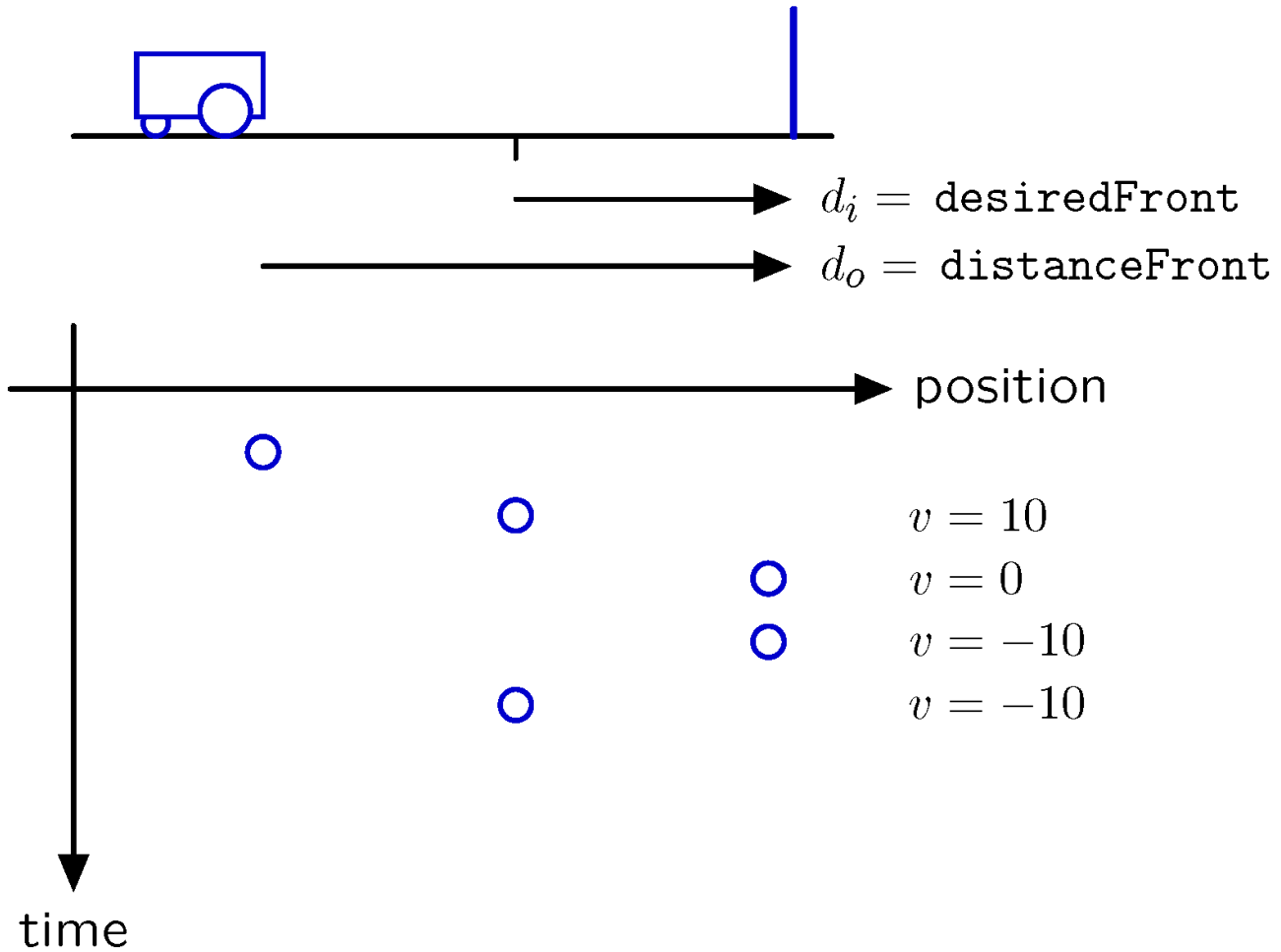
Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



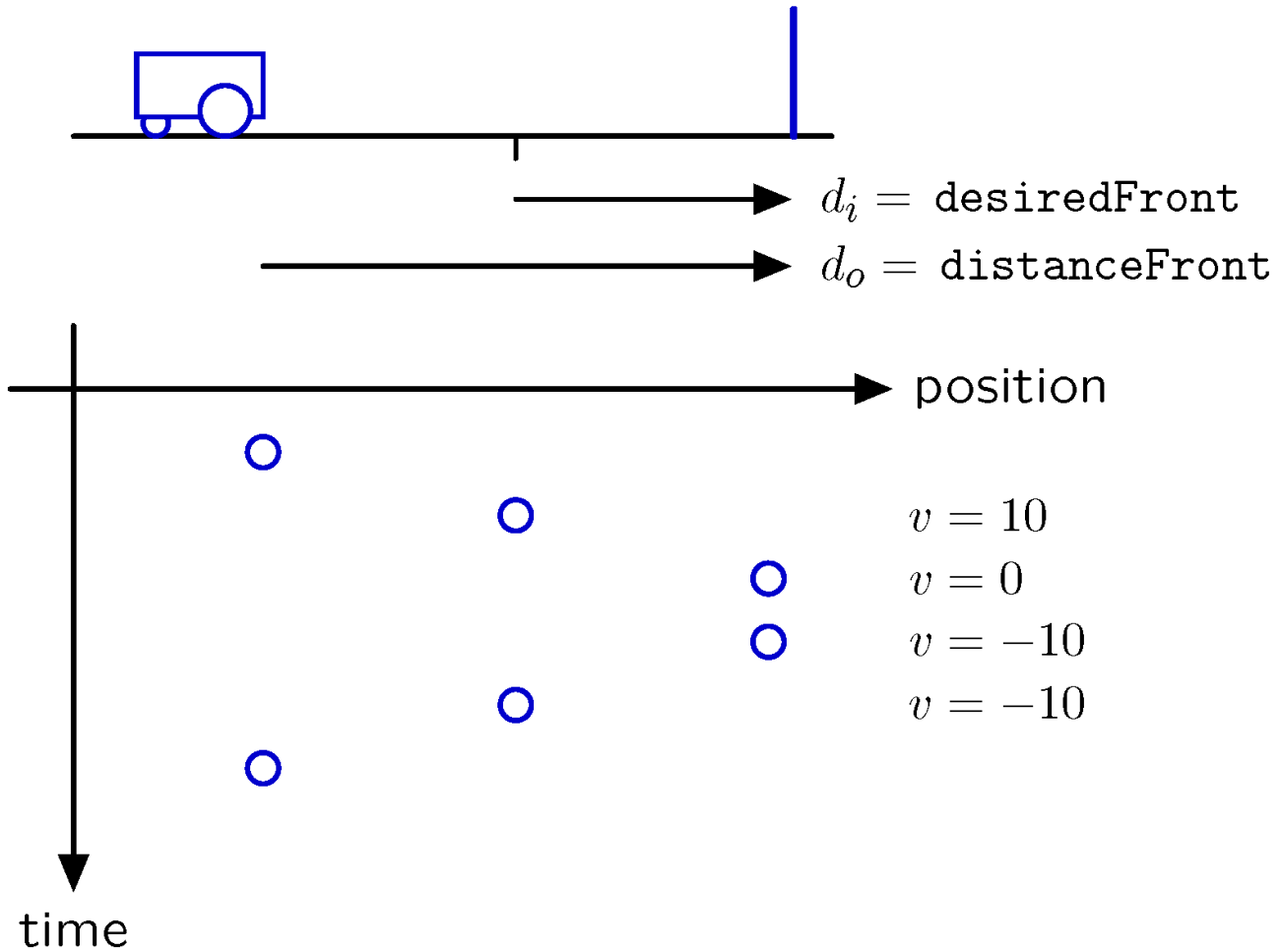
Analysis of wallFinder System: Adding Sensor Delay

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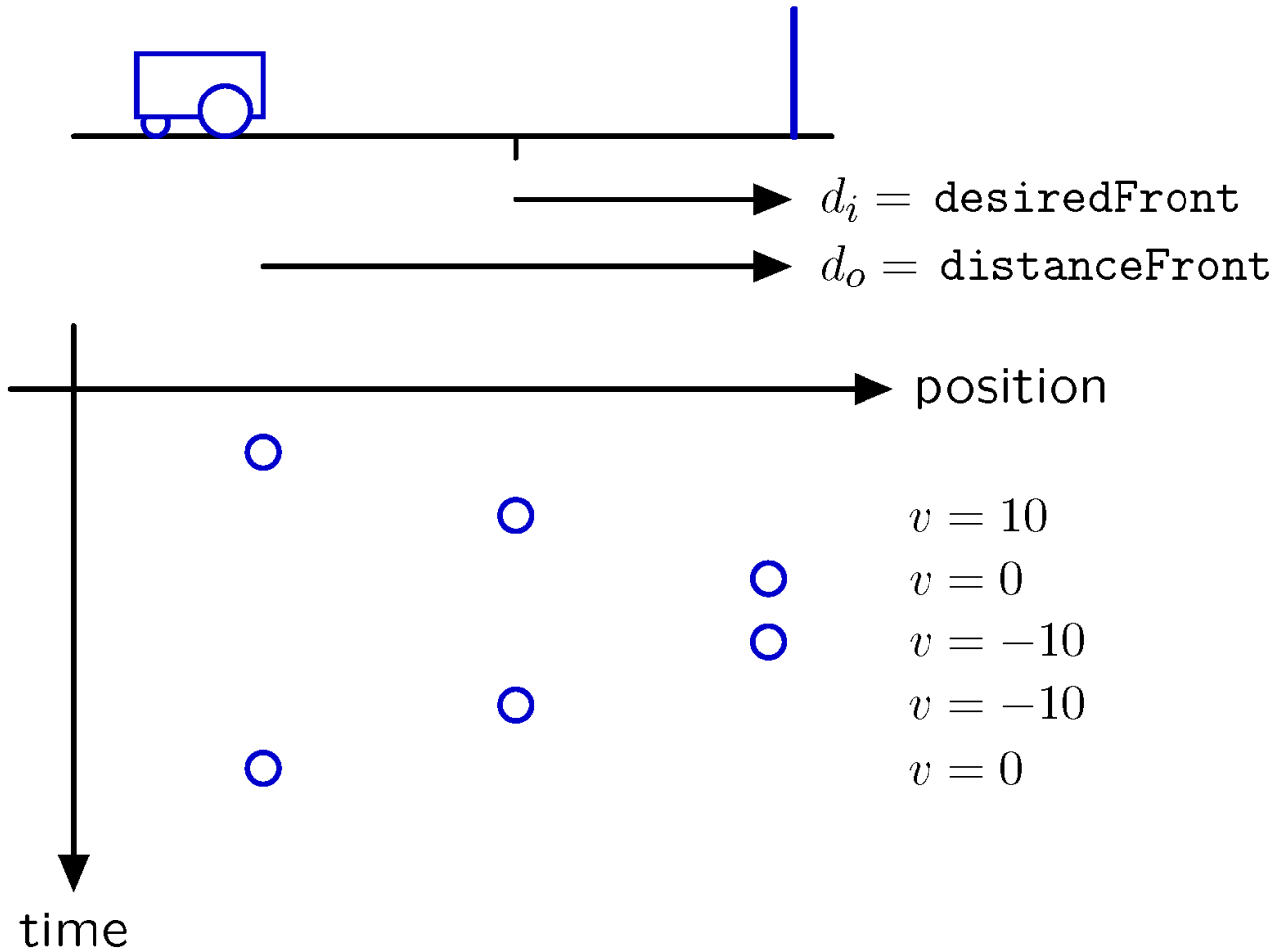
Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



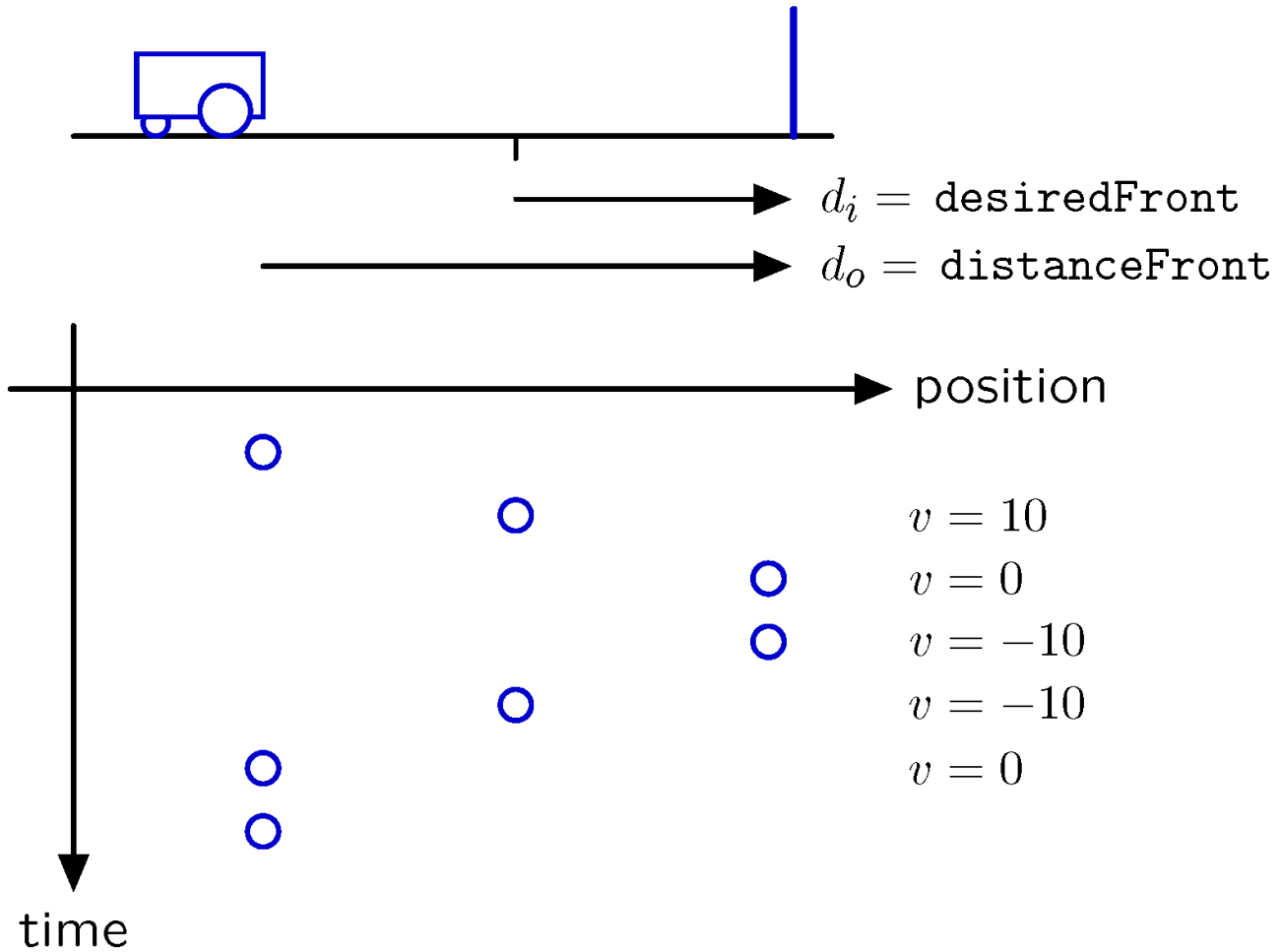
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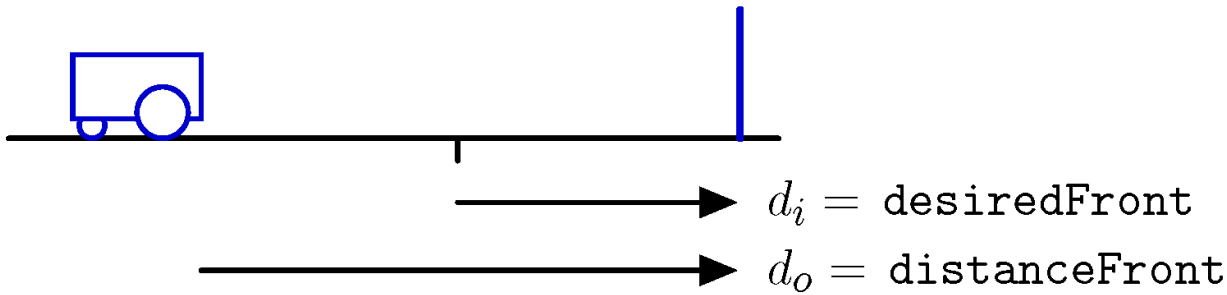
Analysis of wallFinder System: Adding Sensor Delay

Adding delay tends to destabilize control systems.



Analysis of wallFinder System: Block Diagram

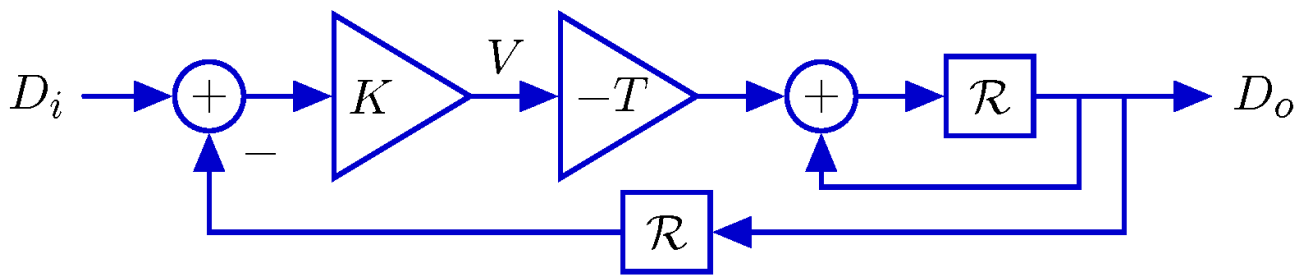
Incorporating sensor delay in block diagram.



proportional controller: $v[n] = Ke[n] = K\big(d_i[n] - d_s[n]\big)$

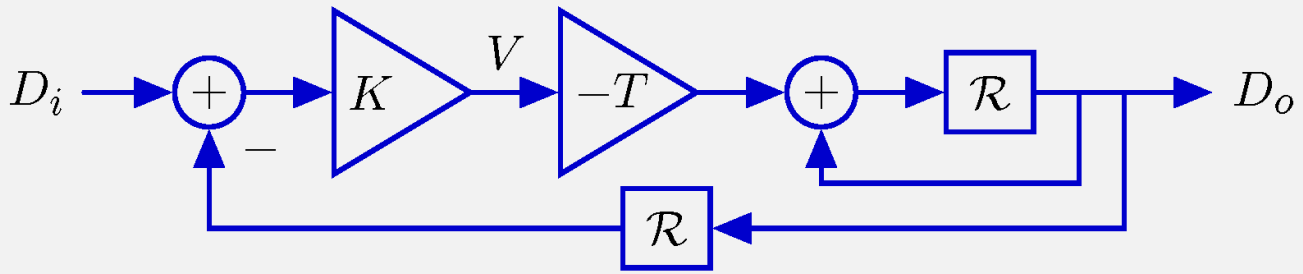
locomotion: $d_o[n] = d_o[n-1] - Tv[n-1]$

sensor with no delay: $d_s[n] = d_o[n-1]$



Check Yourself

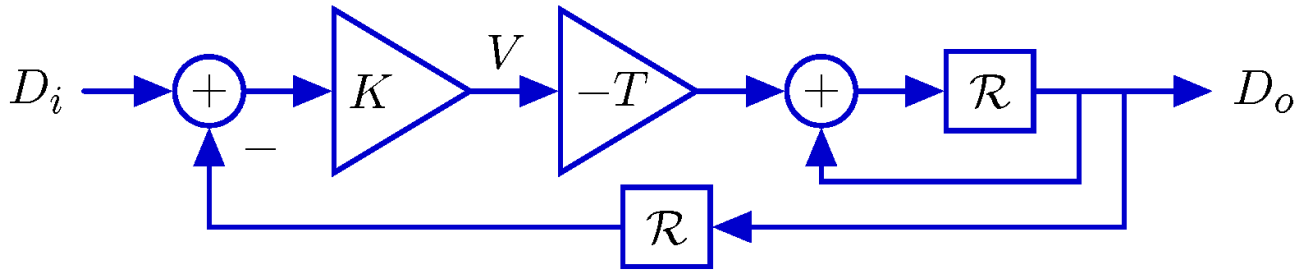
Find the system function $H = \dfrac{D_o}{D_i}$.



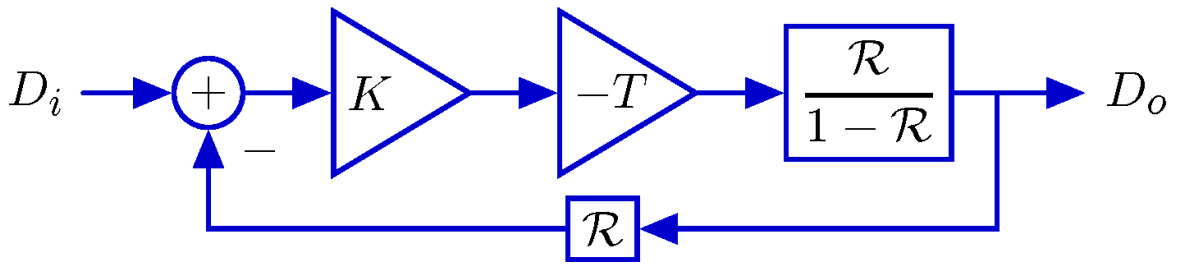
1. $\dfrac{KT\mathcal{R}}{1-\mathcal{R}}$
2. $\dfrac{-KT\mathcal{R}}{1+\mathcal{R}-KT\mathcal{R}^2}$
3. $\dfrac{KT\mathcal{R}}{1-\mathcal{R}} - KT\mathcal{R}$
4. $\dfrac{-KT\mathcal{R}}{1-\mathcal{R}-KT\mathcal{R}^2}$
5. none of the above

Check Yourself

Find the system function $H = \dfrac{D_o}{D_i}$.



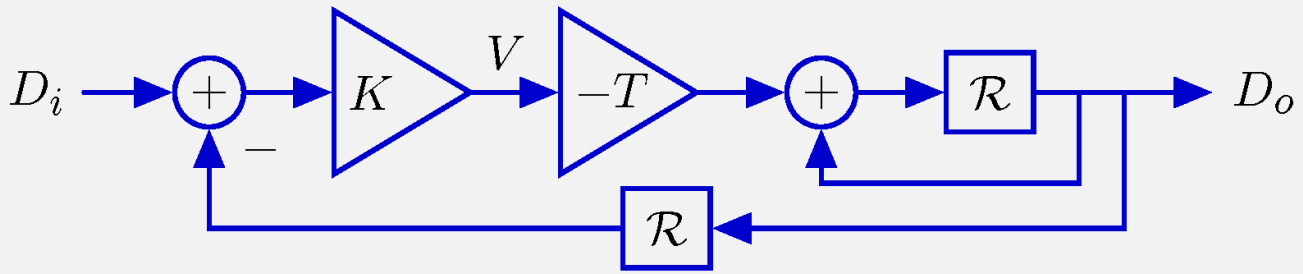
Replace accumulator with equivalent block diagram.



$$\frac{D_o}{D_i} = \frac{\dfrac{-KT\mathcal{R}}{1-\mathcal{R}}}{1 + \dfrac{-KT\mathcal{R}^2}{1-\mathcal{R}}} = \frac{-KT\mathcal{R}}{1-\mathcal{R}-KT\mathcal{R}^2}$$

Check Yourself

Find the system function $H = \dfrac{D_o}{D_i}$.



1. $\dfrac{KT\mathcal{R}}{1-\mathcal{R}}$
2. $\dfrac{-KT\mathcal{R}}{1+\mathcal{R}-KT\mathcal{R}^2}$
3. $\dfrac{KT\mathcal{R}}{1-\mathcal{R}} - KT\mathcal{R}$
4. $\dfrac{-KT\mathcal{R}}{1-\mathcal{R}-KT\mathcal{R}^2}$
5. none of the above

Analyzing wallFinder: Poles

Substitute $\mathcal{R} \to \frac{1}{z}$ in the system functional to find the poles.

$$\frac{D_o}{D_i} = \frac{-KT\mathcal{R}}{1 - \mathcal{R} - KT\mathcal{R}^2} = \frac{-KT\frac{1}{z}}{1 - \frac{1}{z} - KT\frac{1}{z^2}} = \frac{-KTz}{z^2 - z - KT}$$

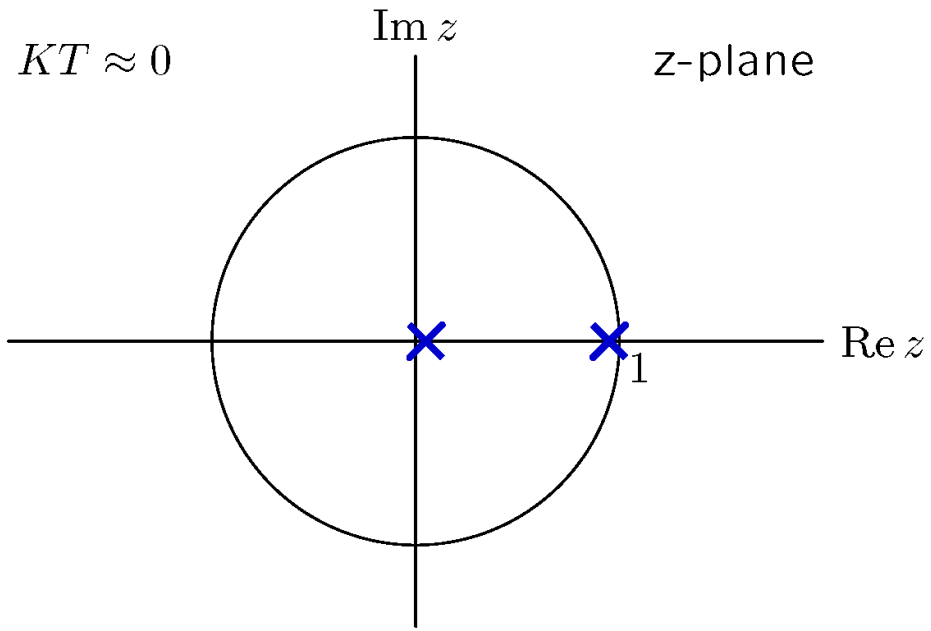
The poles are then the roots of the denominator.

$$z = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 + KT}$$

Feedback and Control: Poles

If KT is small, the poles are at $z \approx -KT$ and $z \approx 1 + KT$.

$$z = \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2}\right)^2 + KT} \approx \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2} + KT\right)^2} = 1 + KT,\ -KT$$



Pole near 0 generates fast response.

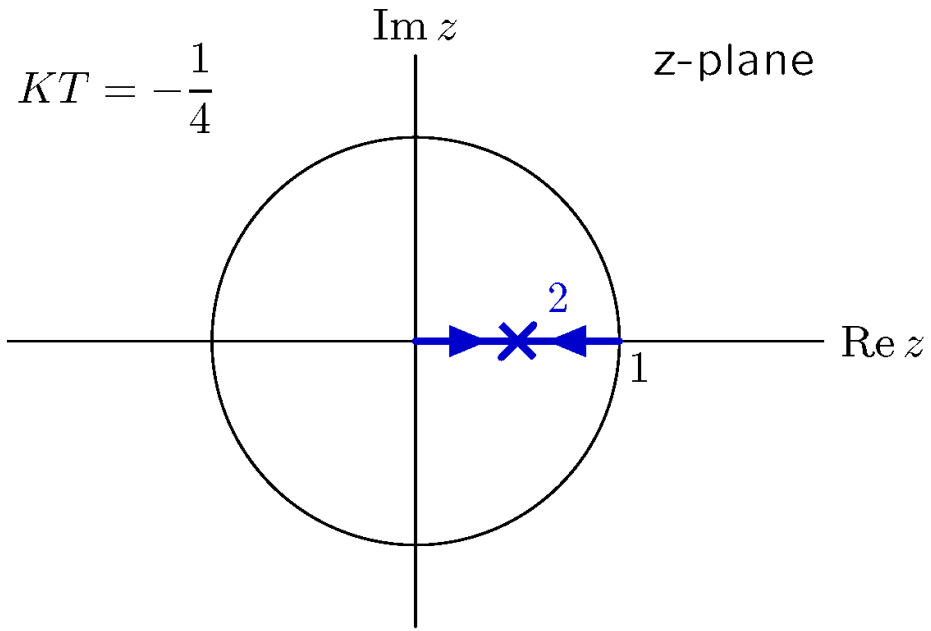
Pole near 1 generates slow response.

Slow mode (pole near 1) dominates the response.

Feedback and Control: Poles

As KT becomes more negative, the poles move toward each other and collide at $z = \frac{1}{2}$ when $KT = -\frac{1}{4}$.

$$z = \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2}\right)^2 + KT} = \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2}\right)^2 - \tfrac{1}{4}} = \tfrac{1}{2}, \tfrac{1}{2}$$

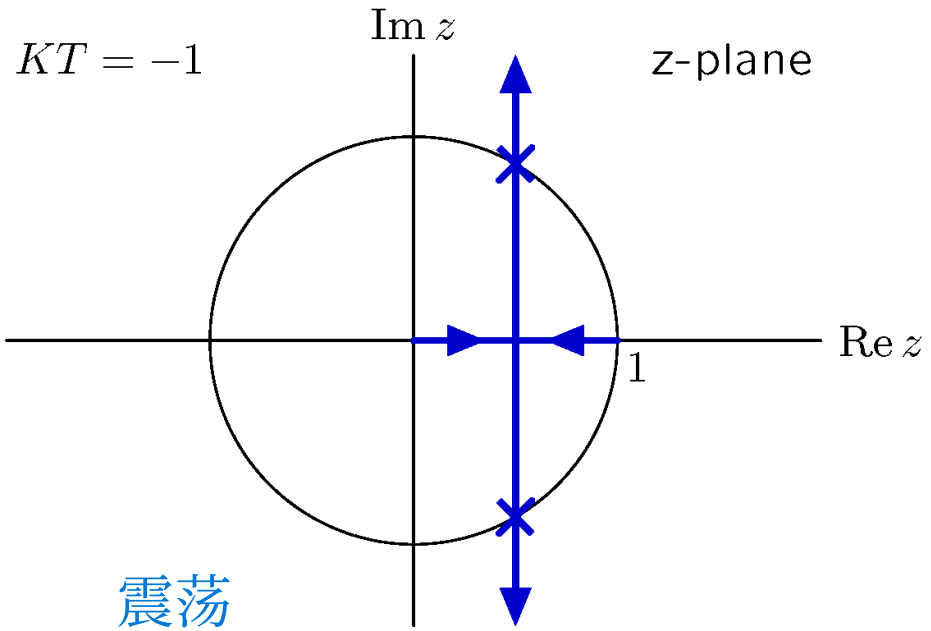


Persistent responses decay. The system is stable.

Feedback and Control: Poles

If $KT < -1/4$, the poles are complex.

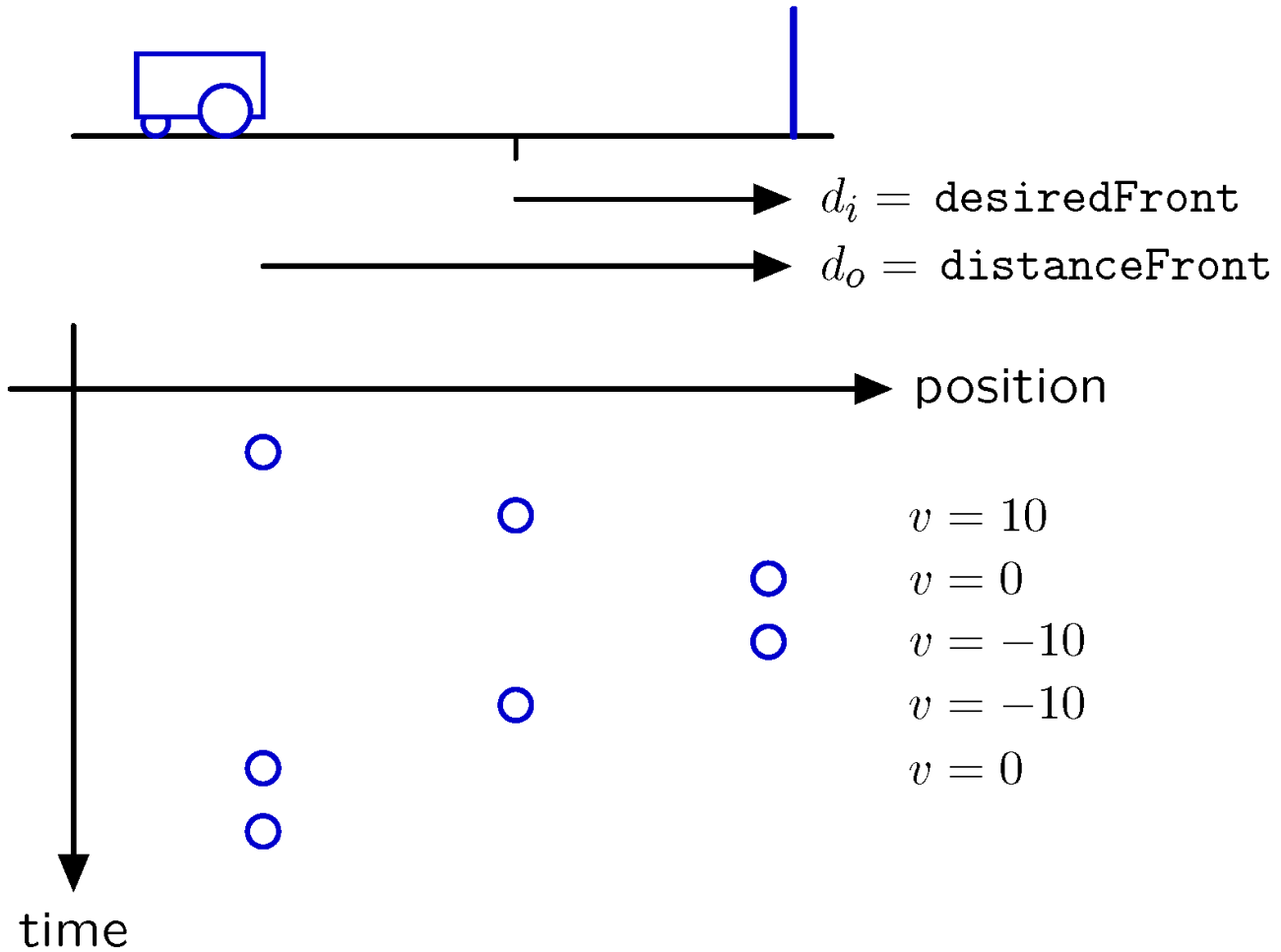
$$z = \tfrac{1}{2} \pm \sqrt{\left(\tfrac{1}{2}\right)^2 + KT} = \tfrac{1}{2} \pm j\sqrt{-KT - \left(\tfrac{1}{2}\right)^2}$$



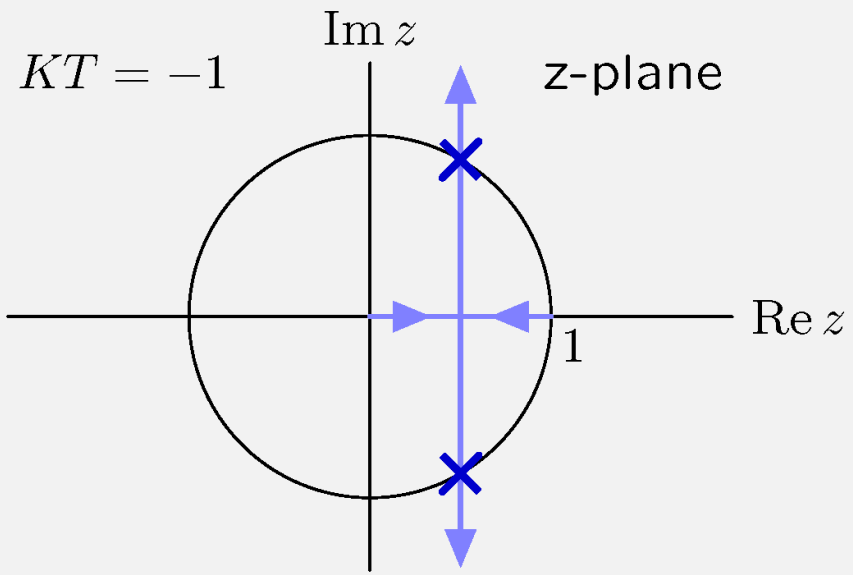
Complex poles $\rightarrow$ oscillations.

Same oscillation we saw earlier!

Adding delay tends to destabilize control systems.



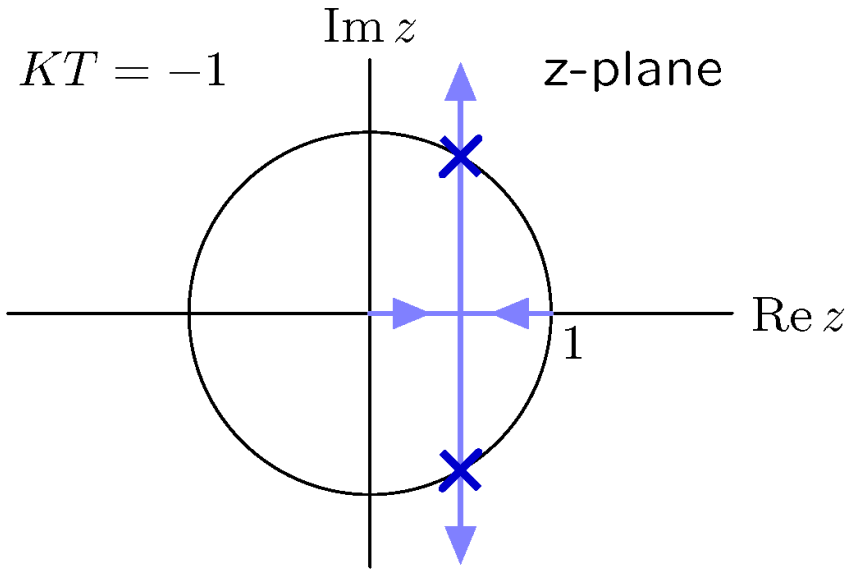
Check Yourself



What is the period of the oscillation?

1. 1 2. 2 3. 3
4. 4 5. 6 0. none of above

Check Yourself

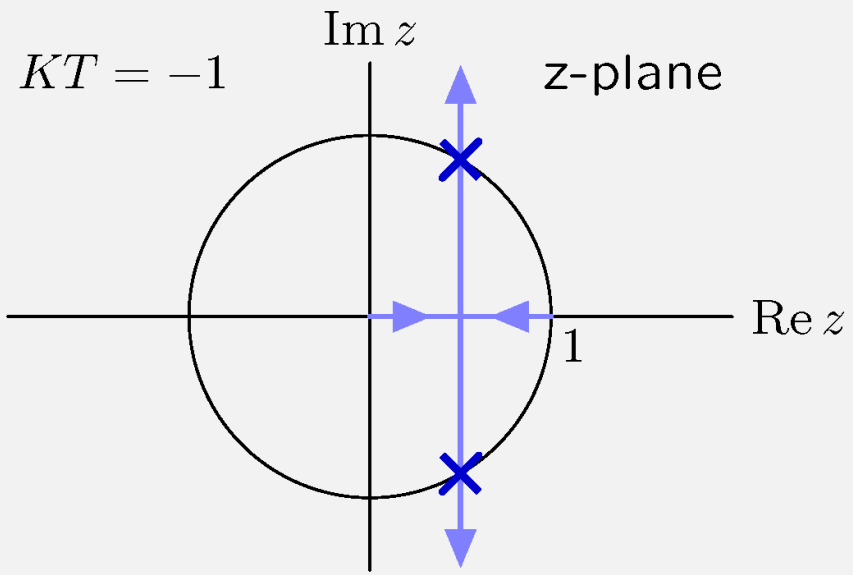


$$p_0 = \frac{1}{2} \pm j\frac{\sqrt{3}}{2} = e^{\pm j\pi/3}$$

$$p_0^n = e^{\pm j\pi n/3}$$

$$\underbrace{e^{\pm j0\pi/3}}_{1},\ e^{\pm j\pi/3},\ e^{\pm j2\pi/3},\ e^{\pm j3\pi/3},\ e^{\pm j4\pi/3},\ e^{\pm j5\pi/3},\ \underbrace{e^{\pm j6\pi/3}}_{e^{\pm j2\pi}=1}$$

Check Yourself

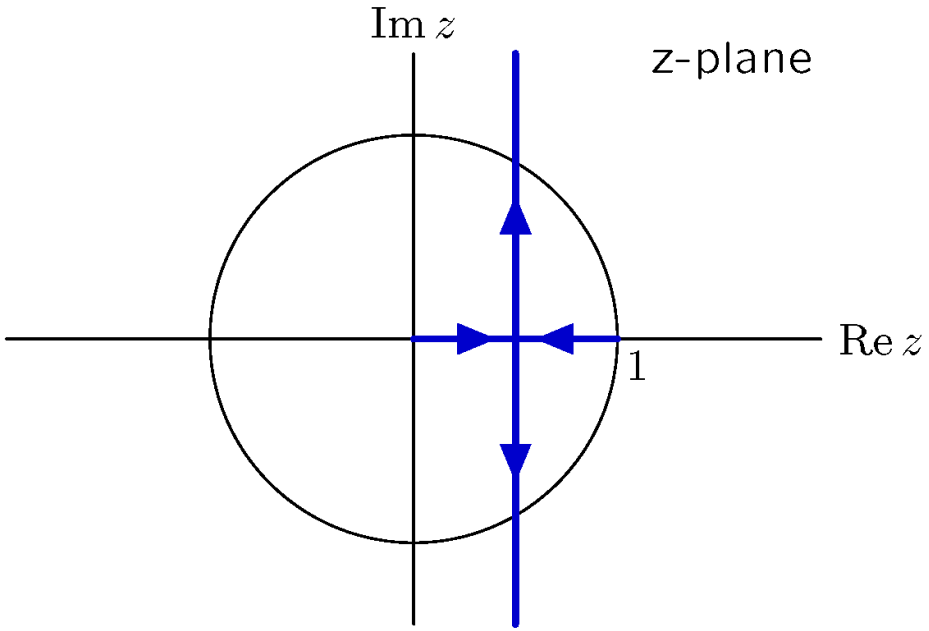


What is the period of the oscillation?

1. 1 2. 2 3. 3
4. 4 5. 6 0. none of above

Feedback and Control: Poles

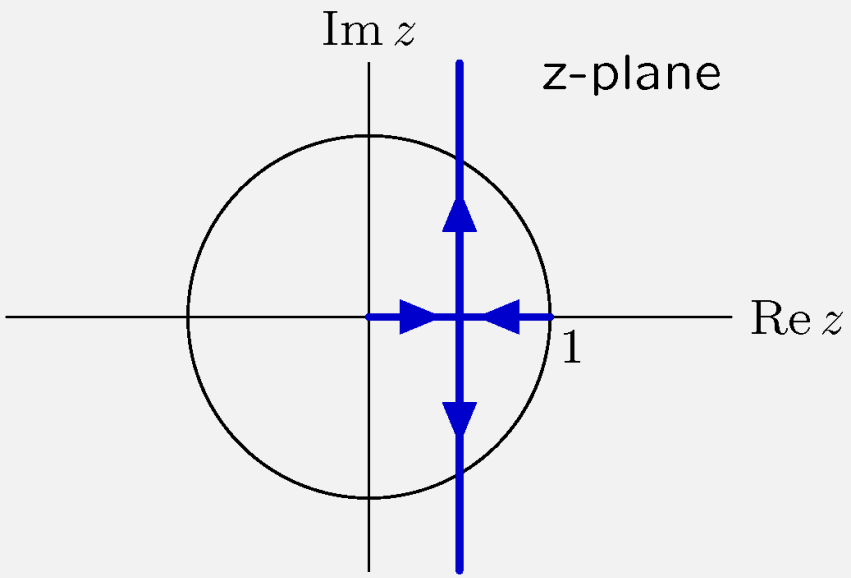
The closed loop poles depend on the gain.



If $KT : 0 \to -\infty$: then $z_1, z_2 : 0, 1 \to \frac{1}{2}, \frac{1}{2} \to \frac{1}{2} \pm j\infty$

Check Yourself

Find KT for fastest response.



closed-loop poles

$$\frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 + KT}$$

1. 0
2. $-\frac{1}{4}$
3. $-\frac{1}{2}$
4. -1
5. $-\infty$
0. none of above

Check Yourself

$$z = \frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 + KT}$$

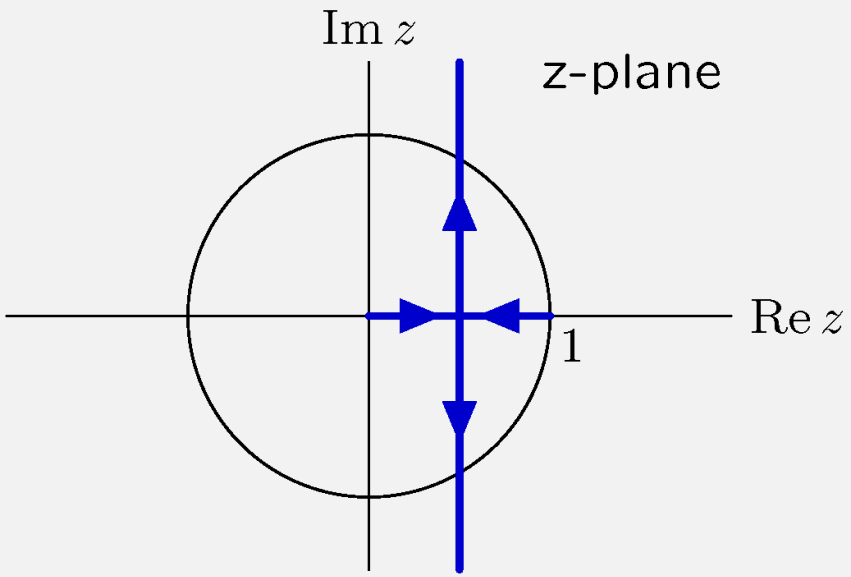
The dominant pole always has a magnitude that is $\geq \frac{1}{2}$.

It is smallest when there is a double pole at $z = \frac{1}{2}$.

Therefore, $KT = -\frac{1}{4}$.

Check Yourself

Find KT for fastest response.



closed-loop poles

$$\frac{1}{2} \pm \sqrt{\left(\frac{1}{2}\right)^2 + KT}$$

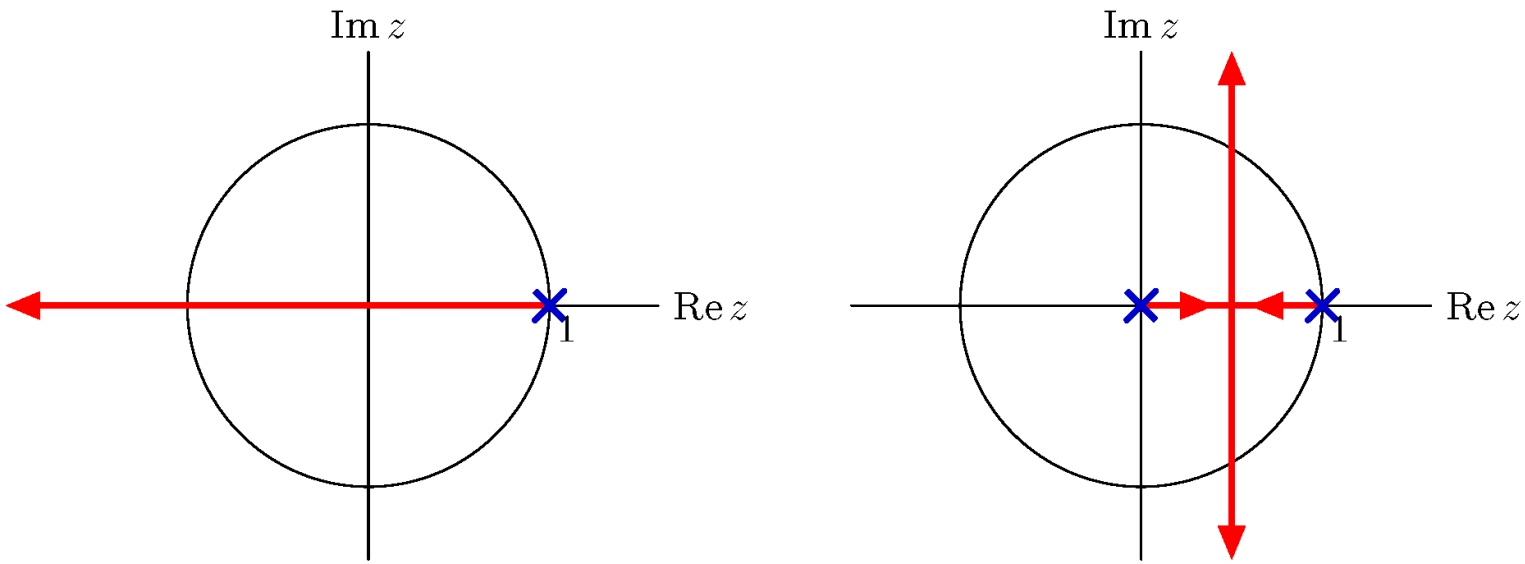
1. 0
2. $-\frac{1}{4}$
3. $-\frac{1}{2}$
4. -1
5. $-\infty$
0. none of above

Destabilizing Effect of Delay

Adding delay in the feedback loop makes it more difficult to stabilize.

Ideal sensor: $d_s[n] = d_o[n]$

More realistic sensor (with delay): $d_s[n] = d_o[n-1]$



Fastest response without delay: single pole at $z = 0$.

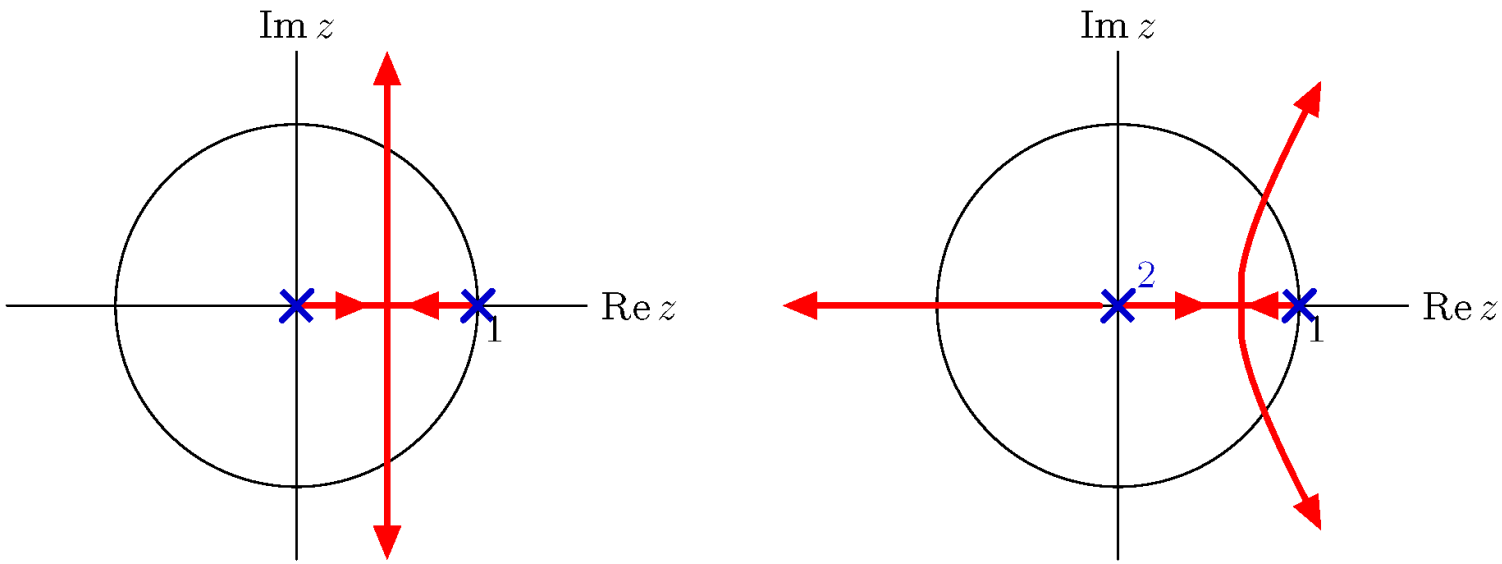
Fastest response with delay: double pole at $z = \frac{1}{2}$. **much slower!**

Destabilizing Effect of Delay

Adding more delay in the feedback loop is even worse.

More realistic sensor (with delay): $d_s[n] = d_o[n-1]$

Even more delay: $d_s[n] = d_o[n-2]$



Fastest response with delay: double pole at $z = \frac{1}{2}$.

Fastest response with more delay: double pole at $z = 0.682$.

→ **even slower**

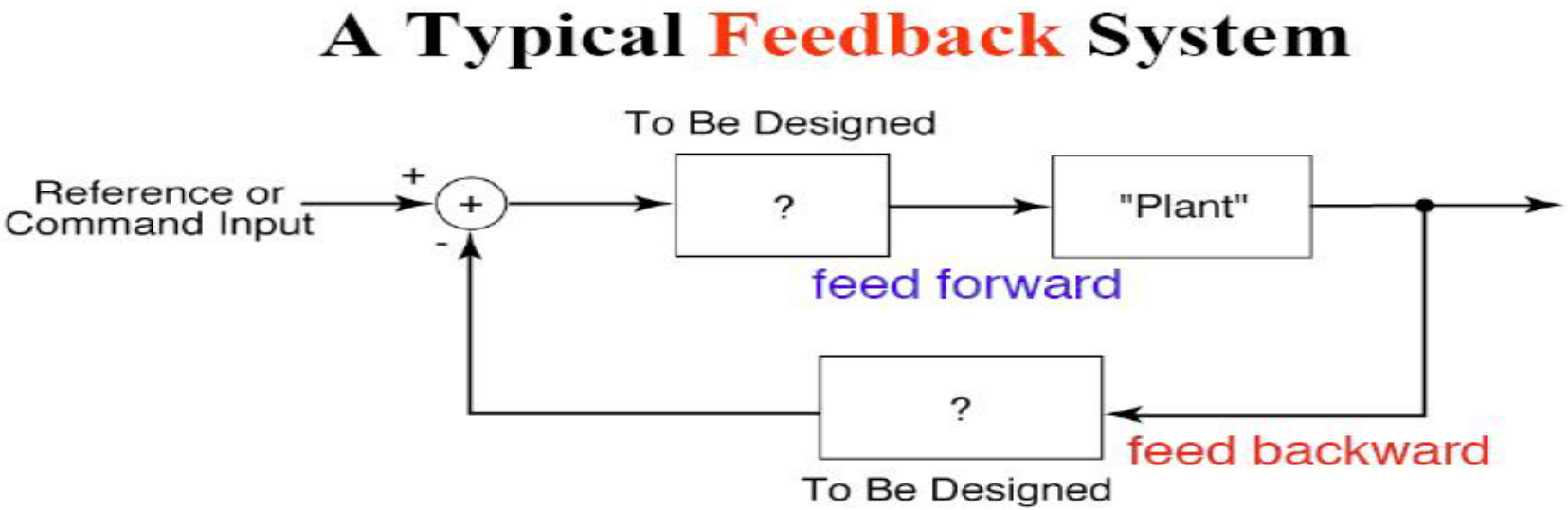
Feedback and Control: Summary

Feedback is an elegant way to design a control system.

Stability of a feedback system is determined by its dominant pole.

Delays tend to decrease the stability of a feedback system.

Feedback and Control: Summary



Why use Feedback?

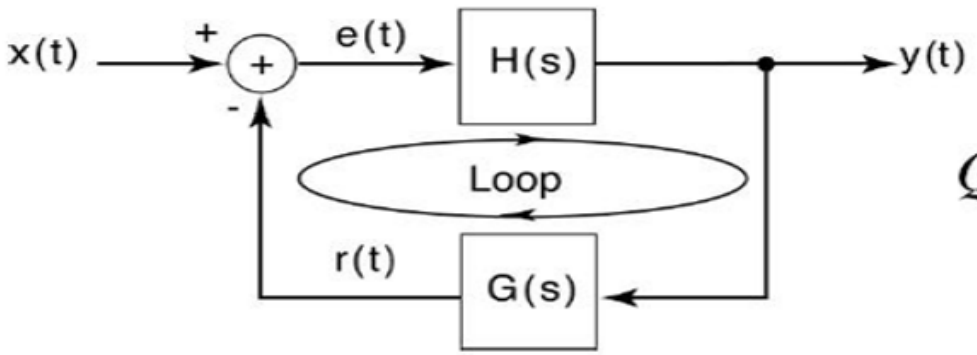
- Reducing Nonlinearities
- Reducing Sensitivity to Uncertainties and Variability
- Stabilizing Unstable Systems
- Reducing Effects of Disturbances
- Tracking
- Shaping System Response Characteristics (bandwidth/speed)

:

Feedback and Control: Summary

General formula for a closed-loop system:

Black's Formula



$$Q(s) = \frac{Y(s)}{X(s)} = \frac{H(s)}{1 + G(s)H(s)}$$

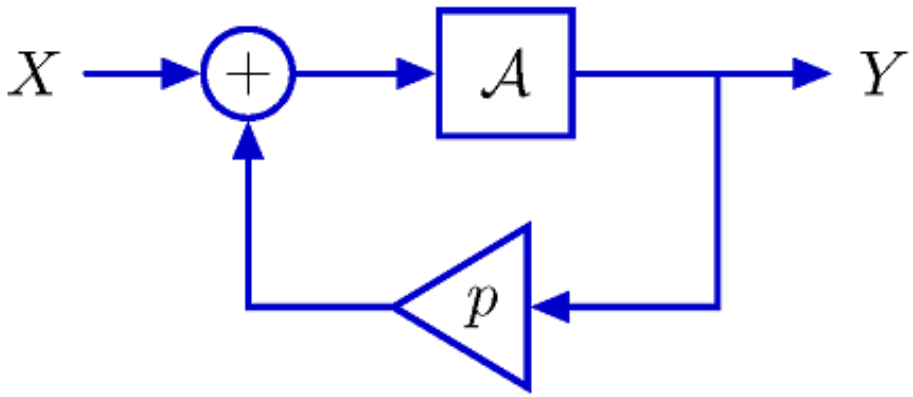
Can show for any closed-loop systems, the system function is given by Black's formula (H. S. Black in the 1920's, along with Nyquist and Bode):

$$\text{Closed - loop system function} = \frac{\text{forward gain}}{1 \text{ - loop gain}}$$

Forward gain — total gain along the forward path from the *input* to the *output*
the gain of an adder is ≡ 1

Loop gain — total gain along the closed loop — shared by all the system functions

Review

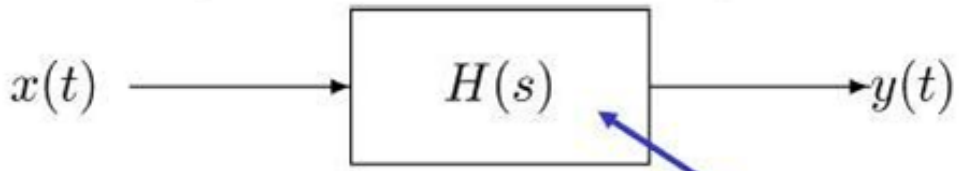


$$\frac{\mathcal{A}}{1-p\mathcal{A}}$$

$$e^{pt}u(t)$$

Pole characteristics	Time function
On real axis	Exponential
On imaginary axis	Sinusoid
In complex s plane	Exponentially modulated sinusoid
Negative real part	Bounded
Positive real parts	Unbounded
Far from origin of s plane	Rapid time course

Review



H(s) = "system function"

$$Y(s) = H(s)X(s)$$

1) System is stable $\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|dt < \infty$ $\Leftrightarrow$ ROC of $H(s)$ includes $j\omega$ axis

2) Causality $\Rightarrow$ $h(t)$ right-sided signal $\Rightarrow$ ROC of $H(s)$ is a right-half plane

Question:

If the ROC of $H(s)$ is a right-half plane, is the system causal?

Ex. $H(s) = \dfrac{e^{sT}}{s+1}, \quad \Re e\{s\} > -1 \Rightarrow h(t)$ right-sided

$$h(t) = \mathcal{L}^{-1}\left\{\frac{e^{sT}}{s+1}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}_{t\to t+T} = e^{-t}u(t)|_{t\to t+T}$$

$$= e^{-(t+T)}u(t+T) \neq 0 \quad \text{at} \quad t < 0$$ Non-causal

Feedback and Control

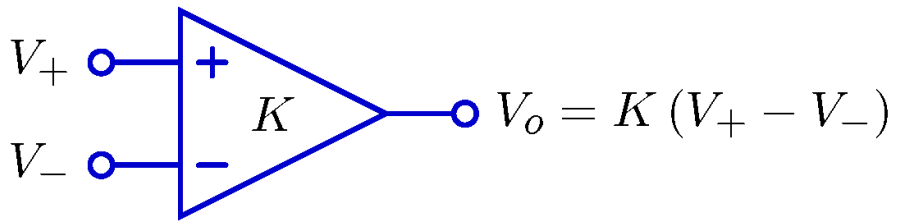
This week: using feedback to enhance performance.

Examples:

- increasing speed and bandwidth
- controlling position instead of speed
- reducing sensitivity to parameter variation
- reducing distortion
- stabilizing unstable systems
  - magnetic levitation
  - inverted pendulum

Op-amps

An "ideal" op-amp has many desireable characteristics.



- high speed
- large bandwidth
- high input impedance
- low output impedance
- ...

It is difficult to build a circuit with all of these features.

LM741 Operational Amplifier

1 Features

- Overload Protection on the Input and Output
- No Latch-Up When the Common-Mode Range is Exceeded

2 Applications

- Comparators
- Multivibrators
- DC Amplifiers
- Summing Amplifiers
- Integrator or Differentiators
- Active Filters

3 Description

The LM741 series are general-purpose operational amplifiers which feature improved performance over industry standards like the LM709. They are direct, plug-in replacements for the 709C, LM201, MC1439, and 748 in most applications.

The amplifiers offer many features which make their application nearly foolproof: overload protection on the input and output, no latch-up when the common-mode range is exceeded, as well as freedom from oscillations.

The LM741C is identical to the LM741 and LM741A except that the LM741C has their performance ensured over a 0°C to +70°C temperature range, instead of −55°C to +125°C.

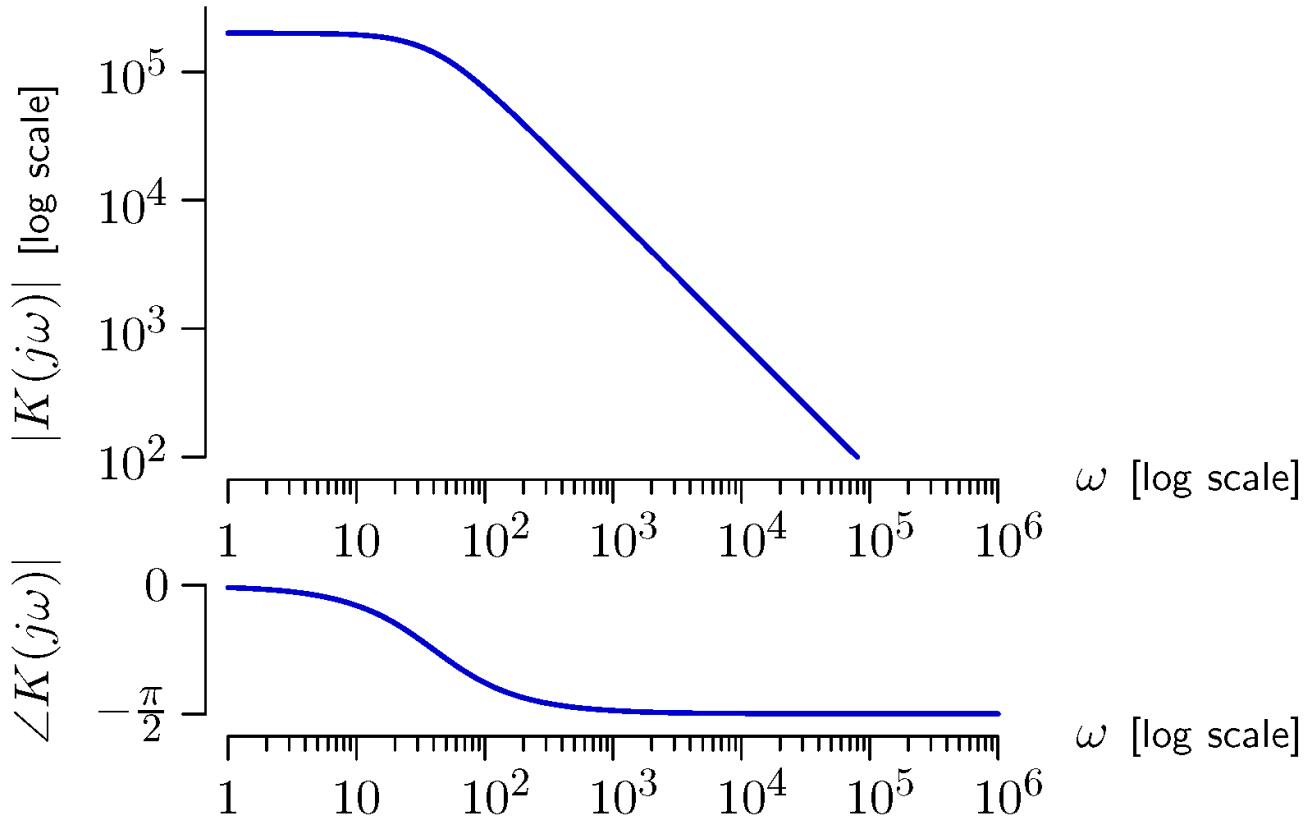
Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
LM741	TO-99 (8)	9.08 mm × 9.08 mm
	CDIP (8)	10.16 mm × 6.502 mm
	PDIP (8)	9.81 mm × 6.35 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Op-Amp

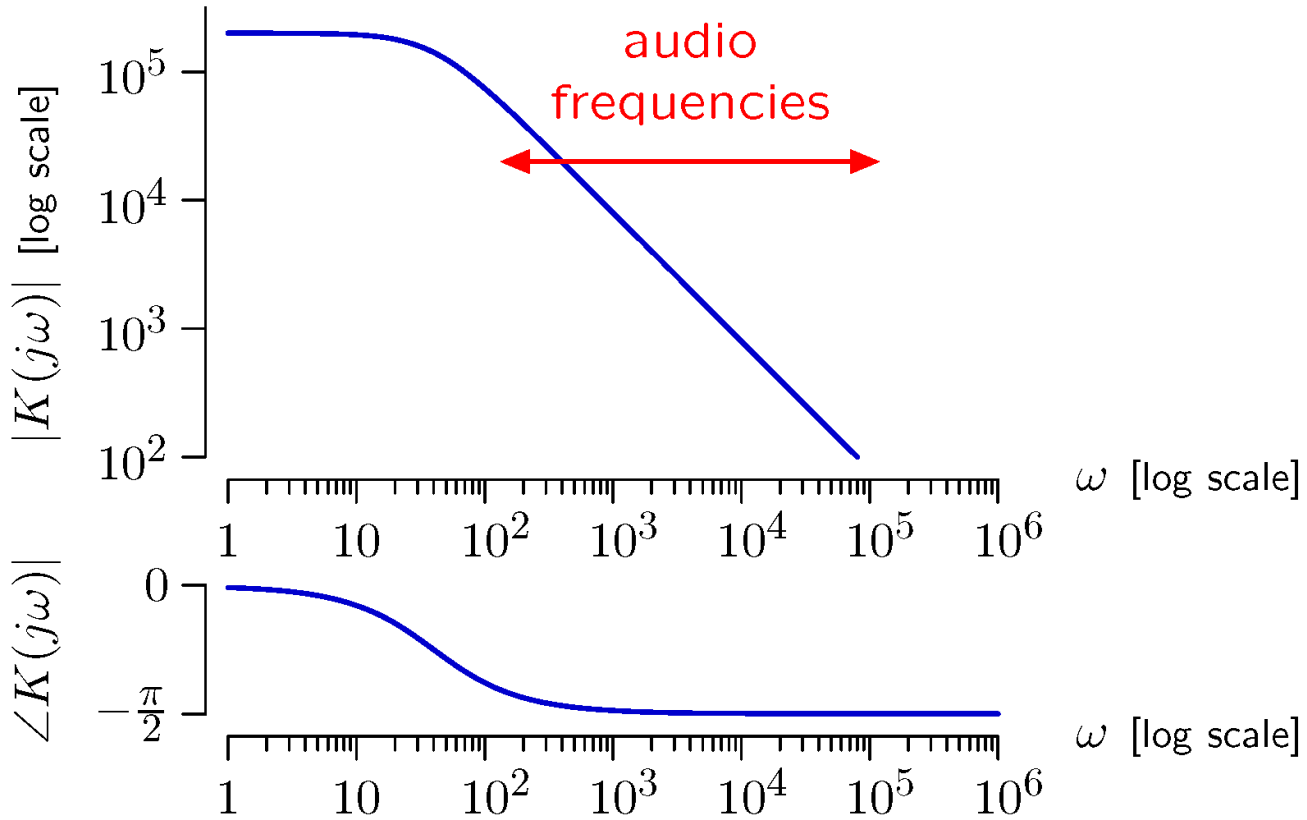
The gain of an op-amp depends on frequency.



Frequency dependence of LM741 op-amp.

Op-Amp

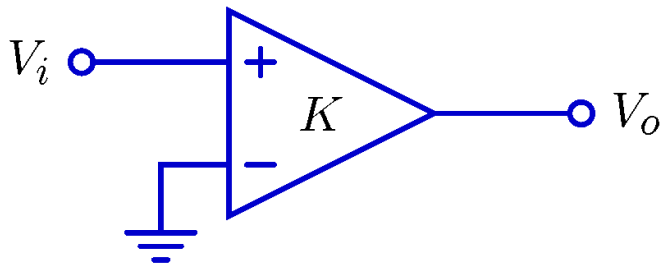
Low-gain at high frequencies limits applications.



Unacceptable frequency response for an audio amplifier.

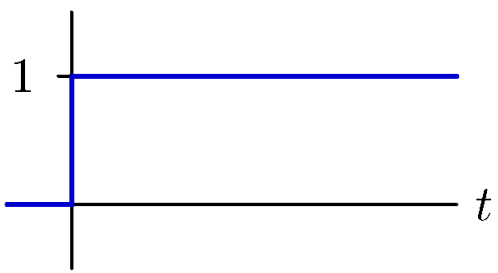
Op-Amp

An ideal op-amp has fast time response.

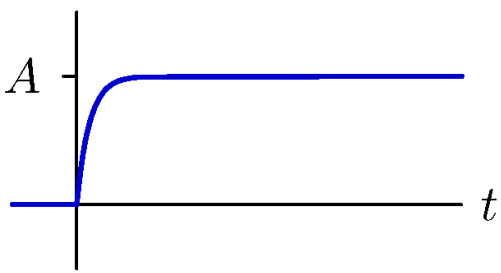


Step response:

$V_i(t) = u(t)$

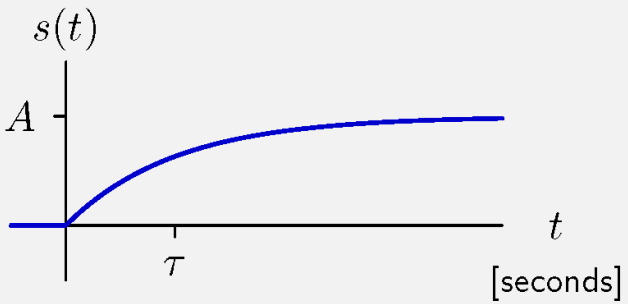


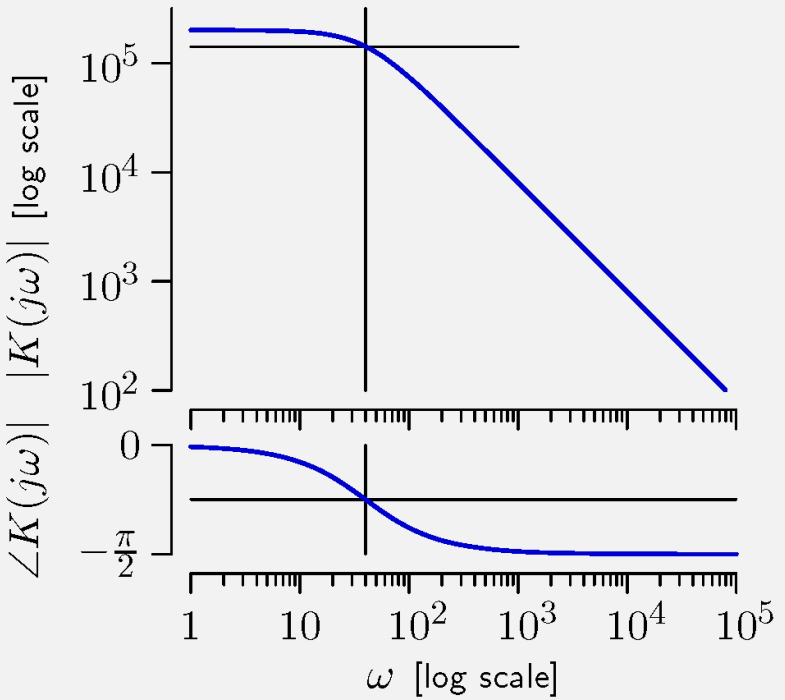
$V_o(t) = s(t)$



Check Yourself

Determine τ for the unit-step response $s(t)$ of an LM741.





1. $40\,\mathrm{s}$ 2. $\frac{40}{2\pi}\,\mathrm{s}$ 3. $\frac{1}{40}\,\mathrm{s}$ 4. $\frac{2\pi}{40}\,\mathrm{s}$ 5. $\frac{1}{2\pi\times 40}\,\mathrm{s}$

0. none of the above

Check Yourself

Determine the step response of an LM741.

System function:

$$K(s) = \frac{\alpha K_0}{s + \alpha}$$

Impulse response:

$$h(t) = \alpha K_0 e^{-\alpha t} u(t)$$

Step response:

$$s(t) = \int_{-\infty}^{t} h(\tau) d\tau = \int_{0}^{t} \alpha K_0 e^{-\alpha \tau} d\tau = \left. \frac{\alpha K_0 e^{-\alpha \tau}}{-\alpha} \right|_{0}^{t} = K_0 (1 - e^{-\alpha t}) u(t)$$

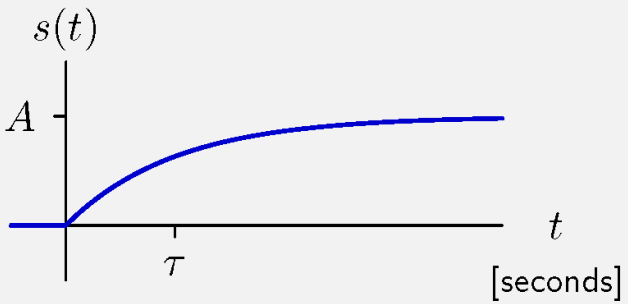
Parameters:

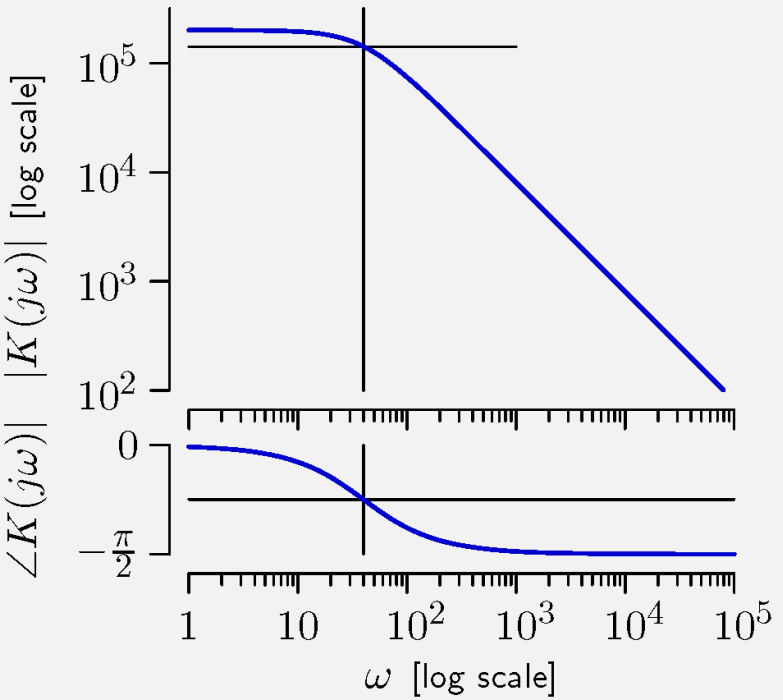
$$A = K_0 = 2 \times 10^5$$

$$\tau = \frac{1}{\alpha} = \frac{1}{40}\,\mathrm{s}$$

Check Yourself

Determine τ for the unit-step response $s(t)$ of an LM741.





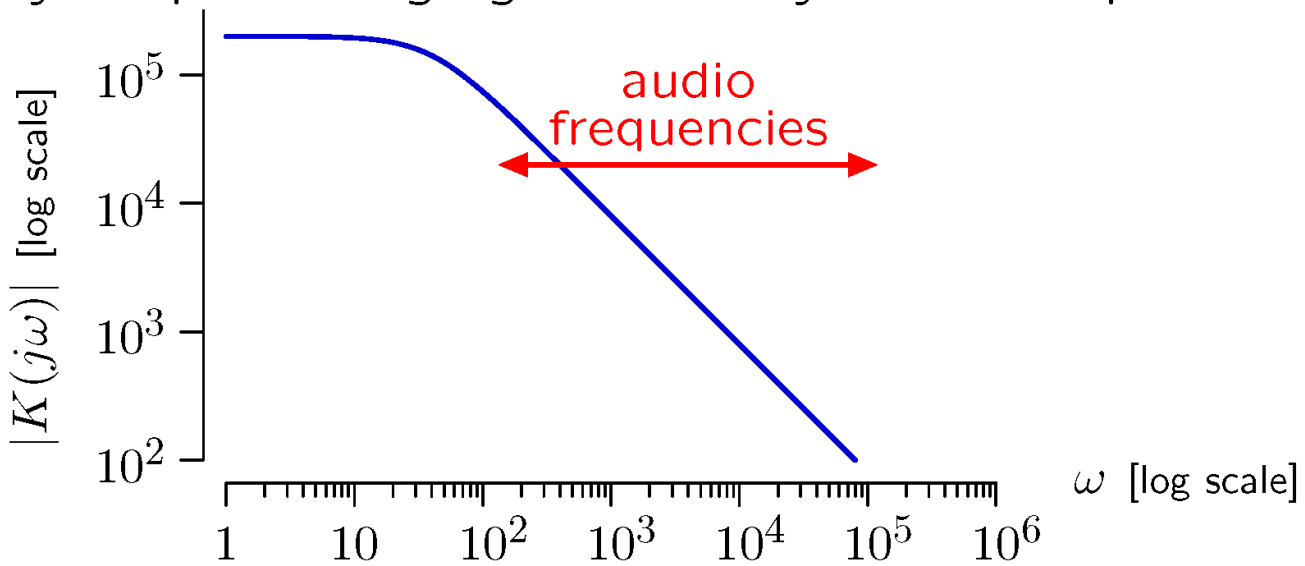
1. $40\,\mathrm{s}$ 2. $\frac{40}{2\pi}\,\mathrm{s}$ 3. $\frac{1}{40}\,\mathrm{s}$ 4. $\frac{2\pi}{40}\,\mathrm{s}$ 5. $\frac{1}{2\pi\times 40}\,\mathrm{s}$

0. none of the above

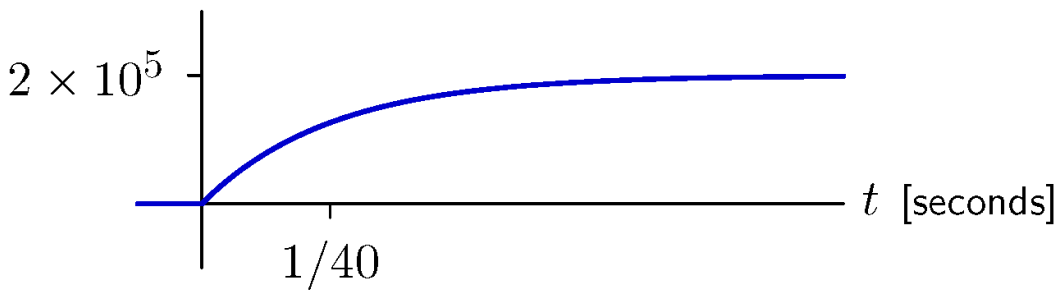
Op-Amp

Performance parameters for real op-amps fall short of the ideal.

Frequency Response: high gain but only at low frequencies.



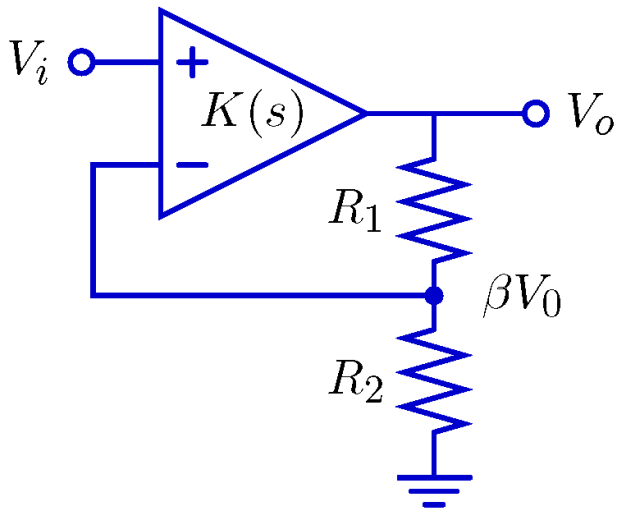
Step Response: slow by electronic standards.



Op-Amp

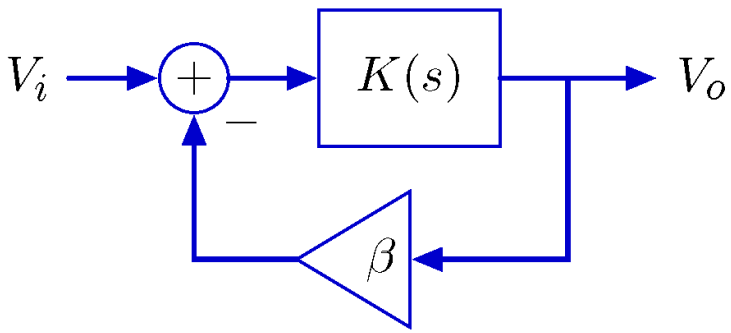
We can use feedback to improve performance of op-amps.

circuit



$$V_- = \beta V_o = \left(\frac{R_2}{R_1 + R_2}\right) V_o$$

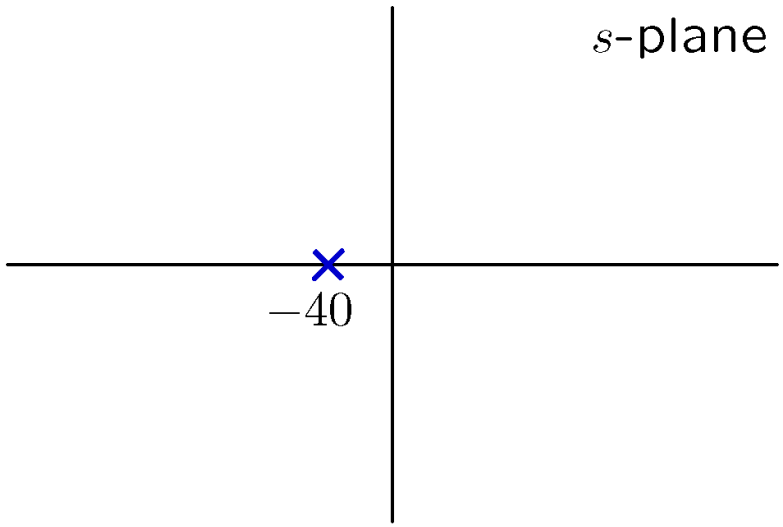
6.003 model



$$\frac{V_o}{V_i} = \frac{K(s)}{1 + \beta K(s)}$$

Dominant Pole

Op-amps are designed to have a dominant pole at low frequencies:
$\rightarrow$ simplifies the application of feedback.

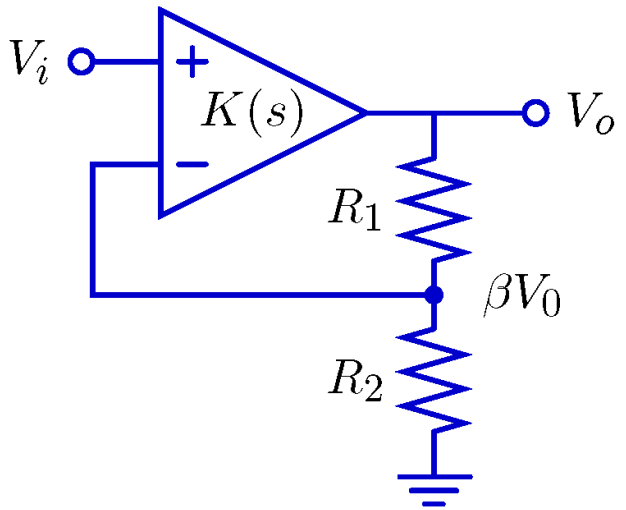


$$\alpha = 40\,\text{rad/s} = \frac{40\,\text{rad/s}}{2\pi\,\text{rad/cycle}} \approx 6.4\,\text{Hz}$$

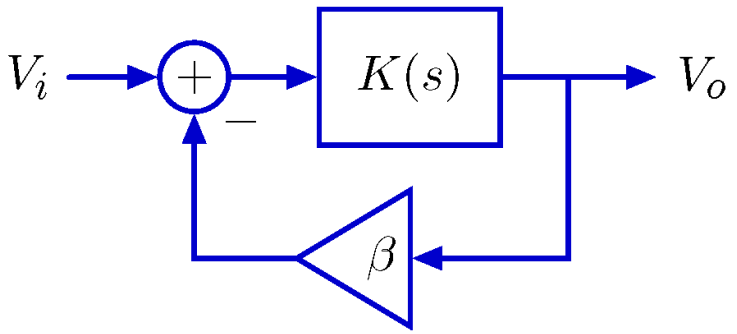
Improving Performance

Using feedback to improve performance parameters.

circuit



6.003 model

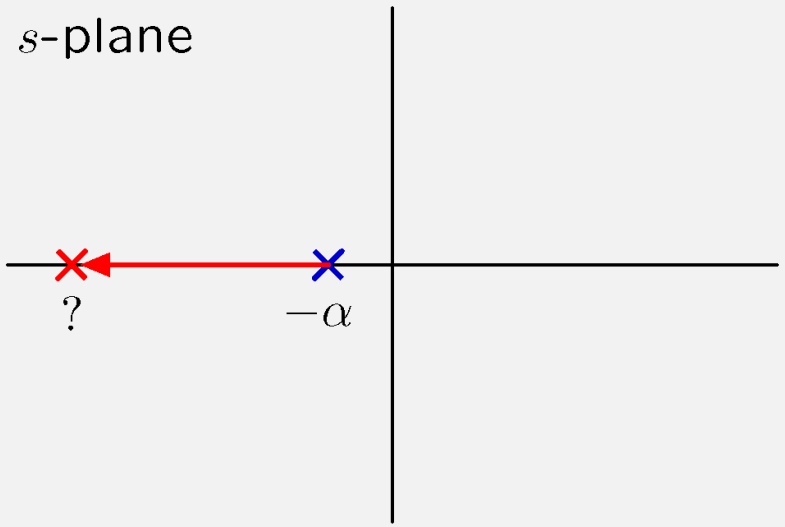


$$V_- = \beta V_o = \left(\frac{R_2}{R_1 + R_2}\right) V_o$$

$$\begin{aligned}\frac{V_o}{V_i} &= \frac{K(s)}{1 + \beta K(s)} \\ &= \frac{\frac{\alpha K_0}{s+\alpha}}{1 + \beta\frac{\alpha K_0}{s+\alpha}} \\ &= \frac{\alpha K_0}{s + \alpha + \alpha\beta K_0}\end{aligned}$$

Check Yourself

What is the most negative value of the closed-loop pole that can be achieved with feedback?

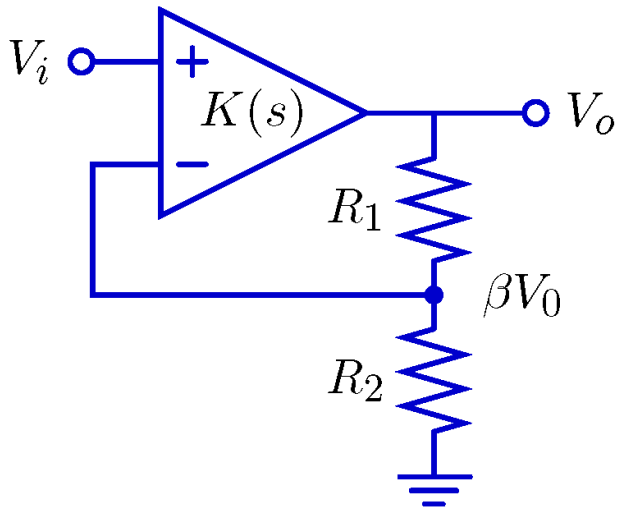


1. $-\alpha(1+\beta)$
2. $-\alpha(1+\beta K_0)$
3. $-\alpha(1+K_0)$
4. $-\infty$
5. none of the above

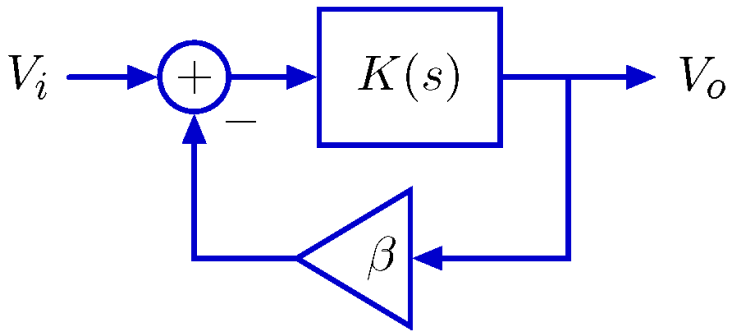
Improving Performance

Using feedback to improve performance parameters.

circuit



6.003 model



$$V_- = \beta V_o = \left(\frac{R_2}{R_1 + R_2}\right) V_o$$

$$\begin{aligned}\frac{V_o}{V_i} &= \frac{K(s)}{1 + \beta K(s)} \\ &= \frac{\frac{\alpha K_0}{s+\alpha}}{1 + \beta\frac{\alpha K_0}{s+\alpha}} \\ &= \frac{\alpha K_0}{s + \alpha + \alpha\beta K_0}\end{aligned}$$

Check Yourself

What is the most negative value of the closed-loop pole that can be achieved with feedback?

Open loop system function: $\dfrac{\alpha K_0}{s+\alpha}$

$\rightarrow$ pole: $s = -\alpha$.

Closed-loop system function: $\dfrac{\alpha K_0}{s+\alpha+\alpha\beta K_0}$

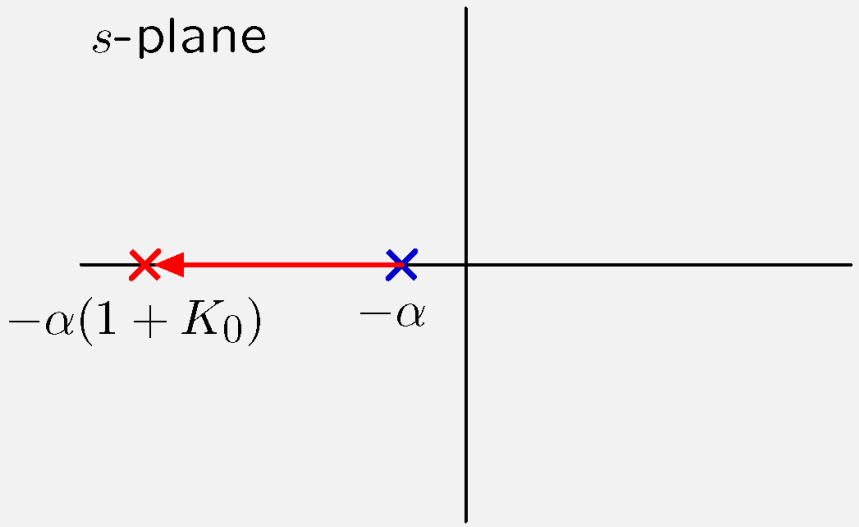
$\rightarrow$ pole: $s = -\alpha(1+\beta K_0)$.

The feedback constant is $0 \leq \beta \leq 1$.

$\rightarrow$ most negative value of the closed-loop pole is $s = -\alpha(1+K_0)$.

Check Yourself

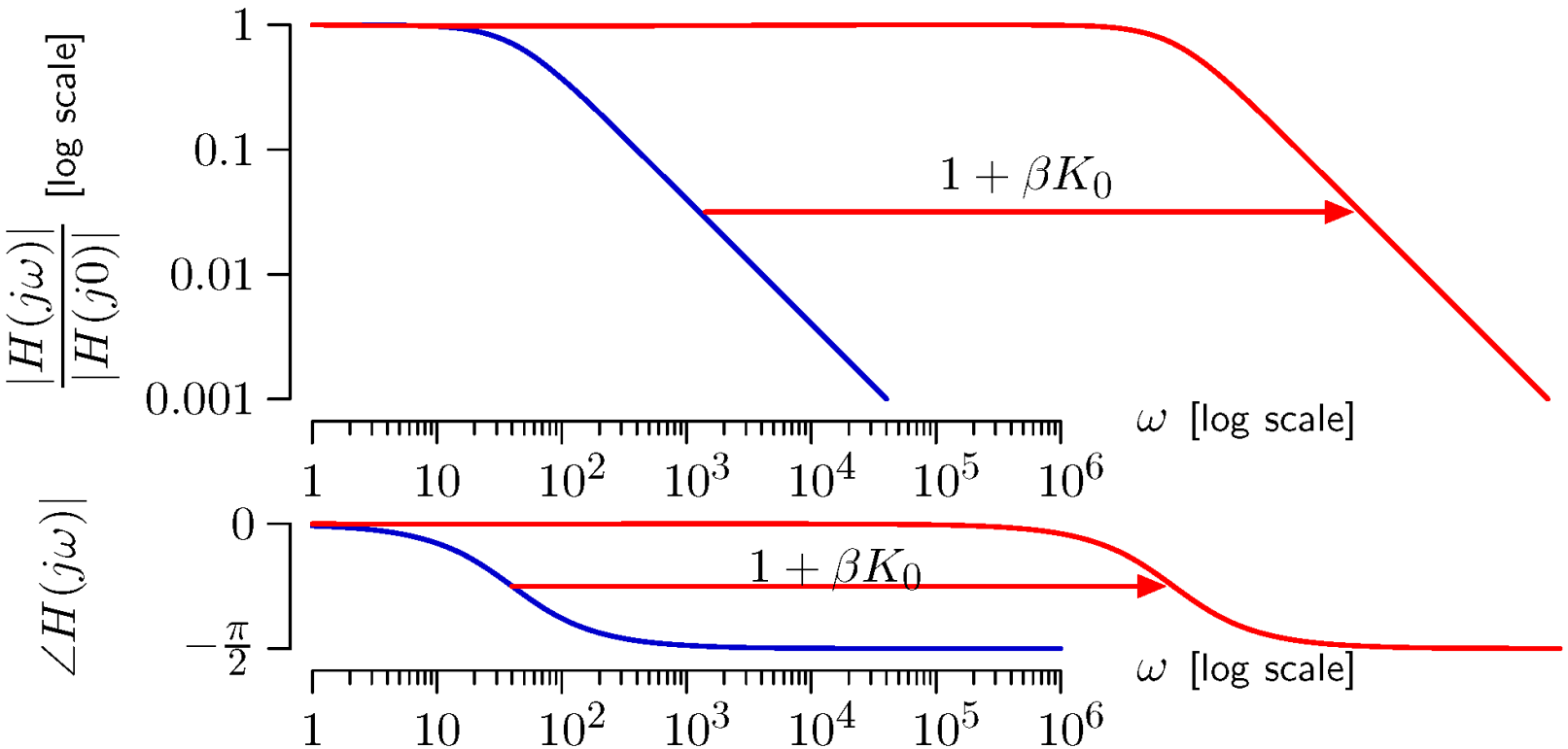
What is the most negative value of the closed-loop pole that can be achieved with feedback? 3



1. $-\alpha(1+\beta)$
2. $-\alpha(1+\beta K_0)$
3. $-\alpha(1+K_0)$
4. $-\infty$
5. none of the above

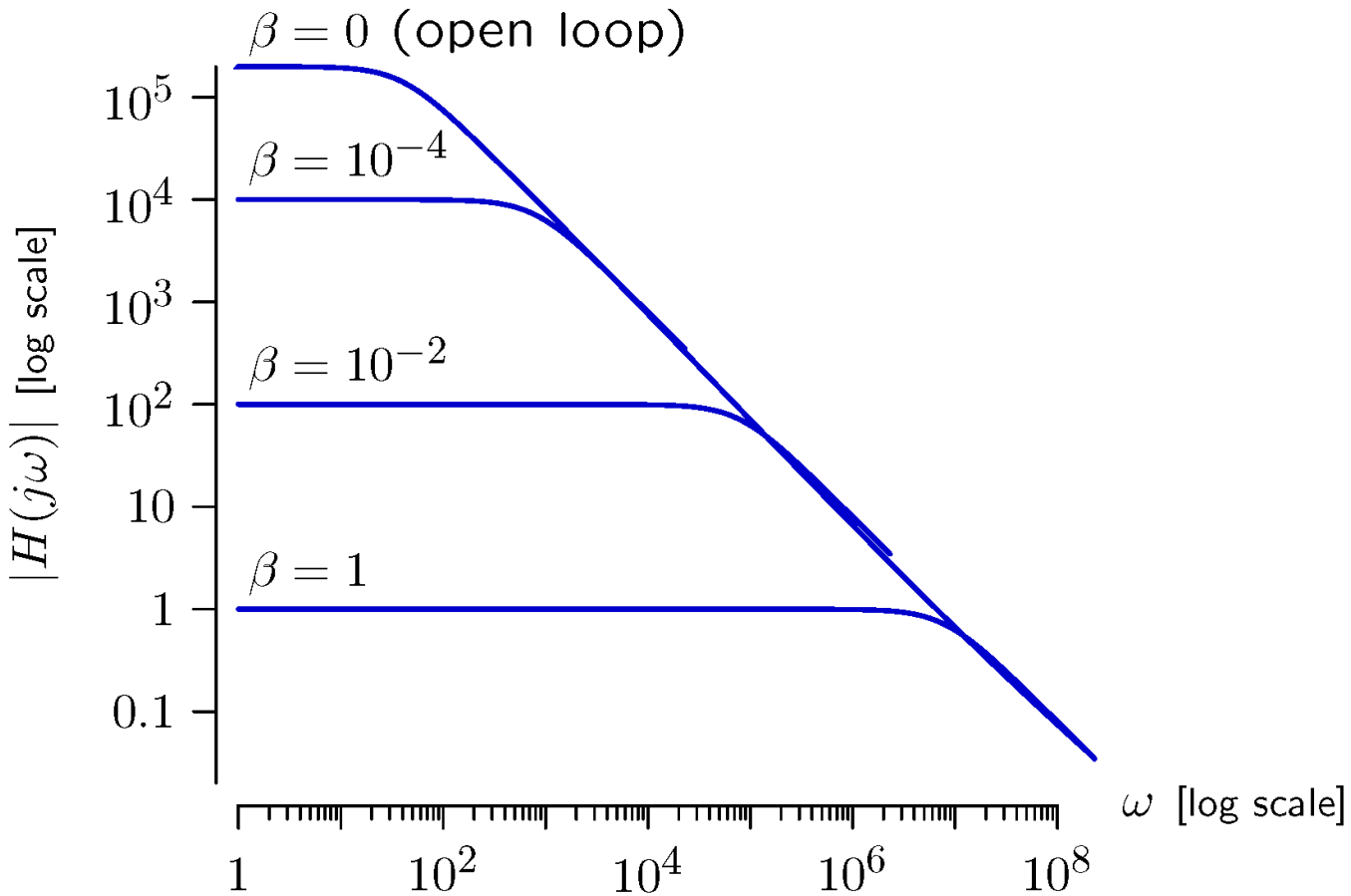
Improving Performance

Feedback extends frequency response by a factor of $1 + \beta K_0$ ($K_0 = 2 \times 10^5$).



Improving Performance

Feedback produces higher bandwidths by **reducing** the gain at low frequencies. It trades gain for bandwidth.

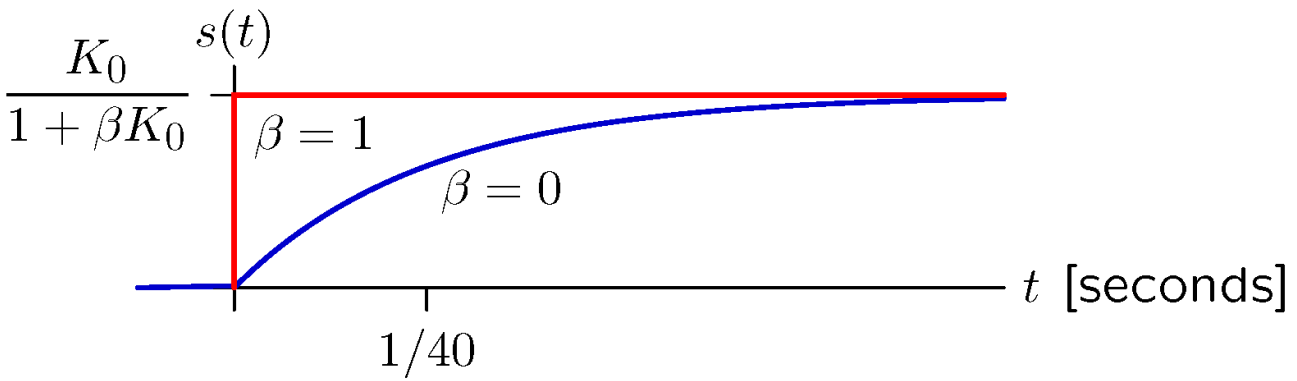


Improving Performance

Feedback makes the time response faster by a factor of $1 + \beta K_0$ ($K_0 = 2 \times 10^5$).

Step response

$$s(t) = \frac{K_0}{1 + \beta K_0}(1 - e^{-\alpha(1+\beta K_0)t})u(t)$$

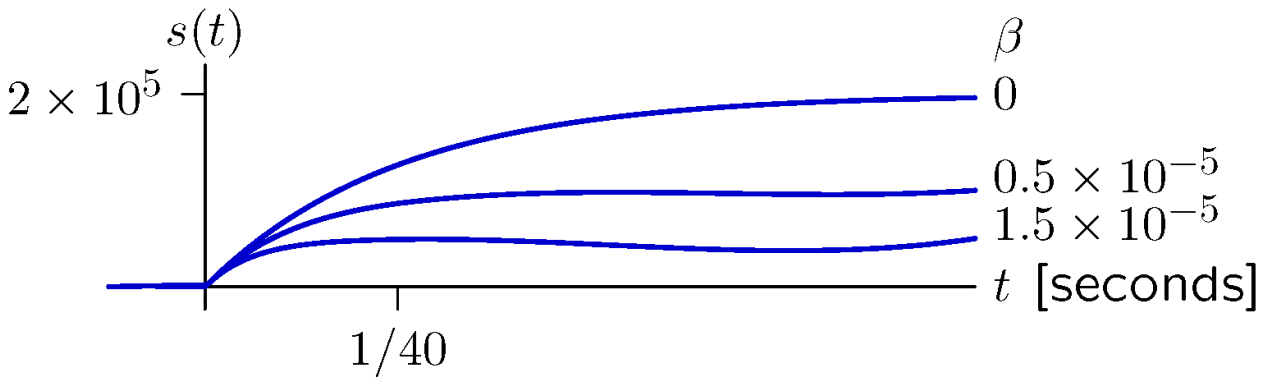


Improving Performance

Feedback produces faster responses by **reducing** the final value of the step response. It trades gain for speed.

Step response

$$s(t) = \frac{K_0}{1+\beta K_0}(1 - e^{-\alpha(1+\beta K_0)t})u(t)$$



The maximum rate of voltage change $\left.\frac{ds(t)}{dt}\right|_{t=0+}$ is not increased.

Improving Performance

Feedback improves performance parameters of op-amp circuits.

- can extend frequency response
- can increase speed

Performance enhancements are achieved through a reduction of gain.

Assignments

- Reading Assignment: Chap. 11.0-11.2
- Homework 4
- Mid-term Exam: 04/18/2025, 19:00-21:00, Location: TBD